S.W.A.R.A.J

System for Warfare Autonomous Reconnaissance and Adaptive Judgement

अहिंसा परमो धर्मस्तथाहिंसा परो दमः।
अहिंसा परमं दानमहिंसा परमं तपः॥
(Mahabharata, Anushasana Parva, 13.117.37)

SWARAJ

The name “SWARAJ” draws inspiration from the concept of “Hindavi Swarajya” Drafted by Chhatrapati Shivaji Maharaj. Rooted in the Ideology of self-rule by the people or self-governance, Swarajya represents the ability to exercise independent authority and make decisions without external dependence. This philosophy forms the conceptual foundation of SWARAJ, where the principle of self-governance is translated into autonomous decision-making within an aerospace engineering framework. The concept therefore represents an amalgamation of India's historical legacy with modern engineering principles, reflecting the pursuit of intelligent, independent and responsible autonomous systems, hence I start this report with the maratha phrase जय भवानी! जय शिवाजी!

Executive Summary

SWARAJ (System for Warfare Autonomous Reconnaissance and Adaptive Judgment) is an autonomous guidance and flight-control framework developed to improve the operational independence and adaptability of unmanned aerial vehicles. The broader project envisions applications in autonomous reconnaissance and information-oriented defence operations. This Preliminary Design Review (PDR) concentrates specifically on establishing the Autonomous Adaptive Judgment capability, investigating how environmental information, behavioural supervision and aircraft-control functions can be coordinated to support independent flight-control decisions under changing operating conditions.

Building upon the earlier Conceptual Design Review (CDR), which demonstrated behavioural control using simplified three-degree-of-freedom aircraft dynamics, the present research advances SWARAJ through integration with a validated nonlinear six-degree-of-freedom Cessna 182 aircraft model. The developed architecture combines a mathematical Environment Map, predictive Look-Ahead guidance, behavioural supervision, Banked Dynamics, coordinated lateral and longitudinal control, energy management and supplementary gust recovery. Together, these functions translate environmental and aircraft-state information into autonomous aerodynamic control-surface and propulsion commands.

The methodology involved aircraft-model verification, controller sensitivity analysis and closed-loop validation under nominal and disturbed flight conditions. Nominal valley-following simulations achieved a maximum active cross-track error of 34.802 m, an RMS error of 22.537 m and a maximum altitude deviation of 1.000 m. Airspeed remained between 66.888 and 67.313 m/s, while maximum bank angle was limited to 27.347°, satisfying the prescribed nominal performance requirements. These results demonstrate a substantial progression from the CDR's recorded peak cross-track error of 175.5 m, although differences in aircraft models and test configurations prevent a direct performance-equivalent comparison.

Independent and combined altitude and airspeed command-deflection tests demonstrated effective longitudinal control coordination while maintaining lateral trajectory tracking. Atmospheric validation across four prescribed gust conditions further established the benefits of supplementary recovery intervention. Enabling the Gust Recovery Controller reduced peak cross-track error by 45.88% under lateral disturbances and 58.68% under combined disturbances. All four GRC-enabled cases maintained peak cross-track error below 50 m and minimum airspeed above 58 m/s. However, vertical and combined gust responses produced transient altitude deviations of 10.502 m and 11.333 m, respectively, exceeding the selected 10 m criterion and identifying opportunities for further longitudinal recovery refinement.

The research demonstrates achievement of the three principal objectives within the defined simulation scope: behavioural flight-control development, quantitative supervisory-response evaluation and nonlinear aircraft validation. Future development will address route adaptability, lateral throttle coordination, predictive gust recovery, aircraft-specific manoeuvring supervision and controller parameterisation, followed by sensor integration and hardware-in-the-loop validation.Overall, the PDR establishes SWARAJ as an integrated and

quantitatively evaluated autonomous flight-control architecture. By demonstrating how environmental interpretation and adaptive supervisory decisions can influence aircraft behaviour through coordinated control, the research provides a validated preliminary foundation for the broader reconnaissance and information-oriented capabilities envisioned under SWARAJ.

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1. Introduction

SWARAJ (System for Warfare Autonomous Reconnaissance and Adaptive Judgment) is an autonomous guidance and flight-control framework developed to improve the operational independence and adaptability of unmanned aerial vehicles. The project is founded on the principle that an autonomous aircraft should be capable of more than following predetermined flight commands; it should also interpret its operating conditions, anticipate manoeuvre requirements and independently coordinate its control responses. SWARAJ seeks to achieve this through the integration of environmental guidance, behavioural supervision, predictive control and aircraft dynamics within a common architecture. Although conventional autopilot systems already employ feedback control to regulate aircraft motion, the proposed framework investigates how environmental and aircraft-state information can influence the priorities of multiple flight-control functions. Within SWARAJ's broader vision, warfare refers primarily to the role of information, reconnaissance and situational awareness in defence operations rather than direct combat engagement. The present research concentrates on establishing the Autonomous Adaptive Judgment component through simulation-based guidance and flight-control development, while autonomous reconnaissance and information-oriented applications remain objectives for subsequent stages.

The relevance of such research extends to defence, general aviation and emergency-response operations, where aircraft may be required to navigate challenging environments with limited continuous operator intervention. For India, these capabilities are particularly significant given the mountainous terrain, difficult atmospheric conditions and geographical inaccessibility associated with its northern border regions. The 1999 Kargil conflict demonstrated the consequences of limitations in intelligence gathering and surveillance, with the subsequent Kargil Review Committee identifying the need to strengthen aerial and space-based reconnaissance capabilities, including the employment of unmanned aerial vehicles (Kargil Review Committee, 2000). More recently, the confrontation in the Yangtse area of Arunachal Pradesh's Tawang sector on 9 December 2022 highlighted the continuing sensitivity of these border regions (Ministry of Defence, 2022). Such circumstances reinforce the importance of reliable information gathering and sustained situational awareness. Beyond defence applications, similar requirements arise in search and rescue, environmental monitoring and other operations where dependable autonomous flight could support access to difficult locations.

The development of SWARAJ began with the Conceptual Design Review (CDR), which investigated the feasibility of behavioural supervision using a simplified three-degree-of-freedom aircraft model. While the conceptual framework demonstrated the ability to modify flight-control priorities in response to prescribed operating conditions, limitations in aerodynamic representation, propulsion response and coupled aircraft motion highlighted the need for further development. This Preliminary Design Review (PDR) advances the research through the integration of SWARAJ with a validated nonlinear six-degree-of-freedom Cessna 182 aircraft model. The resulting architecture incorporates an Environment Map, predictive Look-Ahead guidance, Banked Dynamics, lateral and longitudinal controllers, energy management and supplementary Gust Recovery Control. These subsystems work together to translate route geometry, aircraft-state information and changing flight requirements into coordinated aerodynamic control-surface and propulsion commands.

The research is structured around three principal objectives: developing an integrated behavioural supervisory-control framework, quantitatively evaluating its response and demonstrating its operation under nonlinear aircraft dynamics. To assess these objectives, the PDR examines controller parameter sensitivity, nominal valley-route tracking, independent and simultaneous altitude and speed command variations, and atmospheric disturbances under four prescribed gust conditions. Particular attention is given to cross-track error, altitude and airspeed regulation, aircraft manoeuvring response, control authority and disturbance recovery. The findings provide a quantitative assessment of the capabilities established during this development stage while identifying areas requiring further refinement, including lateral throttle coordination, gust-induced altitude damping, manoeuvring-envelope supervision and adaptation to different aircraft and mission geometries.

The report therefore presents SWARAJ's progression from a conceptual behavioural-control approach towards an integrated and quantitatively evaluated autonomous flight-control architecture. By establishing how environmental guidance, supervisory judgment and aircraft-control execution can function together, the research provides a preliminary engineering foundation for the wider capabilities envisioned under SWARAJ. The following sections examine the relevant research literature and objectives, establish the industrial requirements motivating the project, present the control methodology and evaluate the resulting simulation performance before discussing future development.

2. Literature Review and Objectives

2.1 Literature Review

The development of autonomous aircraft systems has been supported by advances in hierarchical control, behavioural decision-making and environment-responsive guidance. These approaches establish mechanisms through which unmanned aircraft interpret operating conditions, select appropriate manoeuvres and execute flight-control commands. However, the coordination between environmental interpretation, supervisory decisions and the operating priorities of underlying flight-control functions requires further investigation, particularly in conditions where maintaining navigation accuracy, aircraft stability and acceptable flight performance demands simultaneous control actions.

Gutiérrez, Vachtsevanos and Heck (2003) investigated hierarchical UAV control through mission planning, task coordination and adaptive mode transitions. Dong and Sun (2004) developed a behaviour-based architecture connecting autonomous task execution and event handling with vehicle-control functions. Vachtsevanos et al. (2005) examined the integration of mission planning, trajectory generation, navigation, adaptive control and flight-mode management within a hierarchical architecture. Samar and Rehman (2011) demonstrated autonomous terrain-following guidance, translating terrain information into flight references executed through an aircraft-control system. These studies establish the principal research foundations for investigating coordinated behavioural supervision within autonomous aircraft guidance and control. Their respective contributions and the research opportunities identified through critical analysis are summarized in Table 1.

Table 1 Foundational research contributions and identified research gaps

Research reference	Established contribution	Research gap identified
Gutiérrez, Vachtsevanos and Heck (2003)	Hierarchical UAV architecture integrating mission planning, task coordination, stabilization and adaptive mode transitions.	The specific relationship between environmental risk and coordinated changes in the priorities of multiple aerodynamic and propulsion-control functions is not the principal subject of investigation.
Dong and Sun (2004)	Behaviour-based architecture linking autonomous tasks, event-driven behaviour and vehicle-control execution.	Systematic quantitative evaluation of the complete response from environmental condition and behavioural transition to multi-axis actuator action and aircraft recovery is not the principal focus.
Vachtsevanos et al. (2005)	Integrated framework incorporating mission planning, guidance, flight-mode transitions, adaptation, envelope protection and fault management.	The individual contribution of environmental behavioural decisions to changes in underlying control priorities and the resulting aircraft response is not isolated as a separate quantitative investigation.
Samar and Rehman (2011)	Integrated terrain-following guidance converting environmental information into trajectory references and corresponding aircraft-control actions.	Environmental information principally influences trajectory generation, while explicit behavioural modification of the priorities of multiple underlying flight-control functions is not the central research objective.

The identified gaps establish the need to investigate behavioural supervision as a coordinated flight-control function rather than solely as a mechanism for navigation or operational mode selection. Furthermore, the effectiveness of the resulting control actions must be assessed quantitatively through aircraft-state response, actuator demand and recovery characteristics. Aircraft-model fidelity introduces an additional consideration. Meola, Iannelli and Glielmo (2013) demonstrated the application of nonlinear six-degree-of-freedom dynamics to fixed-wing UAV control design and software-in-the-loop validation, supporting the need to evaluate integrated flight-control behaviour using representative aircraft dynamics.

These findings establish three interconnected research requirements: behavioural modification of multiple flight-control priorities, quantitative evaluation of supervisory intervention and progressive validation under nonlinear aircraft dynamics. The corresponding research objectives are defined in the following section.

2.2 Research Objectives

The research objectives are derived from the limitations identified in the literature and establish the intended development, integration and validation of a behavioural supervisory flight-control framework. The objectives address the relationship between environmental interpretation, aircraft-control intervention and the resulting aircraft response, while accounting for the influence of aircraft-model fidelity on the assessment of control performance.

Table 2 Research objectives and intended deliverables

Research objective	Objective description	Intended deliverables
Objective 1 — Behavioural Flight-Control Development	Develop a behavioural supervisory control loop capable of responding to high-stakes environmental and aircraft-state conditions by dynamically modifying aircraft flight-control priorities and generating corresponding aerodynamic control-surface and propulsion commands without pilot input.	1. Behavioural supervisory-control architecture 2. behaviour-state definitions 3. interfaces between supervisory and low-level control systems 4. autonomous elevator, aileron, rudder and throttle control 5. demonstration under representative difficult-environment conditions.
Objective 2 — Quantitative Behavioural-Control Evaluation	Quantitatively evaluate the behavioural-control response by analysing aircraft trajectory, altitude, airspeed, control-surface deflections, throttle response, behavioural transitions, recovery characteristics and disturbance rejection under difficult operating conditions.	1. Defined performance metrics; 2. baseline versus behavioural-control comparisons

		3. actuator-response analysis 4. recovery-time and disturbance-rejection measurements 5. safe-operating and corridor-performance assessment.
Objective 3 — Progressive Nonlinear Aircraft Validation	Validate the behavioural-control framework using progressively higher-fidelity aircraft dynamics, beginning with a nonlinear parameterized aircraft model and subsequently evaluating the architecture using a nonlinear six-degree-of-freedom model.	1. Validated nonlinear aircraft-dynamics model. 2. integration of behavioural control with nonlinear aircraft dynamics 3. six-degree-of-freedom validation 4. comparison of behavioural performance across model fidelities 5. identification of conditions requiring behavioural or control refinement.

Together, these objectives establish the technical basis for developing an autonomous flight-control framework capable of coordinating environmental guidance, behavioural decision-making and aircraft-control responses within a common architecture. The significance of these capabilities extends to operational applications where reliable navigation, reduced dependence on continuous operator intervention and controlled aircraft response under changing environmental conditions are essential. The following section therefore examines the requirements of the Indian defence and wider aviation industries, establishing the relationship between operational needs, the proposed research objectives and their potential applications.

3. Industrial Requirement

3.1 Indian Defense Industry

The development of autonomous aerial systems has become an important area of technological interest within the Indian defence industry, particularly as the Armed Forces seek to improve their surveillance capabilities, operational flexibility and independence from continuous human intervention. The Technology Perspective and Capability Roadmap 2025 identifies requirements for remotely piloted aircraft across the Indian Army, Navy and Air Force, including reconnaissance, maritime surveillance, search and rescue, disaster response and extended duration operations. For medium altitude, long endurance aircraft, the roadmap specifies requirements such as operating ceilings of approximately 30,000 feet, endurance exceeding 24 hours and the ability to accommodate different surveillance payloads (Headquarters Integrated Defence Staff, 2025, p. 100). Similarly, the Indian Army's Compendium of Problem Definition Statements 2025 identifies challenges associated with unmanned aircraft operating in high altitude environments. These include limitations in endurance, operational altitude and the number of pilots required to manage autonomous platforms. The document also identifies autonomous navigation, compatibility with satellite navigation systems and operation under difficult environmental conditions as important technological requirements (Army Design Bureau, 2025, pp. 21, 38). Together, these documents indicate an increasing requirement for reliable, adaptable and independently

developed autonomous technologies that can support different aircraft configurations and operational environments.

SWARAJ is developed with consideration of these broader requirements, concentrating specifically on the guidance, control and behavioural decision making functions of an autonomous aircraft. The project does not attempt to reproduce the complete specifications of the unmanned platforms identified by the Armed Forces. Instead, it investigates the underlying control architecture required to coordinate navigation, aircraft stability, energy management and responses to changes in operating conditions. The current preliminary design integrates an environmental map, predictive look ahead guidance, lateral and vertical control, behavioural supervision and gust response compensation within a nonlinear six degree of freedom aircraft simulation. Its performance is examined through nominal trajectory tracking, altitude and speed command variations, and atmospheric disturbance scenarios. The relationship between the requirements identified in the defence publications and the capabilities demonstrated or developed within SWARAJ is summarised in Table 1.

Table 3Relationship between Indian defence industry requirements and SWARAJ preliminary design capabilities

Industry requirement	**SWARAJ capability at the preliminary design stage**
Autonomous aerial reconnaissance and surveillance support (Headquarters Integrated Defence Staff, 2025, p. 100)	Integrated autonomous flight guidance capable of following a defined mission trajectory within a simulated terrain environment. Surveillance sensors and information processing are outside the present validation scope.
Reliable navigation in difficult geographical environments (Army Design Bureau, 2025, p. 21)	Environmental mapping and predictive look ahead guidance developed to manage aircraft movement along a defined valley trajectory with changing path geometry.
Autonomous flight with reduced dependence on continuous operator intervention (Army Design Bureau, 2025, p. 21)	Integrated guidance, behavioural supervision and flight control demonstrated through closed loop simulations without continuous manual control inputs.
Continued operation under atmospheric disturbances (Army Design Bureau, 2025, p. 38)	Gust Response Compensator and associated supervisory functions evaluated through four atmospheric disturbance cases, including comparisons between enabled and disabled compensation modes.
Coordination of aircraft operation across changing mission conditions (Headquarters Integrated Defence Staff, 2025, pp. 100–101)	Lateral guidance, vertical guidance and energy supervision integrated to manage trajectory tracking, altitude and airspeed within a common simulation architecture.
Development of autonomous technologies suitable for different aircraft classes (Headquarters Integrated Defence Staff, 2025, pp. 100–101)	Modular guidance and control architecture designed for compatibility with multiple aircraft platforms. The current Cessna 182 implementation provides the reference environment for this ongoing research.
Indigenous research and development of defence technologies (Headquarters Integrated Defence Staff, 2025, pp. 2–4; Army Design Bureau, 2025, p. xiv)	Development of an independently structured guidance and control framework using mathematical modelling, system level integration and simulation based performance evaluation.

An important aspect of SWARAJ is the flexibility of its proposed architecture. As established in the Conceptual Design Review, the project begins by defining the system requirements, expected inputs, operational terrain, mission trajectory and the responses required when an aircraft encounters difficult path geometry or changing environmental conditions. This provides a foundation for two possible directions of development. The first, which forms an important objective of the current research, concerns the development of a common guidance and supervisory architecture capable of being integrated with different aircraft and UAV platforms. By separating the autonomous guidance functions from the characteristics of an individual aircraft plant, SWARAJ aims to provide greater flexibility in adapting the system to aircraft with different flight dynamics, control requirements and operational capabilities. The second direction concerns the eventual development of a dedicated aircraft designed around the requirements established by SWARAJ. While this remains a possible future application of the research, the design and development of such an aircraft fall outside the scope of the present study.
The current Preliminary Design Review therefore focuses on the development and evaluation of the autonomous control architecture, using the nonlinear Cessna 182 model as its reference aircraft. This establishes a basis for assessing the system's behaviour while supporting the broader objective of achieving compatibility across multiple aircraft platforms. The following section summarises the objectives, system requirements and architecture established during the Conceptual Design Review, providing the foundation for understanding the progress presented in this report.

3.2 General Aviation and Emergency Departments

SWARAJ also has potential applications in general aviation, environmental research and emergency response. In India, mountainous terrain, remote settlements, extensive forests and large coastal areas create a need for aircraft that can reach difficult locations and provide timely information to response teams. A common guidance and control framework could support different aircraft and mission requirements while retaining human supervision.

Existing Indian programmes demonstrate this requirement. The following table relates these activities to possible future applications of SWARAJ.

Table 4 Non defence industry requirements

Sector	Industry requirement and Indian example	Potential application of SWARAJ
Wildlife research and conservation	Repeated aerial surveys for animal-population assessment, habitat mapping and monitoring inaccessible areas. The Wildlife Institute of India identifies these among its UAV applications (Wildlife Institute of India 2026).	Following planned survey routes while maintaining suitable speed and altitude for an observation payload.
Search and rescue	Rapid assessment of disaster-affected areas and information that helps rescuers locate survivors and plan access. The AGNIi–NDRF advisory identifies unmanned systems, live imagery and gust resistance as important requirements (AGNIi Mission 2023).	Providing a flight-control foundation for aerial searches and repeated coverage of assigned areas under changing wind conditions.
Medical supplies and samples	Timely transport of medicines, vaccines, blood products and diagnostic samples. ICMR's i-DRONE programme includes remote-area delivery and trials in high-altitude Himachal Pradesh (Indian Council of Medical Research n.d.).	Supporting route following and speed management for future medical-delivery aircraft with suitable payload capacity.
Medical evacuation and airlifting	Rapid access to patients where road transport is difficult. Helicopter Emergency Medical Services were launched at AIIMS Rishikesh in October 2024 (Ministry of Health and Family Welfare 2025).	Initially supporting route reconnaissance and situational assessment. Patient transport would require a separately developed aircraft and medical installation.

Sector	Industry requirement and Indian example	Potential application of SWARAJ
Fire and forest emergency services	Timely identification and assessment of fires to guide field teams. Forest Survey of India distributes satellite-based fire alerts to support containment activities (Forest Survey of India n.d.).	Supporting aerial verification, observation of fire extent and information gathering for response teams when fitted with appropriate sensors.
Maritime services	Searching large sea areas, locating persons or vessels in distress and supporting rescue coordination. The Indian Coast Guard identifies drones as useful tools for maritime search and rescue (Indian Coast Guard 2023).	Adapting the guidance framework to planned search routes, wind compensation and observation missions over coastal waters.

These applications share a requirement for dependable navigation, coordinated speed and altitude control, and recovery from disturbances. The present SWARAJ study investigates these underlying functions. Further integration with suitable sensors, communications and mission equipment could extend the framework towards civilian services and contribute to India's emergency-response capability.

4. Conceptual Design Review

The Conceptual Design Review (CDR) presented the initial development of SWARAJ as an autonomous flight control framework incorporating behavioural supervision and adaptive guidance. The system was developed in MATLAB/Simulink using a simplified three degree of freedom aircraft model integrated with an environmental map, adaptive look ahead guidance, lateral and longitudinal controllers, behavioural supervision and supporting telemetry functions. The primary objective was to establish whether an aircraft could interpret changes in its operating environment and adjust its control priorities while continuing to follow a defined mission trajectory. To investigate this concept, two simulation scenarios were developed, representing nominal valley following and flight under a predefined environmental disturbance. These scenarios provided a basis for examining the interaction between navigation, behavioural decision making and aircraft control.

The simulation results demonstrated that the proposed framework could follow a curved valley trajectory while maintaining lateral containment within the defined operating boundaries, as illustrated in figure 1, During nominal operation, the aircraft maintained the prescribed altitude of approximately 100 m, with a maximum sampled cross track error of 175.5 m against the preferred corridor limit of ±250 m.

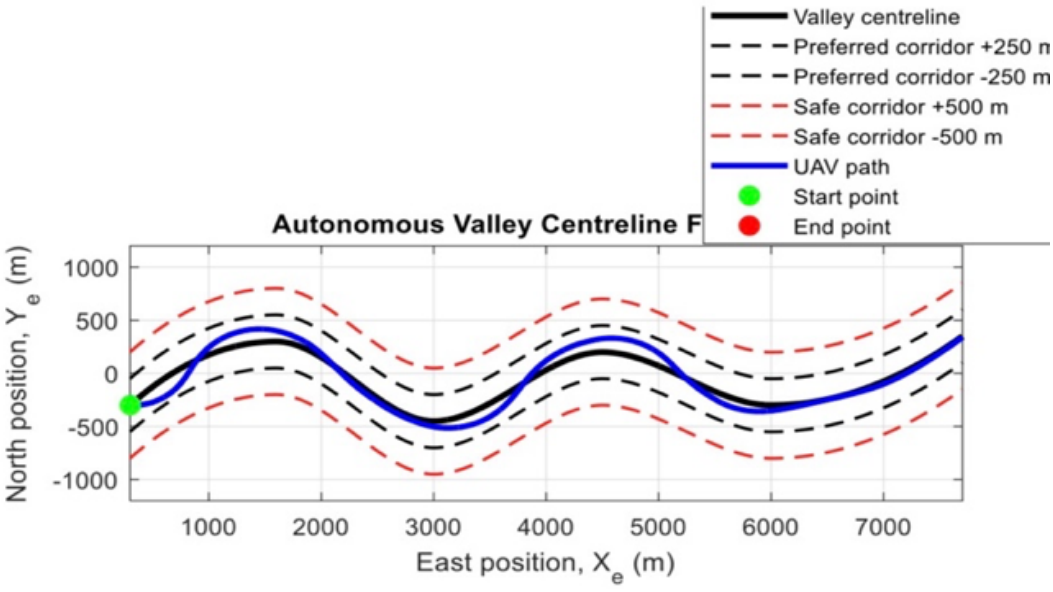


Figure 1 Autonomous Valley Following CDR

When the disturbance was introduced, the behavioural supervisor modified the aircraft's control priorities by increasing altitude demand, reducing speed priority and retaining lateral guidance authority. The corresponding supervisory transition is presented in Figure 2. The aircraft subsequently entered a controlled climb, reaching a maximum altitude of approximately 160.16 m, while maintaining lateral containment with a maximum sampled cross track error of 185.7 m over the valid mission route.

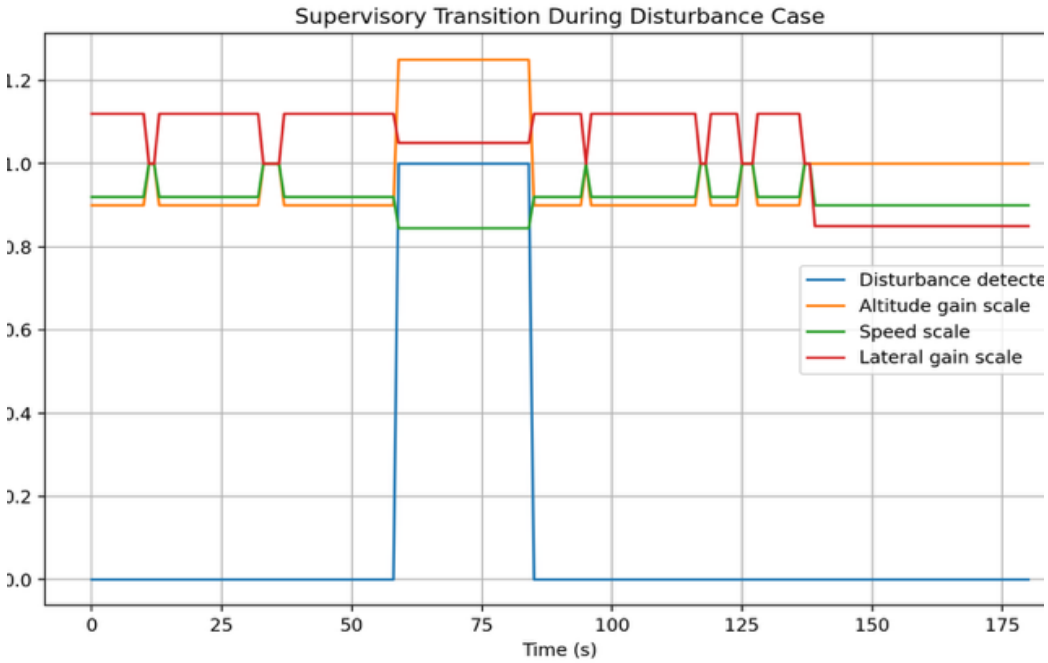


Figure 2 CDR Supervisory Transition

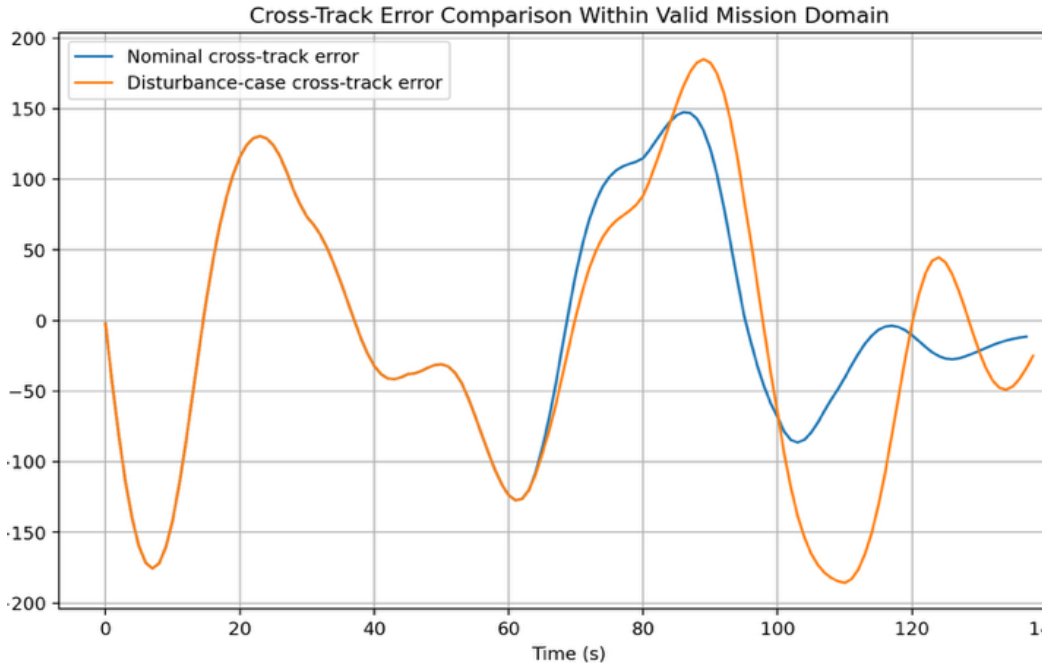


Figure 3 CDR lateral tracking performance for two scenarios

Figure 3 presents the comparison of lateral tracking performance between the two scenarios. The aileron, elevator and rudder commands also remained bounded, indicating that the changes in supervisory behaviour were reflected in the aircraft's control responses, as shown in Figure 4 These observations provided initial validation of the behavioural control concept, with the principal results summarised in Table 2.

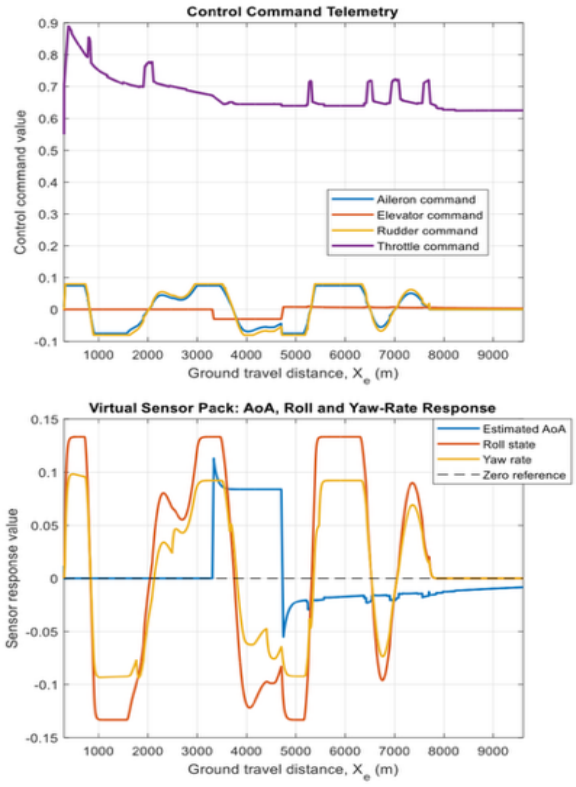


Figure 4 Control Surface Responses CDR

Table 5 Summary of CDR simulation results

Performance parameter	Nominal operation	Disturbance operation
Maximum sampled cross track error within the mapped route	175.5 m	185.7 m
Maximum sampled cross track error during the active disturbance	Not applicable	139.9 m
Preferred lateral corridor	±250 m	±250 m
Lateral containment	Maintained	Maintained
Nominal altitude reference	100 m	100 m
Maximum recorded altitude	Approximately 100.008	160.16 m
Maximum aileron command	0.075	0.075
Maximum elevator command	Approximately 0.003	0.030
Maximum absolute rudder command	0.080	0.080
Forward speed range	45.03 to 65.00 m/s	Approximately 45.03 to 65.00 m/s
Behavioural response	Normal tracking and lateral recovery	Disturbance detection, altered control priorities and lateral recovery

Although the conceptual design established the intended operation of the behavioural supervisory framework, several limitations were identified during its evaluation. These findings provided the basis for further development and helped define the technical objectives

of the Preliminary Design Review. The principal limitations identified during the CDR, together with the developments undertaken to address them, are outlined below.

- Simplified aircraft dynamics: The original three degree of freedom model did not fully represent the aerodynamic forces, moments and rigid body interactions associated with actual aircraft motion.
- Limited representation of aerodynamic behaviour: The conceptual model relied on simplified relationships for parameters such as angle of attack and control surface responses, restricting the physical accuracy of the simulated aircraft behaviour.
- Limited speed and energy interaction: Changes in supervisory speed priorities produced minimal differences in actual airspeed, indicating insufficient representation of the interaction between propulsion, aircraft energy and longitudinal motion.
- Restricted disturbance evaluation: The conceptual design considered only a predefined environmental disturbance, limiting the assessment of aircraft behaviour under different atmospheric conditions and external disturbances.
-

The developments undertaken following the CDR represent a progression from conceptual validation towards a more physically representative autonomous flight control system. While the original study established the feasibility of modifying aircraft control priorities through behavioural supervision, the preliminary design examines these functions within a nonlinear aircraft simulation environment. This progression provides the foundation for the methodology presented in the following section, beginning with the development and validation of the Cessna 182 aircraft model, followed by the integration of the individual guidance, control and supervisory subsystems.

5. Methodology

This section presents the methodology used to develop and evaluate SWARAJ's autonomous flight-control framework. It begins with the aircraft model and C182 validation, followed by the environment map that defines the reference route. The control methodology then explains how look-ahead guidance, Banked Dynamics, lateral control and longitudinal energy management work together to follow the path while regulating altitude and airspeed. The Gust Response Compensator extends this architecture to disturbed flight. Controller tuning parameters and developments since the CDR are subsequently discussed, establishing the basis for the simulation cases, performance measures and validation procedures used to assess the integrated system.

5.1 Aircraft Plant

The Cessna 182 was chosen because its geometry, mass properties and aerodynamic data are readily available through the MathWorks implementation of Roskam's textbook dataset. Established reference conditions also provide a practical basis for calculating trim and comparing the simulated response with published aircraft behaviour (MathWorks n.d.). Its conventional fixed-wing layout and approximately 1,200 kg mass make it a useful starting point for heavier reconnaissance aircraft, including the mass class of the Heron UAV (Israel Aerospace Industries n.d.). A six-degree-of-freedom model was adopted to capture how roll, pitch, yaw and translational motion interact during manoeuvres. These interactions allow SWARAJ's path, altitude and speed control to be assessed together (Stevens, Lewis and Johnson 2015). The parameters below establish the C182 configuration; other aircraft can subsequently be introduced through the same database structure.

Table 6 Cessna 182 parameters implemented in the aircraft database

Cessna 182 aircraft database parameters

Parameter	Symbol	Implemented value	Source or basis
Mass and geometry			
Aircraft mass	m	1,201.0505 kg	MathWorks (n.d.), SI conversion
Wing reference area	S	16.16513 m²	MathWorks (n.d.), SI conversion
Wingspan	b	10.9728 m	MathWorks (n.d.), SI conversion
Mean aerodynamic chord	$\bar{c}$	1.49352 m	MathWorks (n.d.), SI conversion
Inertia properties			
Roll moment of inertia	I_{xx}	1,285.3154 kg·m²	MathWorks (n.d.), SI conversion
Pitch moment of inertia	I_{yy}	1,824.9310 kg·m²	MathWorks (n.d.), SI conversion
Yaw moment of inertia	I_{zz}	2,666.8939 kg·m²	MathWorks (n.d.), SI conversion
Roll-yaw product of inertia	I_{xz}	0 kg·m²	MathWorks (n.d.)
Lift coefficients			
Zero-angle lift coefficient	$C_{L,0}$	0.307	MathWorks (n.d.)
Angle-of-attack derivative	$C_{L,\alpha}$	4.41	MathWorks (n.d.)
Pitch-rate derivative	$C_{L,q}$	3.90	MathWorks (n.d.)
Elevator derivative	C_{L,δ_e}	0.43	MathWorks (n.d.)
Drag coefficients			
Zero-angle drag coefficient	$C_{D,0}$	0.027	MathWorks (n.d.)
Angle-of-attack derivative	$C_{D,\alpha}$	0.121	MathWorks (n.d.)
Pitch-rate derivative	$C_{D,q}$	0	MathWorks (n.d.)
Elevator derivative	C_{D,δ_e}	0	MathWorks (n.d.)
Pitching-moment coefficients			
Zero-angle pitching-moment coefficient	$C_{m,0}$	0.040	MathWorks (n.d.)
Angle-of-attack derivative	$C_{m,\alpha}$	−0.613	MathWorks (n.d.)
Pitch-rate derivative	$C_{m,q}$	−12.4	MathWorks (n.d.)
Elevator derivative	C_{m,δ_e}	−1.122	MathWorks (n.d.)
Angle-of-attack-rate derivative	$C_{m,\dot{\alpha}}$	0	Model simplification; MathWorks specifies −7.27
Side-force coefficients			
Zero-sideslip side-force coefficient	$C_{Y,0}$	0	MathWorks (n.d.), default zero
Sideslip derivative	$C_{Y,\beta}$	−0.393	MathWorks (n.d.)
Roll-rate derivative	$C_{Y,p}$	−0.075	MathWorks (n.d.)
Yaw-rate derivative	$C_{Y,r}$	0.214	MathWorks (n.d.)
Aileron derivative	C_{Y,δ_a}	0	MathWorks (n.d.)
Rudder derivative	C_{Y,δ_r}	0.187	MathWorks (n.d.)
Rolling-moment coefficients			
Zero-sideslip rolling-moment coefficient	$C_{l,0}$	0	MathWorks (n.d.), default zero
Sideslip derivative	$C_{l,\beta}$	−0.0923	MathWorks (n.d.)
Roll-rate derivative	$C_{l,p}$	−0.484	MathWorks (n.d.)
Yaw-rate derivative	$C_{l,r}$	0.0798	MathWorks (n.d.)
Aileron derivative	C_{l,δ_a}	0.229	MathWorks (n.d.)
Rudder derivative	C_{l,δ_r}	0.0147	MathWorks (n.d.)
Yawing-moment coefficients			
Zero-sideslip yawing-moment coefficient	$C_{n,0}$	0	MathWorks (n.d.), default zero
Sideslip derivative	$C_{n,\beta}$	0.0587	MathWorks (n.d.)
Roll-rate derivative	$C_{n,p}$	−0.0278	MathWorks (n.d.)
Yaw-rate derivative	$C_{n,r}$	−0.0937	MathWorks (n.d.)
Aileron derivative	C_{n,δ_a}	−0.0216	MathWorks (n.d.)
Rudder derivative	C_{n,δ_r}	−0.0645	MathWorks (n.d.)
Reference operating condition			
Reference airspeed	V_{ref}	67.08773 m/s	MathWorks (n.d.), SI conversion
Reference altitude above mean sea level	h_{ref}	1,524 m	MathWorks (n.d.), SI conversion
Reference air density	ρ_{ref}	1.055 kg/m³	Adopted reference-atmosphere approximation
Propulsion parameters			
Maximum engine power	P_{max}	171.511 kW (230 hp)	Textron Aviation (n.d.)
Fixed-thrust parameter	T_{max}	0 N	Database setting for the power-based configuration
Simulation limits and stall reference			
Assumed stall angle of attack	α_{stall}	16° (0.279253 rad)	Simulation assumption
Maximum absolute elevator deflection	$\delta_{e,max}$	20° (0.349066 rad)	MathWorks (n.d.), example limit
Maximum absolute aileron deflection	$\delta_{a,max}$	20° (0.349066 rad)	MathWorks (n.d.), example limit
Maximum absolute rudder deflection	$\delta_{r,max}$	20° (0.349066 rad)	MathWorks (n.d.), example limit
Adopted stall-speed reference	V_s	27.8 m/s	Database assumption pending configuration-specific reference

Notes: Angular derivatives use radians; rate derivatives multiply nondimensional angular rates. Zero fixed-thrust parameter is a database configuration setting, not zero available aircraft thrust.
The implemented angle-of-attack-rate pitching derivative is zero; the MathWorks reference gives −7.27. The 16° stall angle and 27.8 m/s stall-speed reference are adopted simulation values.

References

MathWorks (n.d.) Determine static stability of fixed-wing aircraft, Aerospace Toolbox documentation, accessed 5 November 2026. https://www.mathworks.com/help/aeroblks/determine-nonlinear-dynamics-and-static-stability-fixed-wing-aircraft.html
Textron Aviation (n.d.) Cessna Skylane, accessed 5 November 2026. https://cessna.txtav.com/en/piston/cessna-skylane.
National Institute of Standards and Technology (n.d.) NIST guide to the SI, Appendix B.9: factors for units listed by kind of quantity or field of science, accessed 5 November 2026. https://www.nist.gov/pml/special-publication-811/nist-guide-si-appendix-b-conversion-factors/nist-guide-si-appendix-b9

Longitudinal validation examined whether the aircraft maintained its calculated equilibrium with fixed trim controls, following the equilibrium assessment used in aircraft simulation (Stevens, Lewis and Johnson 2015). Over 600 s, the simulated airspeed changed by only 3.968×10^{-6} m/s and altitude by 1.168 mm. Figure 5 shows that the angular deviations also remained small. This establishes that the plant preserves its calculated trim without meaningful drift before SWARAJ applies corrective commands.

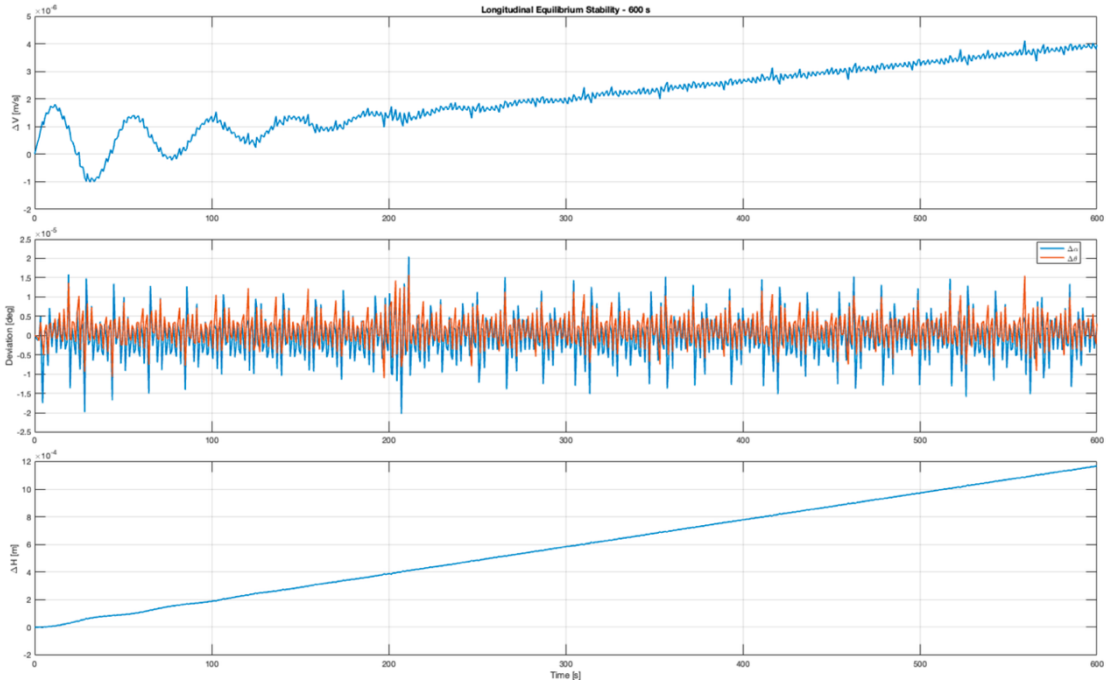


Figure 5 Cessna 182 longitudinal equilibrium over 600 s.

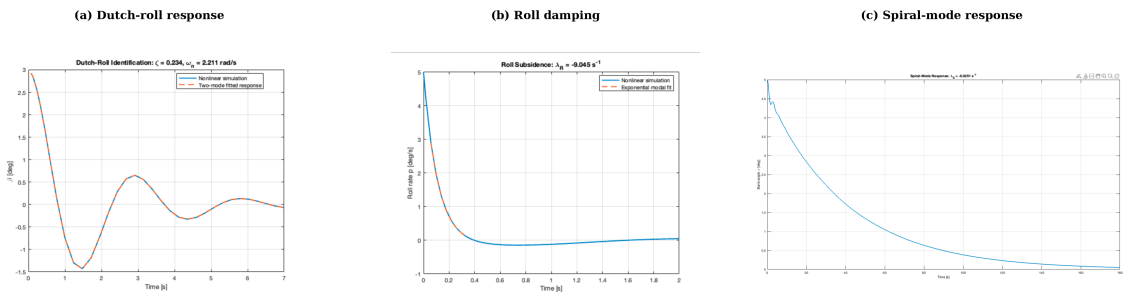


Figure 6 Cessna 182 lateral validation: (a) Dutch-roll response, (b) roll damping and (c) spiral-mode response

Lateral validation compared the developed C182 model with the published approach-condition modes of Kornfeld, Hansman and Deyst (1999). Figure 6 (a) shows decaying Dutch-roll oscillations following a 3° sideslip disturbance. The identified natural frequency was 2.211 rad/s and the damping ratio was 0.234, compared with reference values of 2.69 rad/s and 0.250, giving differences of 17.79% and 6.41%, respectively. Figure 6(b) shows rapid roll-rate decay following a 5°/s disturbance, with a roll pole of -9.045 s^{-1} against -9.40 s^{-1}, a difference of 3.78%. Figure 6 (c) shows stable spiral decay with a pole of -0.02509 s^{-1}, whereas the reference predicts weak divergence at $+0.004$ s^{-1}. This is recorded as a model

difference between the independently developed nonlinear plant and the published linear reference. Together, these results demonstrate agreement in the principal damped lateral responses while identifying a difference in spiral stability.
The aircraft-plant assessment establishes the C182 model as the basis for evaluating SWARAJ. The following section will deal with understanding the controlling systems that provide autonomy through integrated physics and behaviour systems.

5.2 Control Methodology

5.2.1 Environment Map

The environment map defines the fixed centreline that SWARAJ must follow. Six east–north coordinates are connected using a clamped cubic spline, forming five sections sampled at 501 points. The alternating bends represent a valley-type route and test repeated changes in turning direction while maintaining altitude and speed. Smooth connections avoid abrupt changes in the required path direction.

Each section follows the standard cubic-polynomial form used in spline interpolation (MathWorks n.d.-a):

Equation 1

$$N_i(E) = a_i + b_i\xi + c_i\xi^2 + d_i\xi^3, \qquad \xi = E - E_i.$$

Here, E and N are east and north positions. The coefficients ensure that adjoining sections meet with continuous first and second derivatives. Endpoint slopes follow the first and last waypoint connections.

For the eastward-progressing route, heading is measured clockwise from north using the tangent direction and four-quadrant inverse tangent (MathWorks n.d.-c):

Equation 2

$$\chi_{\text{path}} = \operatorname{atan2}\left(1, \frac{dN}{dE}\right).$$

The sharpness of each bend is described by the standard planar-curvature relationship (Weisstein n.d.):

Equation 3

$$|\kappa| = \frac{\left|\frac{d^2N}{dE^2}\right|}{\left[1 + \left(\frac{dN}{dE}\right)^2\right]^{3/2}}.$$

The geometry remains unchanged during flight. Aircraft position is used to measure tracking error and progress, without modifying the route or directly commanding actuators. Future geographical waypoints could be converted from latitude, longitude and height into local

coordinates (MathWorks n.d.-b). A parametric path, E, N would extend this approach to routes containing straight sections, curves and changes in eastward direction.

The figure below showcases the mathematical map generated:

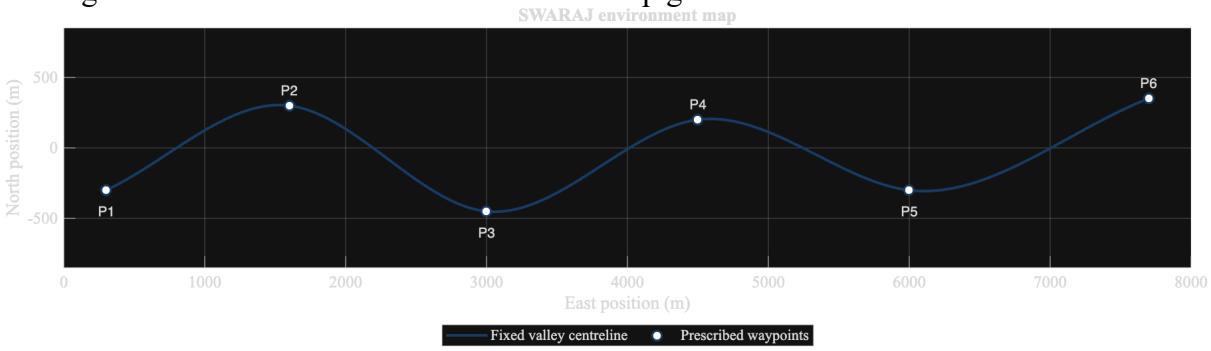


Figure 7 mathematical representation of environment map

The map remains fixed and independent of the control system, supplying route geometry to the look-ahead controller. This controller relates the aircraft's current position and motion to the path ahead, allowing it to anticipate bends and determine the required direction of travel. In a future physical implementation, geographical waypoints could be converted into a local Earth-referenced frame, with relevant directions then expressed relative to the aircraft. GPS could provide position information, while LiDAR, range sensors and onboard cameras could support environmental mapping and position estimation.

Storing the route mathematically would also allow it to remain available during a loss of GPS reception. Continued autonomous tracking would require an alternative position estimate, obtained through inertial navigation and, where available, camera- or LiDAR-based localisation. Combining these measurements could provide navigation redundancy and reduce dependence on continuous GPS availability; coordinate conversion alone would not provide this capability.

5.2.2 Look Ahead Controller and Banked Dynamics

The look-ahead controller converts the mapped centreline into a heading requirement using the aircraft's present tracking condition and the route ahead. Figures X(a) and X(b) illustrate the geometric basis of this process. The use of forward path information follows the principle of anticipatory guidance described by Park, Deyst and How (2004). SWARAJ develops this principle through variable near and far preview times, mode-dependent heading selection and feedback from Banked Dynamics.

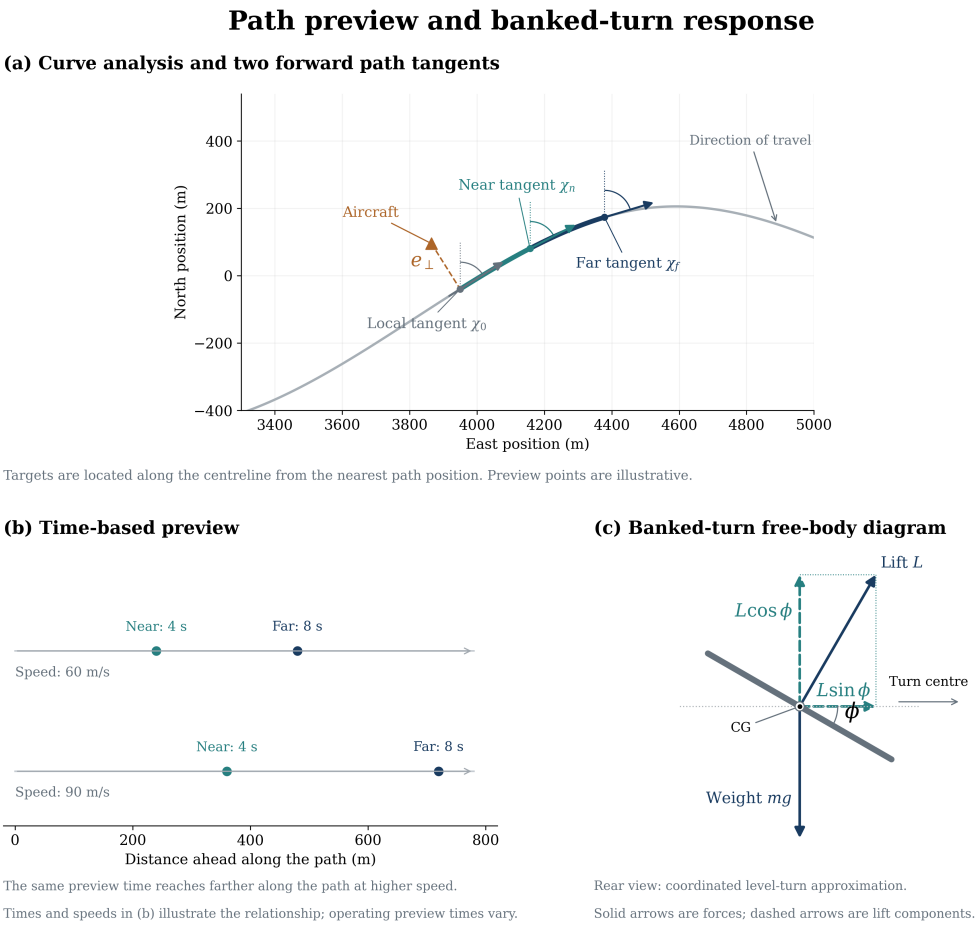


Figure 8 Look ahead Guidance and banked dynamics

Curve analysis and path tangents

The controller identifies the nearest position on the centreline by projecting the aircraft position onto the mapped path segments. The tangent at this position defines the local path heading, $\chi_0$while the perpendicular displacement defines the signed cross-track error, $e_\perp$. The near and far reference points provide the future tangent headings $\chi_n$and $\chi_f$respectively.

For a continuous path, signed curvature is

Equation 4

$$\kappa = \frac{d\chi}{ds},$$

where s is distance along the centreline. Assuming negligible wind and approximately constant speed over the preview interval, $ds/dt \approx V$ Therefore,

Equation 5

$$\dot{\chi} = \frac{d\chi}{ds}\frac{ds}{dt} = V\kappa.$$

For a circular section, $|\kappa| = 1/R$, where R is the turn-radius magnitude. Thus, the required directional rate increases with both speed and curvature.

The implementation estimates curvature from successive segment headings:

Equation 6

$$\kappa_j \approx \frac{\mathrm{wrap}(\chi_j - \chi_{j-1})}{\Delta s_j}.$$

Here, Δs_j is segment length, and

Equation 7

$$\mathrm{wrap}(x) = \mathrm{atan2}(\sin x, \cos x).$$

Mean curvature describes the overall bend, while peak curvature identifies locally demanding sections. Changes in curvature sign identify opposing bends. This distinction matters because successive left and right turns can produce a small net heading change while still requiring substantial aircraft response.

Implemented near and far preview logic

The preview distances shown in Figure X(b) are calculated from selected time horizons:

Equation 8

$$L_n = VT_n, \qquad L_f = VT_f.$$

The distances are measured along the centreline from the nearest path position. Time-based selection means that increasing speed extends the route examined, preserving an approximate interval for manoeuvre preparation.

The implemented time schedules are

Equation 9

$$T_{n,0} = 4.0 - 1.50U_e - 0.20U_\psi - 0.10S + \Delta T,$$

Equation 10

$$T_n = \mathrm{sat}\left[K_n\,\mathrm{sat}\left(T_{n,0}; 2.5, 4.0\right); 2.0, 5.5\right]$$
$$T_f = \mathrm{sat}\left[K_f\left(6.0 + 2.0U_p - 0.25U_e\right); 5.0, 12.0\right].$$

All times are in seconds. The operator $\mathrm{sat}(x; a, b)$ bounds x between a and bThe multipliers K_n and K_f are tuning parameters.

The dimensionless terms U_e U_ψ $U_p$and S lie between zero and one:

- $U_e$represents tracking urgency. Its base schedule increases from zero to one as $|e_\perp|/C$ rises from 0.10 to 0.30, where C is the scheduled internal cross-track allowance.
- U_ψ Increases with the difference between aircraft heading and the local path tangent, from a 5°threshold to full urgency at 30°
- U_p combines approaching heading change, curve severity, turn radius, curvature trend, direction changes and supervisory feedback.
- S describes geometric severity using peak curvature, accumulated heading change and turn reversals.

The response-dependent adjustment ΔT is applied before the time limits:

Table 7 design choice for anticipation and recovery

Condition	Near-horizon adjustment	Intended effect
Correct turning direction, normalised acceleration mismatch below 0.30 and tracking urgency below 0.50	+0.15 s	Soften correction when the aircraft is responding adequately
Geometry classified as tight or more complex	−0.10 s	Place greater attention on the immediate route
Path-direction discontinuity detected	−0.15 s	Reduce dependence on distant geometry
Configured bank-authority indicator below its threshold	+0.20 s	Reduce the aggressiveness of local steering

These schedules are design choices. They implement the intended balance between anticipation and recovery; their coefficients are not derived from the banked-turn force balance.

Angular prediction and heading selection

For either preview horizon,

Equation 11

$$\Delta\chi_i = \mathrm{wrap}(\chi_i - \chi_0), \qquad \omega_i = \frac{\Delta\chi_i}{T_i}, \quad i \in \{n, f\}.$$

The quantity ω_i estimates the average directional rate required over the preview interval. Using path tangents keeps prediction of route geometry separate from the aircraft's existing heading error.

The Master Controller combines the near and far headings through circular averaging:

Equation 12

$$\chi_b = \mathrm{atan2}\big(w_n \sin\chi_n + w_f \sin\chi_f\,,\; w_n \cos\chi_n + w_f \cos\chi_f\big),$$

where w_n and w_f are non-negative heading weights satisfying $w_n + w_f = 1$

Their selection changes with the operating condition. Turn preparation increases the far-heading contribution, while recovery increases the near-heading contribution. Opposing near and far turn-rate predictions identify a reversal, allowing preparation for the next bend while retaining the current turning requirement.

The final heading demand is

Equation 13

$$\psi_{\mathrm{cmd}} = \mathrm{wrap}\left(\chi_b + \delta_{\mathrm{CTE}} + \delta_{\mathrm{pred}}\right).$$

The centreline correction is

Equation 14

$$\delta_{\mathrm{CTE}} = \mathrm{sat}\left[K_I \,\mathrm{atan2}\left(e_{\perp,\mathrm{eff}}, L_n\right); -20^\circ, 20^\circ\right],$$

with

Equation 15

$$e_{\perp,\mathrm{eff}} = \mathrm{sgn}(e_\perp)\,\max(|e_\perp| - 0.05C, 0).$$

Here, K_I is the scheduled intercept gain. The deadband reduces repeated correction close to the centreline. The predictive correction, δ_{pred}, advances the heading demand according to approaching geometry and is also bounded to $\pm 20^\circ$Its influence is reduced during strong recovery.

The internal allowance C supports controller scheduling; it does not replace the fixed corridor boundaries used to assess validation results.

Banked Dynamics

Banked Dynamics relates the guidance requirement to the aircraft's current bank, angular motion and altitude condition. Figure X(c) provides the physical basis for estimating the required turning response and its effect on vertical support.

Physical derivation and assumptions:

For a coordinated, approximately steady level turn, assume negligible sideslip, approximately constant speed and negligible vertical thrust contribution. Resolving lift vertically and towards the turn centre gives

Equation 16

$$L\cos\phi = mg,$$
$$L|\sin\phi| = \frac{mV^2}{R}.$$

Here, L is lift, ϕ bank angle, m aircraft mass and g gravitational acceleration. The horizontal lift component supplies the centripetal force.

Dividing the horizontal balance by the vertical balance gives

Equation 17

$$g|\tan\phi| = \frac{V^2}{R}.$$

Retaining turn direction through signed turn rate yields

Equation 18

$$g\tan\phi = V\omega.$$

The bank estimate corresponding to either preview demand is therefore

Equation 19

$$\phi_i = \tan^{-1}\left(\frac{V\omega_i}{g}\right).$$

This connects the tangent changes in Figure X(a), the available response interval in Figure X(b), and the force balance in Figure X(c). These relationships follow from coordinated-turn mechanics. The nonlinear aircraft model subsequently determines the response under aerodynamic, inertial and actuator effects.

Implemented tracking and bank corrections

The demanded lateral acceleration combines route geometry with tracking corrections:

Equation 20

$$a_{\perp,\mathrm{d}} = V^2\kappa_{\mathrm{eff}} + K_e e_{\perp,\mathrm{eff}} + K_\psi V e_\psi,$$

where

Equation 21

$$\kappa_{\mathrm{eff}} = \bar{\kappa} + 0.35\Delta\kappa,$$
$$e_\psi = \mathrm{wrap}(\psi_{\mathrm{cmd}} - \psi).$$

The mean curvature $\bar{\kappa}$ is the accumulated tangent change divided by the examined path length. The trend signal $\Delta\kappa$ is the final sampled curvature minus the initial sampled curvature. The factor 0.35 is an implemented weighting, rather than a geometric identity.

The gains $K_e$and K_ψ scale cross-track and heading-error corrections into acceleration. Their respective units are $\mathrm{s}^{-2}$and s^{-1}The three acceleration terms therefore represent the bend itself, displacement from the centreline and directional misalignment.

The current turning response is estimated as

Equation 22

$$\hat{a}_{\perp} = g \tan \phi,$$

and compared with the demand:

Equation 23

$$e_a = a_{\perp,\mathrm{d}} - \hat{a}_{\perp}.$$

The estimate uses measured bank angle and the coordinated-turn approximation; it is not a direct measurement of acceleration from the plant forces.

The bank target blends current tracking and predictive requirements:

Equation 24

$$\phi_{\mathrm{track}} = (1 - P_b) \tan^{-1}\left(\frac{a_{\perp,\mathrm{d}}}{g}\right) + P_b \phi_{\mathrm{pred}},$$

where $P_b \in [0,1]$ is bank-preparation priority. In the implementation,

Equation 25

$$\phi_{\mathrm{pred}} = 0.30\phi_M + 0.70 \tan^{-1}\left(\frac{V\omega_M}{g}\right).$$

The terms ϕ_M and ω_M are the Master Controller's bank and turn-rate demands. Their weighting reconciles the two predictive inputs.

The supervisory bank correction is

Equation 26

$$\Delta\phi = K_\phi(\phi_{\mathrm{track}} - \phi) - K_p p + K_{\mathrm{pred}} P_b(\phi_{\mathrm{pred}} - \phi).$$

Here, p is body roll rate. Bank-error feedback supplies correction, roll-rate feedback provides damping, and the predictive term increases preparation as the approaching manoeuvre becomes more important. The gains are selected controller parameters.

Feedback to look-ahead and longitudinal control

Banked Dynamics closes the guidance loop by returning the requested and estimated lateral accelerations to the look-ahead supervisor. Their signs indicate whether the aircraft is turning in the required direction, while their normalised difference indicates response mismatch. This feedback contributes to the near-horizon adjustment described above and softens the intercept correction when the aircraft is responding adequately near the centreline.

The vertical force balance also gives

Equation 27

$$L = \frac{mg}{\cos\phi}, \qquad n = \frac{L}{mg} = \frac{1}{\cos\phi},$$

where n is the level-turn load factor. If sufficient additional lift is unavailable, the simplified vertical acceleration is

Equation 28

$$\ddot{h} = \frac{L\cos\phi - mg}{m}.$$

Both turn directions follow the same vertical balance because $\cos(-\phi) = \cos\phi$. Banked Dynamics therefore passes an estimated vertical-support requirement and altitude error to longitudinal control. This coordination reflects the increased lift and propulsion requirements associated with maintaining altitude and speed during turns (Federal Aviation Administration 2021).

5.2.3 Lateral Controller

The lateral controller converts the heading requirement from look-ahead guidance and the corrections from Banked Dynamics into aileron and rudder commands. Its role is to execute the requested manoeuvre while continuously correcting the aircraft's roll and yaw response. The architecture separates route guidance from attitude and angular-rate control, consistent with established fixed-wing autopilot practice (Beard and McLain 2012). The command sequence and feedback connections are shown in Figures 9 and 10

Generation of roll and yaw demands

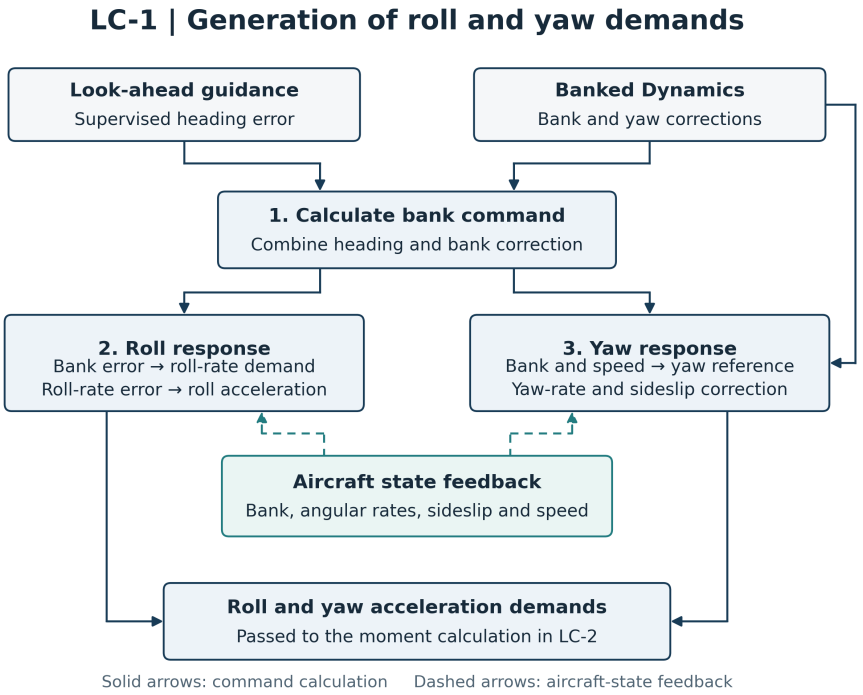


Figure 9 : Conversion of guidance and supervisory inputs into roll and yaw acceleration demands.

Figure 9 shows how the lateral controller translates the guidance requirement into a coordinated roll–yaw response. The heading error from look-ahead is combined with the bank correction from Banked Dynamics to establish the commanded bank angle. The difference between commanded and measured bank generates a roll-rate demand, while roll-rate feedback determines the acceleration required to approach that demand with damping. In parallel, commanded bank and airspeed establish an approximate coordinated-turn yaw reference. The supervisory yaw correction modifies this reference, and yaw-rate and sideslip feedback determine the yaw-acceleration demand. This arrangement allows the aircraft to respond to approaching bends while correcting its current motion. During turn entry, rollout or reversal, the demands change continuously with guidance and aircraft feedback, allowing the manoeuvre to proceed without an operator prescribing individual surface movements.

Heading requirement and bank command

The heading error is the wrapped difference between the commanded and measured aircraft headings:

Equation 29

$$e_{\psi} = \operatorname{atan2}[\sin(\psi_{\mathrm{cmd}} - \psi), \cos(\psi_{\mathrm{cmd}} - \psi)].$$

The bank command combines this error with the correction supplied by Banked Dynamics:

Equation 30

$$\phi_{\mathrm{cmd}} = \mathrm{sat}\left[K_h e_{\psi} + K_b \Delta\phi; -\phi_{\mathrm{lim}}, \phi_{\mathrm{lim}}\right].$$

Here, K_h scales the heading response, K_b scales the supervisory bank correction $\Delta\phi$ and ϕ_{lim} is the incoming bank limit. The saturation operator bounds its first argument between the stated limits.

This arrangement preserves the division of responsibility established in the preceding sections. Look-ahead selects the required direction, Banked Dynamics modifies the manoeuvre using tracking and aircraft-response information, and the lateral controller determines the corresponding control action.

Bank tracking and roll damping

The difference between commanded and measured bank produces a body roll-rate demand:

Equation 31

$$p_{\mathrm{cmd}} = \mathrm{sat}\left[\omega_{\phi}(\phi_{\mathrm{cmd}} - \phi); -p_{\mathrm{lim}}, p_{\mathrm{lim}}\right].$$

The roll-rate error then produces the acceleration demand:

Equation 32

$$\dot{p}_{\mathrm{cmd}} = \mathrm{sat}\left[2\zeta_{\phi}\omega_{\phi}(p_{\mathrm{cmd}} - p); -\dot{p}_{\mathrm{lim}}, \dot{p}_{\mathrm{lim}}\right].$$

The parameters ω_{ϕ} and ζ_{ϕ} set the roll-response characteristics. Bank error requests the movement, while roll-rate feedback moderates the acceleration as the aircraft approaches the demanded response. The implemented limits are $35°/\mathrm{s}$ for roll-rate demand and $65°/\mathrm{s}^2$ for roll-acceleration demand.

Body roll rate p is used as bank-rate feedback in this reduced control law. The full attitude kinematics remain represented in the aircraft plant.

3. Turn coordination and sideslip correction

From the coordinated-turn relationship derived in the Banked Dynamics section,

Equation 33

$$\dot{\psi} \approx \frac{g \tan \phi}{V}.$$

The controller uses this relationship to construct a yaw-rate reference and adds the supervisory yaw correction:

Equation 34

$$r_{\mathrm{ref}} = \mathrm{sat}\left[\frac{g \tan \phi_{\mathrm{cmd}}}{V_{\mathrm{safe}}} + K_y \Delta r; -r_{\mathrm{lim}}, r_{\mathrm{lim}}\right],$$

where $V_{\mathrm{safe}} = \max(|V|, 10)$ provides numerical protection, and K_y scales the Banked Dynamics correction Δr

The resulting yaw-acceleration demand is

Equation 35

$$\dot{r}_{\mathrm{cmd}} = \mathrm{sat}\left[K_r(r_{\mathrm{ref}} - r) - K_\beta \beta; -\dot{r}_{\mathrm{lim}}, \dot{r}_{\mathrm{lim}}\right].$$

Here, r is measured body yaw rate, β is sideslip angle, and K_r and K_β scale their corrective contributions. The implemented limits are $12°/\mathrm{s}$ and $45°/\mathrm{s}^2$.

The reference is a coordinated-turn approximation used within the body-rate controller. Euler heading rate and body yaw rate are not generally identical, so this approximation becomes less representative during large-bank, large-pitch or substantial sideslip manoeuvres.

Conversion into aileron and rudder commands

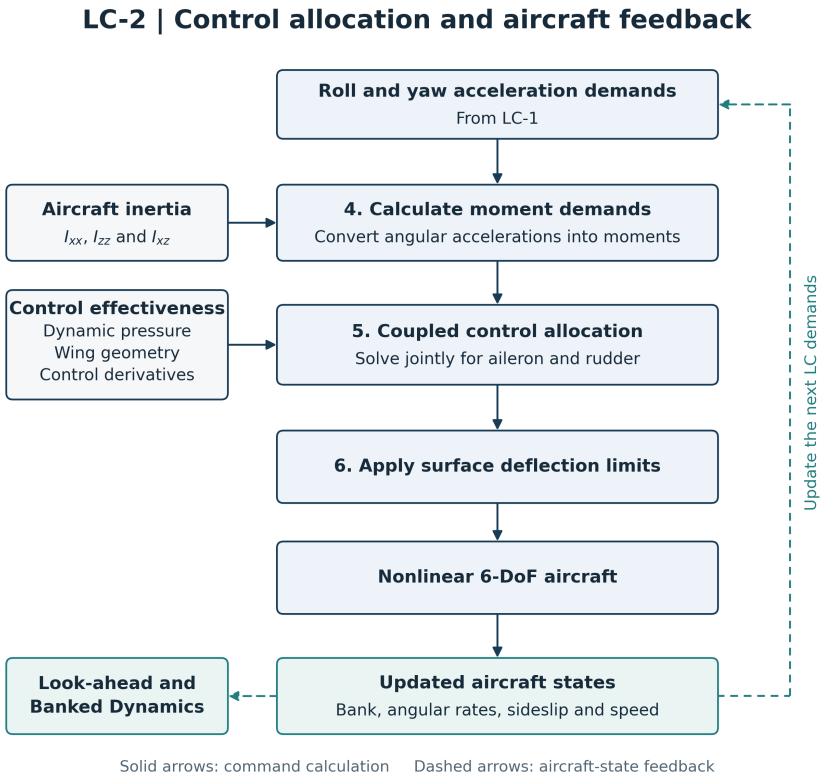


Figure 10 : Moment calculation, coupled control allocation and aircraft feedback.

Figure 10 shows how the acceleration demands become aileron and rudder commands. The controller first uses the aircraft's inertia parameters to calculate rolling and yawing moment demands, retaining acceleration coupling through the product of inertia. It then solves for both surface deflections using dynamic pressure, wing geometry and aerodynamic control derivatives. Solving the commands together accounts for each surface influencing both roll and yaw. Aircraft-specific deflection limits bound the resulting commands before application to the nonlinear plant. The updated aircraft states return to the lateral controller and upstream guidance, closing the feedback process shown across both figures. Autonomous execution therefore depends on the observed response: remaining bank, rate or sideslip errors change subsequent commands. Supplying aircraft properties as parameters also supports extension to other configurations, subject to appropriate tuning and validation.

Inertia-based moment demand

Using the inertia convention of the aircraft model, the controller calculates

Equation 36

$$\begin{bmatrix} L_{\mathrm{cmd}} \\ N_{\mathrm{cmd}} \end{bmatrix} = \begin{bmatrix} I_{xx} & -I_{xz} \\ -I_{xz} & I_{zz} \end{bmatrix} \begin{bmatrix} \dot{p}_{\mathrm{cmd}} \\ \dot{r}_{\mathrm{cmd}} \end{bmatrix}.$$

The quantities L_{cmd} and N_{cmd} are rolling and yawing moment demands. The diagonal terms represent resistance to angular acceleration, while I_{xz} retains acceleration coupling between the two axes.

This is the implemented reduced moment mapping. Existing aerodynamic moments and the remaining nonlinear rotational coupling terms are represented in the plant but are not explicitly cancelled by this calculation.

Coupled aerodynamic control allocation

The control-generated moments are related to surface deflections through

Equation 37

$$\begin{bmatrix} L_{\mathrm{ctrl}} \\ N_{\mathrm{ctrl}} \end{bmatrix} = \bar{q}Sb \begin{bmatrix} C_{l\delta_a} & C_{l\delta_r} \\ C_{n\delta_a} & C_{n\delta_r} \end{bmatrix} \begin{bmatrix} \delta_a \\ \delta_r \end{bmatrix}.$$

Here, $\bar{q}$ is dynamic pressure, S wing area, b wingspan, and δ_a and δ_r are aileron and rudder deflections. The coefficient derivatives describe how each surface changes the rolling and yawing moments.

Setting the control-generated moments equal to the commanded moments gives a simultaneous solution for both surfaces. The cross-derivatives retain aileron-generated yaw and rudder-generated roll. This connects the differential aerodynamic forces illustrated in the free-body sketches to the aircraft-level coefficient model.

Distributing moment demands through a control-effectiveness matrix follows established control-allocation practice (Miller and Pei 2025). SWARAJ uses a direct matrix solution, with a pseudoinverse fallback when the matrix is nearly singular.

Surface limits and aircraft response

The aileron and rudder commands form the physical connection between the lateral controller and the aircraft model. Before application, each command is bounded by the deflection limit supplied through the aircraft database:

Equation 38

$$\delta_{a,\text{applied}} = \text{sat}\left(\delta_{a,\text{calculated}}; -\delta_{a,\max}, \delta_{a,\max}\right),$$
$$\delta_{r,\text{applied}} = \text{sat}\left(\delta_{r,\text{calculated}}; -\delta_{r,\max}, \delta_{r,\max}\right).$$

These limits constrain the surface positions available to execute the manoeuvre. When a calculated command exceeds its limit, the applied deflection remains at the boundary. The aircraft must therefore respond using the available control authority, even if the controller requests a greater moment.

Within the aerodynamic model, the applied deflections contribute to the rolling and yawing moment coefficients:

Equation 39

$$\Delta C_l = C_{l\delta_a}\delta_a + C_{l\delta_r}\delta_r,$$
$$\Delta C_n = C_{n\delta_a}\delta_a + C_{n\delta_r}\delta_r.$$

Their corresponding aerodynamic moment contributions are

Equation 40

$$\Delta L = \bar{q}Sb\,\Delta C_l, \qquad \Delta N = \bar{q}Sb\,\Delta C_n.$$

The same surface command therefore produces a different moment as dynamic pressure changes. Surface position alone does not determine the response; aerodynamic effectiveness, aircraft inertia and the existing flight condition also influence the resulting motion.

Aileron deflection changes the aerodynamic loading across the wings, producing a rolling moment that changes bank angle. Banking rotates the lift vector, allowing part of the aerodynamic force to turn the aircraft while changing the vertical support available. Rudder deflection contributes side force and yawing moment, affecting sideslip and directional motion. The cross-derivatives retain the interaction between these responses, so aileron and rudder effects are evaluated together.

The resulting forces and moments enter the nonlinear six-degree-of-freedom equations. Rotational motion follows

Equation 41

$$\mathrm{I}\dot{\omega} = \mathrm{M}_{\text{total}} - \omega \times (\mathrm{I}\omega),$$

where $\boldsymbol{\omega} = [p, q, r]^T$ The plant integrates angular acceleration to obtain angular rates and updates attitude through the rotational kinematics. Its translational equations update velocity, and the coordinate transformation and position integration determine the evolving flight path. Consequently, changes in surface deflection influence aircraft position through the intervening aerodynamic forces, moments and attitude response. Elevator and throttle control provide the accompanying pitch and propulsion response.

The controller's moment outputs are retained for diagnosis; they are not applied directly to the plant. The realised motion follows from the applied surface deflections and the complete

aircraft dynamics. If saturation prevents the requested response, the remaining bank, rate and sideslip errors continue to influence subsequent commands through feedback.

The deflection limits constrain surface position, while the preceding roll and yaw limits constrain requested aircraft motion. Physical actuator movement-rate limits describe how quickly a surface can reach its target and require separate representation. In validation, sustained surface saturation should therefore be examined alongside tracking errors to determine whether performance is constrained by control demand or available authority.

5.2.4 Longitudinal controller

The longitudinal controller coordinates altitude, pitch and airspeed during route following. It comprises vertical look-ahead, the Master Energy Controller and model-based pitch control. These functions connect the approaching turn requirement to the elevator and throttle commands, allowing vertical recovery to account for the aircraft's available speed and developing lateral manoeuvre.

Vertical displacement during a banked manoeuvre

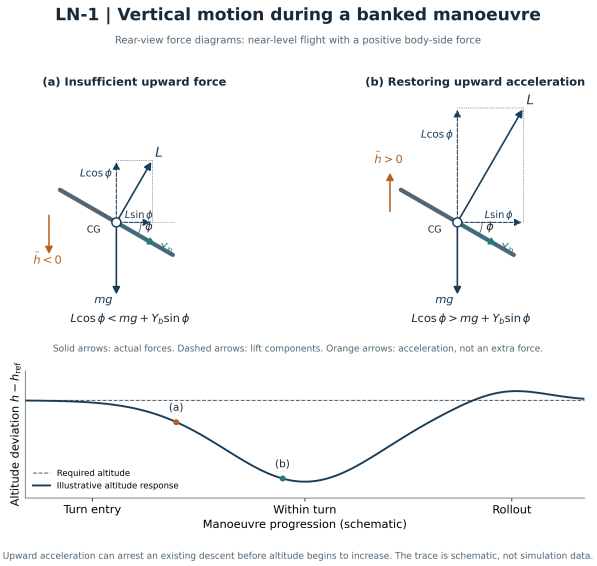


Figure 11 : Vertical-force imbalance and the resulting altitude response during a banked manoeuvre. The altitude trace is schematic

Figure 11 distinguishes downward acceleration from the restoring acceleration required to arrest a descent. During turn entry, banking changes the vertical projection of aerodynamic forces. If the upward force is insufficient, the aircraft accelerates downwards. Increasing lift subsequently produces an upward acceleration, but the aircraft may continue descending until its existing downward velocity has been removed. The force balance must therefore retain the acceleration term throughout the manoeuvre.

With altitude h positive upwards, the plant's body-to-Earth transformation gives

Equation 42

$$\dot{h} = u \sin\theta - v \sin\phi \cos\theta - w \cos\phi \cos\theta.$$

The corresponding vertical acceleration is

Equation 43

$$m\ddot{h} = X_b \sin\theta - Y_b \sin\phi \cos\theta - Z_b \cos\phi \cos\theta - mg.$$

Here, u, v, w are body-axis velocity components; ϕ and θ are bank and pitch angles. The forces X_b, Y_b, Z_b contain aerodynamic and propulsion contributions, excluding gravity.

For near-level flight with small angle of attack, the illustrative force balance reduces to

Equation 44

$$m\ddot{h} \approx L\cos\phi - Y_b \sin\phi - mg.$$

This shows that a body-side force can contribute vertically when the aircraft is banked. Rudder and sideslip effects therefore enter vertical motion through the force transformation, while angular coupling also changes aircraft attitude. Resolving forces into inertial vertical acceleration follows Newton's second law; equilibrium is only the special condition where the resultant is zero (NASA n.d.; Beard and McLain 2012).

Altitude displacement depends on the accumulated acceleration:

Equation 45

$$h(t) = h_0 + \dot{h}_0(t - t_0) + \int_{t_0}^{t} (t - \tau)\, \ddot{h}(\tau)\, d\tau.$$

Consequently, restoring the force balance does not immediately remove an altitude deviation or an existing vertical velocity.

Vertical look-ahead and lift preparation

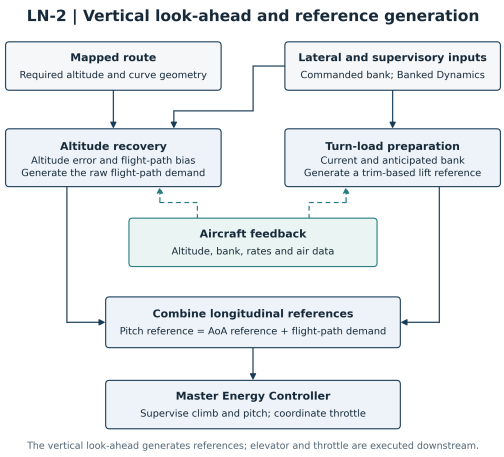


Figure 12 Altitude recovery and anticipated turn-load compensation in the vertical look-ahead.

Figure 12 shows two connected requirements. Altitude feedback determines the flight-path correction, while actual and anticipated bank determine the additional lift reference. The

commanded bank from the lateral controller carries the upstream look-ahead decision into the vertical channel. Banked Dynamics supplies altitude error, flight-path bias and supervisory context. Vertical look-ahead therefore prepares for the developing turn without waiting for the complete bank or altitude departure to occur.

Altitude error is

Equation 46

$$e_h = h_{\mathrm{req}} - h.$$

Before command limiting and relief logic, the active flight-path demand principally follows

Equation 47

$$\gamma_{\mathrm{raw}} = K_{h,\mathrm{state}} e_{h,\mathrm{eff}} + \Delta\gamma_{\mathrm{BD}},$$

where $e_{h,\mathrm{eff}}$ and $\Delta\gamma_{\mathrm{BD}}$ are supplied by Banked Dynamics. The altitude gain changes with its supervisory state.

The lift preparation uses current and future bank magnitudes, bounded by the configured prediction limits:

Equation 48

$$n_a = \frac{1}{\cos|\phi|}, \qquad n_f = \frac{1}{\cos\phi_f},$$
$$n_{\mathrm{FF}} = n_a + \lambda \max\left(n_f - n_a, 0\right).$$

The future magnitude ϕ_f considers commanded and predicted bank. The shared initialisation selects $\lambda = 0.20$ introducing part of the anticipated increase in turn load. The steady level-turn relationship provides this planning estimate of required lift (Federal Aviation Administration 2021).

The required change in lift coefficient is anchored to the reference trim condition:

Equation 49

$$\bar{q}_{\mathrm{ref}} = \bar{q}\left(\frac{V_{\mathrm{ref}}}{V_{\mathrm{safe}}}\right)^2,$$
$$\Delta C_{L,\mathrm{req}} = \frac{mg}{S}\left(\frac{n_{\mathrm{FF}}}{\bar{q}} - \frac{1}{\bar{q}_{\mathrm{ref}}}\right).$$

Using the model's lift derivatives, the corresponding unsaturated angle-of-attack adjustment is

Equation 50

$$\Delta\alpha = \frac{\Delta C_{L,\mathrm{req}} - C_{Lq}\hat{q} - C_{L\delta_e}\left(\delta_e - \delta_{e,\mathrm{trim}}\right)}{C_{L\alpha}},$$

where

Equation 51

$$\hat{q} = \frac{q\bar{c}}{2V_{\mathrm{safe}}}.$$

This accounts for changed speed, turn load, pitch rate and elevator contribution. The implementation applies separate limits to the bank and remaining increments, using the selected lift-schedule gain of 0.75. The resulting reference is combined with flight-path demand:

Equation 52

$$\theta_{\mathrm{raw}} \approx \alpha_{\mathrm{cmd}} + \gamma_{\mathrm{raw}}.$$

The approximation forms the guidance reference; the plant retains the full attitude kinematics. Additional turn lift is introduced through the angle-of-attack reference, avoiding a duplicate climb command simply because the aircraft is banking.

In the present connection, the Banked Dynamics signal named VerticalSupportRatio represents a geometric load multiplier. It does not measure available upward force. Its below-unity support-deficit branch therefore remains inactive; altitude feedback and trim-based lift preparation provide the active correction.

Energy supervision before pitch execution

The Master Energy Controller checks the vertical look-ahead demands against the aircraft's current speed and altitude condition before passing the pitch reference to the elevator controller. This is necessary because increasing pitch to recover altitude can increase the lift requirement and redirect available energy into climbing while airspeed is already deficient. The supervisor therefore moderates positive climb demand according to speed deficit, while preserving recovery authority when the aircraft is also below its required altitude.

Determining the recovery requirement:

The effective speed error and positive speed deficit are

Equation 53

$$e_V = V_{\mathrm{eff}} - V, \qquad d_V = \max(e_V, 0).$$

The speed deficit is normalised using

Equation 54

$$x_V = \mathrm{sat}\left[\frac{d_V}{\max(0.13V_{\mathrm{eff}}, 8)}; 0{,}1\right].$$

The denominator establishes a reference deficit that scales with commanded speed, with a minimum of $8\ \mathrm{m/s}$ The resulting speed urgency is

Equation 55

$$U_V = x_V^2(3 - 2x_V).$$

This smooth transition increases the supervisory response progressively as the speed deficit grows. The implementation increases this urgency by 5% in Bank Dynamics states 3–4 and by 10% in states 5 and above, retaining an upper bound of one.

Altitude urgency is calculated separately:

Equation 56

$$d_h = \max\big(h_{\mathrm{req}} - h, 0\big),$$
$$x_h = \mathrm{sat}\left[\frac{d_h - 5}{15}; 0{,}1\right],$$
$$U_h = x_h^2(3 - 2x_h).\backslash]$$

This additional altitude-priority schedule begins above a 5 m deficit and reaches full urgency at 20 m. **Altitude correction remains active through vertical look-ahead below these thresholds**; the schedule determines how strongly that correction is protected during simultaneous speed recovery.

Moderating climb without abandoning altitude recovery

The climb-authority scale is

Equation 57

$$A_{\mathrm{climb}} = \mathrm{sat}[\max(1 - 0.75U_V,\ 0.35 + 0.50U_h)\,; 0.25{,}1].$$

The first term reduces climb authority as speed recovery becomes more urgent. The second establishes an altitude-dependent lower bound. For example, full speed urgency with zero altitude urgency retains 35% of the positive climb demand. When both urgencies reach one, the altitude-dependent floor retains 85%. These values are selected supervisory settings that balance the two recovery requirements.

The flight-path demand becomes

Equation 58

$$\gamma^* = \begin{cases} A_{\mathrm{climb}}\gamma_{\mathrm{raw}}, & \gamma_{\mathrm{raw}} > 0, \\ \gamma_{\mathrm{raw}}, & \gamma_{\mathrm{raw}} \le 0. \end{cases}$$

The final supervised command is

Equation 59

$$\gamma_{\mathrm{sup}} = \mathrm{sat}(\gamma^{*}; -6^{\circ}, 8^{\circ}).$$

The speed-dependent scaling acts only on positive climb demand. Descent commands pass through this scaling stage unchanged before the common flight-path limits are applied.

Preserving the lift reference in the pitch command

Vertical look-ahead has already combined its angle-of-attack reference with the requested flight-path angle. The energy supervisor therefore adjusts pitch by the same amount that it changes flight-path demand:

Equation 60

$$\theta^{*}_{\mathrm{sup}} = \theta_{\mathrm{raw}} + \gamma_{\mathrm{sup}} - \gamma_{\mathrm{raw}}.$$

Consequently,

Equation 61

$$\theta^{*}_{\mathrm{sup}} - \gamma_{\mathrm{sup}} = \theta_{\mathrm{raw}} - \gamma_{\mathrm{raw}}.$$

Before final pitch limiting, this preserves the angle-of-attack contribution embedded in the incoming references. The supervisor can therefore moderate the climb while retaining the lift preparation generated for the turn.

The final pitch reference and tracking error are

Equation 62

$$\theta_{\mathrm{sup}} = \mathrm{sat}\left(\theta^{*}_{\mathrm{sup}}; -20^{\circ}, 20^{\circ}\right),$$
$$e_{\theta} = \theta_{\mathrm{sup}} - \theta.$$

The pitch controller receives this supervised error directly; flight-path error is not added again. Meanwhile, the throttle channel responds to the combined speed and altitude energy requirement. This coordinates propulsion recovery with the attitude demand executed by the elevator.

Pitch control and elevator execution

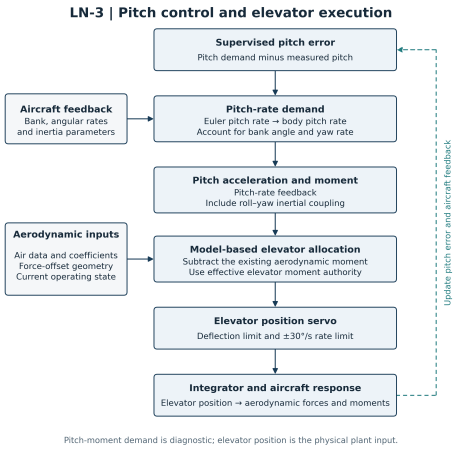


Figure 13 : Conversion of supervised pitch error into a model-based elevator target and rate-limited elevator movement

Figure 13 follows the physical execution of the pitch requirement. Attitude error generates an Euler pitch-rate demand, which is converted into a body pitch-rate requirement using bank angle and yaw rate. Pitch-rate feedback then determines angular acceleration. The controller calculates the required pitching moment, subtracts the moment already represented by the current aerodynamic condition, and solves for elevator position.

The outer demand is

Equation 63

$$e_\theta = \theta_{\text{sup}} - \theta, \qquad \dot{\theta}_d = A_\theta \omega_\theta e_\theta,$$

where $\omega_\theta = 1.80\ \text{s}^{-1}$ and A_θ varies with the supervisory state.

Euler pitch kinematics give

Equation 64

$$\dot{\theta} = q\cos\phi - r\sin\phi.$$

Therefore,

Equation 65

$$q_d = \frac{\dot{\theta}_d + r\sin\phi}{\cos\phi}.$$

The yaw contribution is significant to the intended coordination: during banked motion, the same body pitch rate does not always produce the same pitch-attitude rate. The implementation guards the denominator and applies state-dependent rate limits.

Pitch acceleration and required moment are

Equation 66

$$\dot{q}_{\mathrm{cmd}} = 2\zeta_\theta \omega_\theta (q_d - q),$$
$$M_{\mathrm{cmd}} = I_{yy}\dot{q}_{\mathrm{cmd}} + (I_{xx} - I_{zz})pr + I_{xz}(p^2 - r^2),$$

with $\zeta_\theta = 0.90$ and acceleration limiting applied. The final two terms retain the pitch-axis inertial contribution of simultaneous roll and yaw.

The aerodynamic moment is expressed as

Equation 67

$$M_{\mathrm{total}} = M_{\mathrm{base}} + B_e \delta_e.$$

The base moment includes angle-of-attack and rate contributions together with aerodynamic force-offset moments. The effective elevator authority B_e includes both the direct pitching-moment derivative and elevator-dependent force-offset effects. The target is consequently

Equation 68

$$\delta_{e,\mathrm{target}} = \frac{M_{\mathrm{cmd}} - M_{\mathrm{base}}}{B_e}.$$

This is an absolute position, so trim elevator is not added again. After deflection limiting, the position servo produces

Equation 69

$$\dot{\delta}_{e,\mathrm{cmd}} = \mathrm{sat}\left[4\left(\delta_{e,\mathrm{target}} - \delta_e\right); -30^\circ/\mathrm{s}, 30^\circ/\mathrm{s}\right].$$

The external integrator generates elevator position within the aircraft-specific deflection bounds. If calculated elevator authority becomes insufficient for division, the implementation falls back to trim position.

Throttle and speed–altitude coordination

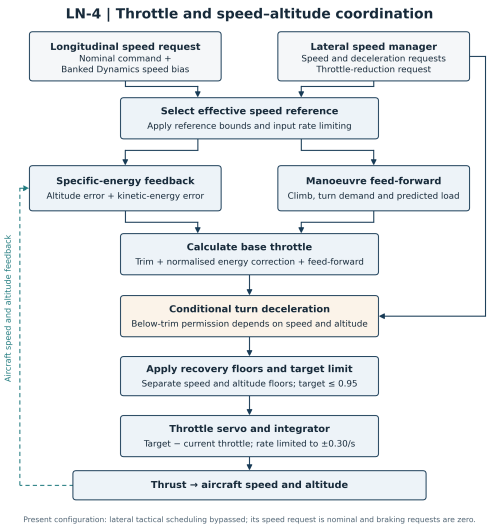


Figure 14 Speed-reference selection, energy-based throttle demand, conditional deceleration and recovery protection.

Figure 14 shows why throttle cannot depend on speed error alone. Altitude recovery requires potential energy, while speed recovery requires kinetic energy. Turn preparation can introduce an additional propulsion requirement before either tracking error becomes large. The Master Energy Controller combines these demands and retains separate recovery floors so that opposing speed and altitude errors cannot completely conceal an individual deficit. The use of combined energy requirements follows established longitudinal-control principles, while the scheduling and limits shown here are SWARAJ's implementation choices (Lambregts 1984).

The bounded longitudinal speed reference V_{sup} includes the Banked Dynamics speed bias. It is combined with the lateral supervisor's request V_{lat}

Equation 70

$$V_{\mathrm{eff}} = \max\left[\min\left(V_{\mathrm{sup}}, V_{\mathrm{lat}}\right), \max\left(0.70V_{\mathrm{sup}}, 45\right)\right].$$

The numerical speed bounds are in $\mathrm{m/s}$ The lateral input passes through the external rate limiter at $+4/-8\ \mathrm{m/s^2}$ These are reference-command rates, not guaranteed aircraft accelerations.

The effective speed and specific-energy errors are

Equation 71

$$e_V = V_{\mathrm{eff}} - V,$$
$$e_E = g e_h + \frac{V_{\mathrm{eff}}^2 - V_{\mathrm{safe}}^2}{2}.$$

The energy error is normalised and bounded before calculating preliminary throttle:

Equation 72

$$e_{E,n} = \mathrm{sat}\left[\frac{e_E}{\max\left(0.20V_{\mathrm{eff}}^2, 450\right)}; -1.5, 1.5\right],$$
$$\delta_{t,\mathrm{base}} = \delta_{t,\mathrm{trim}} + 0.55e_{E,n} + \delta_{t,\mathrm{FF}}.$$

The feed-forward contribution is

Equation 73

$$\delta_{t,\mathrm{FF}} = 0.055G + 0.050C_T + 0.080\Delta n_p.$$

Here, G is normalised positive climb demand, C_T is the bounded turn-compensation demand, and Δn_p is the bounded predicted load-factor increase. This provides propulsion support from the manoeuvre requirement as well as from accumulated error.

When lateral deceleration is requested, the controller can subtract a further throttle contribution and lower its permitted floor below trim. This permission decreases as altitude deficit grows or speed margin reduces. The prototype schedule withdraws permission across a 0–8 m altitude-deficit range and introduces speed permission between 53 and 58 m/s. These are tuning thresholds, not certified operating limits.

Separate speed and altitude recovery floors are then applied:

Equation 74

$$\delta_{t,\mathrm{target}} = \mathrm{sat}\left[\max\left(\delta_{t,\mathrm{base}} - \Delta\delta_{t,\mathrm{decel}}, \delta_{t,V}, \delta_{t,h}\right); 0, 0.95\right].$$

The floors $\delta_{t,V}$ and $\delta_{t,h}$ rise with positive speed deficit and altitude urgency, respectively. This ensures that either condition can independently require additional throttle.

The target is executed through

Equation 75

$$\dot{\delta}_{t,\mathrm{cmd}} = \mathrm{sat}\left[1.25\left(\delta_{t,\mathrm{target}} - \delta_t\right); -0.30, 0.30\right].$$

The throttle integrator starts at trim and is bounded between zero and one. In the active C182 propulsion block,

Equation 76

$$T = (300 \times 4.4482216152605)\delta_t \approx 1334.47\delta_t \text{ N}.$$

Thus, throttle position changes thrust, which changes the aircraft's subsequent speed and vertical response.

The lateral speed manager is connected to this process, but tactical scheduling is bypassed in the present simulation configuration. It supplies nominal speed and zero supplementary braking requests. Energy feedback, manoeuvre feed-forward and Banked Dynamics speed bias remain active.

5.2.5 Gust Response Compensator

The Gust Response Compensator (GRC) extends SWARAJ's route-following framework to account for atmospheric disturbances. Its development follows the reasoning presented in the original sketches: longitudinal, lateral and vertical wind components alter the airflow experienced by the aircraft, producing changes in aerodynamic forces and moments. These changes can affect speed, altitude and centreline tracking simultaneously.

The GRC therefore evaluates both the estimated disturbance and the aircraft's developing response. It supplies additional bank guidance and bounded control corrections while the established lateral and longitudinal controllers continue operating. The route remains fixed, allowing the assessment to examine how the aircraft responds to its environment rather than to an artificially displaced centreline.

Physical interpretation of the gust disturbance

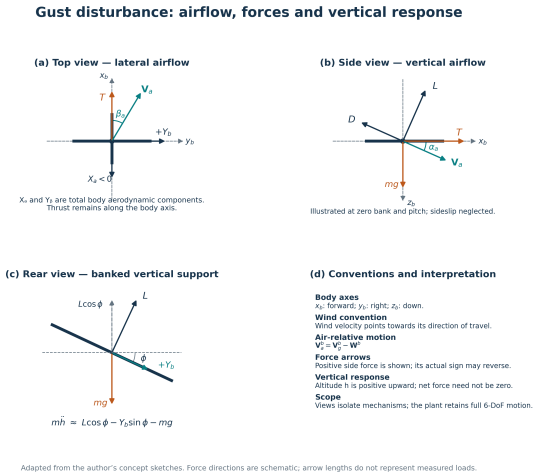


Figure 15 Air-relative velocity, aerodynamic forces and vertical support during gust exposure. Redrawn from the author's conceptual sketches using consistent body-axis conventions.

Figure 15 separates three connected effects. The top view relates lateral airflow to sideslip and side force. The side view shows how vertical airflow changes the angle of attack and the direction of the aerodynamic resultant. The rear view demonstrates how bank and side force influence vertical support. These are complementary views of the same aircraft; their force balances are resolved within a common coordinate system.

The body axes are defined as forward x_b right y_b and downward z_b. Wind expressed in North–East–Down coordinates is transformed into body coordinates before calculating the air-relative velocity:

Equation 77

$$\mathbf{W}^b = R_n^b \begin{bmatrix} W_N \\ W_E \\ W_D \end{bmatrix}, \qquad \begin{bmatrix} u_a \\ v_a \\ w_a \end{bmatrix} = \begin{bmatrix} u \\ v \\ w \end{bmatrix} - \mathbf{W}^b.$$

Here, R_n^b is the navigation-to-body rotation matrix, u, v, w describe ground-relative aircraft velocity resolved in body axes, and u_a, v_a, w_a describe motion relative to the surrounding air. This distinction follows the standard treatment of atmospheric disturbances in aircraft dynamics (Beard and McLain 2012).

The aerodynamic operating condition becomes

Equation 78

$$V_a = \sqrt{u_a^2 + v_a^2 + w_a^2}, \qquad \bar{q}_a = \frac{12}{\rho} V_a^2,$$
$$\alpha_a = \operatorname{atan2}(w_a, u_a), \qquad \beta_a = \operatorname{atan2}\left(v_a, \sqrt{u_a^2 + w_a^2}\right).$$

Consequently, wind changes both dynamic pressure and the angles governing aerodynamic loading. For an unchanged instantaneous ground velocity, a headwind increases airspeed and a tailwind reduces it. Vertical and lateral wind components change angle of attack and sideslip respectively. The subsequent aircraft response depends on the resulting forces, moments and control action.

Using the force representation adopted in the plant,

Equation 79

$$X_b = T - D\cos\alpha_a + L\sin\alpha_a,$$
$$Y_b = \bar{q}_a S C_Y, \qquad Z_b = -D\sin\alpha_a - L\cos\alpha_a,$$

where T, L, D are thrust, lift and drag, and C_Y is the body side-force coefficient. These quantities exclude gravity.

Their contribution to vertical motion is

Equation 80

$$m\ddot{h} = X_b \sin\theta - Y_b \sin\phi\cos\theta - Z_b \cos\phi\cos\theta - mg.$$

Here, h is altitude, positive upward; ϕ and θ are bank and pitch angles. For small pitch and angle of attack, the principal relationship illustrated in Figure 15(c) is

Equation 81

$$m\ddot{h} \approx L\cos\phi - Y_b \sin\phi - mg.$$

The force balance is deliberately **not equated to zero**. A gust can produce vertical acceleration, and recovery requires the controller to change that acceleration while arresting the existing climb or descent. This also explains why a lateral disturbance can affect altitude: bank changes the vertical component of lift, while side force acquires a vertical component.

Estimation of disturbance forces and moments

The GRC compares the current air-relative condition with the condition obtained from the same aircraft velocity without subtracting wind. Defining

Equation 82

$$\alpha_g = \operatorname{atan2}(w, u), \qquad \beta_g = \operatorname{atan2}\left(v, \sqrt{u^2 + w^2}\right),$$

the estimated angular changes are

Equation 83

$$\Delta\alpha_w = \operatorname{wrap}\left(\alpha_a - \alpha_g\right), \qquad \Delta\beta_w = \operatorname{wrap}\left(\beta_a - \beta_g\right).$$

The wrap operation returns the equivalent angle between \(-\pi\) and \(\pi\). This comparison isolates the immediate effect of wind on the relative airflow; it is not simply a comparison against the original trim attitude.

The implemented first-order moment estimates are

Equation 84

$$\begin{aligned} L_g &= \bar{q}_a S b C_{l\beta} \Delta\beta_w, \\ N_g &= \bar{q}_a S\left(b C_{n\beta} + d_{x\mathrm{cp}} C_{Y\beta}\right)\Delta\beta_w, \\ M_g &= \bar{q}_a S\left(\bar{c} C_{m\alpha} + d_{x\mathrm{cp}} C_{L\alpha}\right)\Delta\alpha_w. \end{aligned}$$

Here, L_g, M_g, N_g denote estimated rolling, pitching and yawing disturbances; $S, b, \bar{c}$ are wing area, span and mean aerodynamic chord. The coefficient derivatives describe sensitivity to sideslip or angle of attack, while $d_{x\mathrm{cp}}$ is the signed longitudinal force offset used by the model.

The force-offset terms follow the moment relationship

Equation 85

$$\mathbf{M}_{\mathrm{offset}} = \mathbf{r}_{\mathrm{cp}} \times \mathbf{F}_a.$$

Thus, a change in aerodynamic force can influence rotation through both the aerodynamic moment coefficient and its point of application. The GRC retains selected contributions from this relationship. Its estimates are approximations used to guide compensation; the nonlinear plant calculates the aircraft's resulting motion.

The corresponding drag increment is

Equation 86

$$\Delta D = \bar{q}_a S C_D(\alpha_a) - \bar{q}_g S C_D\left(\alpha_g\right),$$

where

Equation 87

$$\bar{q}_g = \frac{12}{\rho}(u^2 + v^2 + w^2), \qquad C_D(\alpha) = \max(C_{D0} + C_{D\alpha}\alpha, 0).$$

This provides the throttle channel with an estimate of the additional propulsion requirement arising from the changed aerodynamic condition.

Disturbance assessment and anticipated tracking response

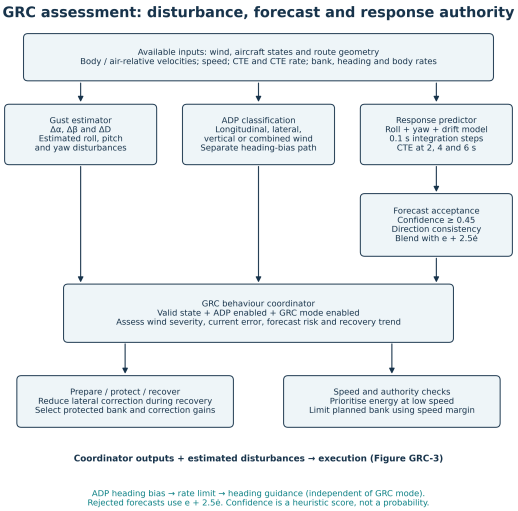


Figure 16 Disturbance estimation, response prediction and selection of GRC correction authority.

Figure 16 develops the "simulator within the loop" concept proposed in the original notes. A reduced response model projects the developing cross-track error using current bank, angular motion, route direction and estimated drift. In parallel, Atmospheric Disturbance Protection (ADP) identifies the wind condition. The coordinator combines this information to determine whether additional action is needed, how strongly to intervene and when to reduce that intervention.

The predictor advances the response in 0.1 s steps and reports cross-track error at two, four and six seconds. Its basic tracking relationship is

Equation 88

$$\dot{e}_{\mathrm{pred}} = -V_a \sin\left(\psi_{\mathrm{pred}} - \chi_{\mathrm{path,pred}}\right) + v_{\mathrm{drift}},$$

where e is signed cross-track error, ψ_{pred} is predicted aircraft heading, and $\chi_{\mathrm{path,pred}}$ is predicted local path direction. The residual drift is obtained from the current response:

Equation 89

$$v_{\mathrm{drift}} = \dot{e} + V_a \sin\left(\psi - \chi_{\mathrm{path}}\right),$$

subject to the implemented bounds. The forecast holds this residual drift, airspeed and the current bank command while propagating the reduced roll and yaw response.

A simpler projection provides a fallback:

Equation 90

$$e_{\mathrm{linear}} = e + T_p \dot{e}, \qquad T_p = 2.5 \text{ s}.$$

When the forecast passes the confidence and direction checks, the coordinator blends it with this projection:

Equation 91

$$e_{\text{forecast}} = (1 - w_p)e_{\text{linear}} + w_p(0.65e_{2s} + 0.35e_{4s}),$$

where $0 \leq w_p \leq 0.35$. The six-second prediction also contributes to the assessment of peak tracking risk.

The forecast therefore estimates where the **current response may lead**; it does not assume advance knowledge of future gusts. Its confidence value is a design-based reliability score rather than a statistical probability.

Intervention increases when the aircraft is moving away from the centreline or the accepted forecast indicates worsening tracking. When the error is already decreasing, correction authority is moderated to avoid opposing the recovery. Low-speed conditions also reduce lateral authority and increase the emphasis on energy recovery. The forecast horizons, weights and switching thresholds are SWARAJ design choices whose effectiveness is assessed through simulation.

Corrective commands and connection to the aircraft

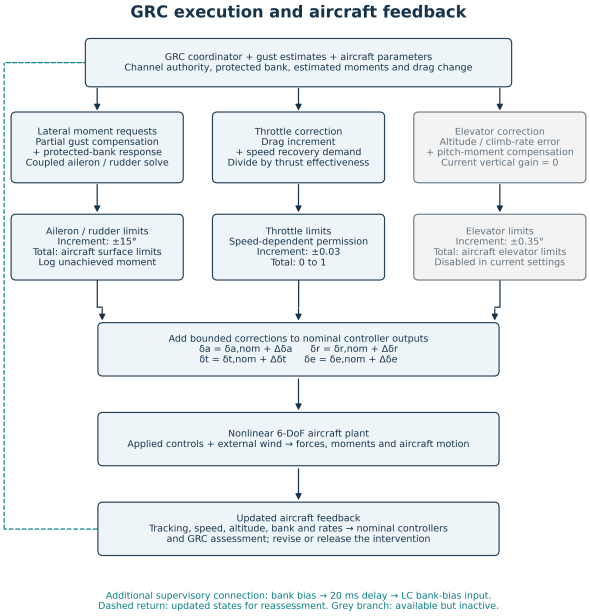


Figure 17 Conversion of GRC decisions into bounded control increments and their addition to the existing aircraft-control commands.

Figure 17 follows the additive arrangement in the original sketch. The nominal controllers continue generating aileron, rudder, elevator and throttle commands. The GRC calculates supplementary corrections and limits them against the remaining control range. Updated aircraft states then return to both the nominal controllers and the GRC assessment, allowing the intervention to change with the response.

The protected-bank demand combines the existing manoeuvre requirement with additional tracking recovery. A delayed bank-bias signal enters the lateral controller, while supplementary rolling and yawing moment requests provide a second correction path.

For the present yaw-compensation settings, the moment requests can be expressed as

Equation 92

$$L_{\mathrm{req}} = \lambda_L\left[-0.25L_g + I_{xx}\Delta\dot{p} - I_{xz}\Delta\dot{r}\right],$$
$$N_{\mathrm{req}} = \lambda_L[-I_{xz}\Delta\dot{p} + I_{zz}\Delta\dot{r}].$$

Here, λ_L is the scheduled lateral correction gain, and $\Delta\dot{p}, \Delta\dot{r}$ are additional roll and yaw acceleration requests associated with the protected-bank response. The inertia terms convert these acceleration requests into moments.

Direct gust-yaw compensation is set to zero in the present configuration, which is why a term proportional to $-N_g$ is absent from the second equation. Rudder correction remains available through the bank-response terms and coupled allocation.

The required surface increments are obtained from

Equation 93

$$\begin{bmatrix} L_{\mathrm{req}} \\ N_{\mathrm{req}} \end{bmatrix} = \bar{q}_a S b \begin{bmatrix} C_{l\delta_a} & C_{l\delta_r} \\ C_{n\delta_a} + \dfrac{d_{x\mathrm{cp}}}{b} C_{Y\delta_a} & C_{n\delta_r} + \dfrac{d_{x\mathrm{cp}}}{b} C_{Y\delta_r} \end{bmatrix} \begin{bmatrix} \Delta\delta_a \\ \Delta\delta_r \end{bmatrix}.$$

The matrix retains the contribution of each surface to both axes. This connects the differential aerodynamic forces in the original sketches to the aircraft-level rolling and yawing response. Expressing moment demand through a control-effectiveness matrix follows established control-allocation practice (Miller and Pei 2025).

Aileron and rudder increments are individually bounded to \(\pm15^\circ\), with the combined nominal and corrective commands restricted by the aircraft-specific deflection limits. If the matrix cannot support a reliable solution, the additional surface commands are withheld. The difference between requested and allocated moments is retained for diagnosis.

The supplementary throttle demand addresses both the estimated drag change and the measured speed error:

Equation 94

$$a_{V,\mathrm{req}} = \mathrm{sat}\left[K_V\left(V_{\mathrm{target}} - V_a\right); -1,\ 1\right],$$
$$K_V = 0.25\ \mathrm{s}^{-1}, \Delta T_{\mathrm{req}} = \Delta D + m a_{V,\mathrm{req}},$$
$$\Delta\delta_{t,\mathrm{raw}} = \lambda_E \frac{\Delta T_{\mathrm{req}}}{\partial T/\partial\delta_t}.$$

The acceleration limits are in $\mathrm{m/s^2}$, λ_E is the scheduled energy correction gain, and $\partial T/\partial\delta_t$ represents thrust effectiveness.

This allows the correction to depend on the aircraft's actual airspeed rather than assuming that every headwind requires increased throttle. Speed-dependent permissions prevent inappropriate acceleration or throttle reduction. The additional command is limited to ± 0.03 equivalent to three percentage points of throttle, and the total command remains between zero and one.

Vertical correction. The implemented elevator branch uses altitude error and vertical speed to request a recovery flight path:

Equation 95

$$\dot{h}_{\mathrm{req}} = \mathrm{sat}\left[0.25 e_h - 0.70 \dot{h}; -2,\ 2\right],$$

where $e_h = h_{\mathrm{ref}} - h$ and the vertical-speed limits are in $\mathrm{m/s}$ The resulting pitch-acceleration request and estimated gust pitching moment give

Equation 96

$$\Delta\delta_{e,\mathrm{raw}} = \lambda_H \frac{I_{yy}\dot{q}_{\mathrm{req}} - 0.20 M_g}{\bar{q}_a S\left(\bar{c} C_{m\delta_e} + d_{x\mathrm{cp}} C_{L\delta_e}\right)}.$$

The branch limits the additional elevator correction to ±0.35° before applying the aircraft's total elevator deflection limit. A tuneable gain controls the strength of this correction and is set to zero in the baseline configuration, which will be discussed in the Controller Tuning Parameters section. The additional GRC elevator branch is therefore implemented but inactive, with vertical gust response handled by the existing vertical look-ahead and longitudinal controllers.

For each control channel, the applied command follows the original additive concept:

Equation 97

$$\delta_{i,\mathrm{applied}} = \mathrm{sat}\left[\delta_{i,\mathrm{nominal}} + \Delta\delta_i; \delta_{i,\mathrm{min}}, \delta_{i,\mathrm{max}}\right], \qquad i \in \{a, r, e, t\}.$$

The estimated gust moments are used to calculate these corrections. They are not added again as separate physical disturbances to the plant, because the wind already affects the plant through its air-relative aerodynamic calculations.

The GRC constitutes a secondary flight-control layer with its own disturbance assessment, response prediction and corrective-command generation. Its architecture relates to established control-augmentation methods, such as the supplementary force and moment commands used by Patterson et al. (2023), while its predictive element shares the objective of anticipating disturbance consequences found in gust-alleviation research (Liu, Sun and Cooper 2017). SWARAJ develops these principles around valley-route tracking: its reduced model predicts cross-track deviation from the developing aircraft response, and its supervisory logic adjusts intervention according to departure, recovery and available speed margin. Unlike the cited predictive gust-load controller, it does not require advance gust measurements or solve an online optimisation problem. Its contribution is the coordinated use of recovery-dependent bank guidance, bounded surface corrections and throttle support while the primary controllers

remain active. The gust-case simulations evaluate this implementation, supporting its treatment as a distinct supplementary control methodology within SWARAJ rather than merely an additional fixed-gain correction.

5.3 Tuning Parameters

SWARAJ's tuning parameters determine how the control architecture responds to the route, aircraft motion and atmospheric disturbances. The governing equations establish the physical relationships, while the gains, prediction weights and supervisory settings provide freedom to shape the response around the project's requirements. The objective is to maintain close centreline tracking while preserving altitude and airspeed, with sufficient preparation for successive bends. Tables in figure18–22 record the initialised configuration and provide a reference for interpreting the simulation results.

SWARAJ / TABLE 01

Guidance and supervisory tuning parameters

Controller	Script parameter	Initialised value	Function and design rationale
Behaviour Supervisor	K_NEAR_TIME, k1_BS	1.365	Scales the near preview-time schedule, controlling the balance between immediate correction and response preparation. This is a multiplier, not a time in seconds.
Behaviour Supervisor	K_FAR_TIME, k2_BS	1.40	Scales the far preview-time schedule to provide preparation for approaching geometry. Excessive preview can introduce premature steering through successive bends.
Behaviour Supervisor	k3_BS	0.60	Scales tracking urgency, moderating how strongly existing route error changes supervisory behaviour.
Behaviour Supervisor	k4_BS	0.60	Scales prediction urgency, controlling the influence of anticipated manoeuvre requirements.
Behaviour Supervisor	k5_BS	0.60	Scales intercept authority, balancing centreline recovery against abrupt directional correction.
Behaviour Supervisor	k6_BS	1.45	Scales curve feed-forward, strengthening preparation based on the mapped geometry.
Lateral Manoeuvre Coordinator	k1_LM	0.85	Scales heading prediction, controlling its contribution to the manoeuvre requirement.
Lateral Manoeuvre Coordinator	k2_LM	0.60	Scales prediction authority to balance anticipation with the current tracking condition.
Lateral Manoeuvre Coordinator	k3_LM	1.15	Scales recovery authority, increasing emphasis on correcting departure from the route.
Lateral Manoeuvre Coordinator	k4_LM	0.70	Scales far-heading authority, limiting how strongly distant geometry influences the present manoeuvre.
Angular Predictor	k_AP	0.625	Scales angular prediction, allowing anticipatory heading adjustment to be moderated.
Master Look-Ahead	k1	0.70	Scales predictive consensus within the combined guidance decision.
Master Look-Ahead	k2	0.70	Scales tracking consensus, determining the influence of current tracking information.
Master Look-Ahead	k3	0.70	Scales far-heading authority during heading selection.
Master Look-Ahead	k4	0.70	Scales intercept authority when returning towards the centreline.
Master Look-Ahead	k5	1.00	Retains the baseline scaling of curve feed-forward.
Master Look-Ahead	k6	1.40	Scales predictive heading contribution to support manoeuvre preparation.

Figure 18 Guidance and supervisory tuning parameters

Table 01, figure 18 groups the parameters governing route anticipation, tracking urgency and heading selection. Separate near and far preview multipliers allow immediate correction and preparation for approaching bends to be adjusted independently. Prediction and curve feed-forward settings influence how early the aircraft begins preparing, while intercept and recovery gains determine the strength of its return towards the centreline. This separation is useful on a winding route, where excessive anticipation can cause premature steering and insufficient anticipation can delay turn entry. The selected values establish the balance assessed in the results; they are controller design choices rather than universal flight-dynamics constants.

SWARAJ / TABLE 02

Banked Dynamics, lateral control and vertical look-ahead tuning

Controller	Script parameter	Initialised value	Function and design rationale
Banked Dynamics	k1_BD	**1.40**	Scales the cross-track-error contribution to demanded lateral acceleration.
Banked Dynamics	k2_BD	**1.40**	Scales the heading-error contribution to demanded lateral acceleration.
Banked Dynamics	k3_BD	**1.40**	Scales bank feedback to correct mismatch between the required and measured banking response.
Banked Dynamics	k4_BD	**1.40**	Scales roll damping, moderating continued rotational motion during correction.
Banked Dynamics	k5_BD	**0.60**	Scales supervisory bank-bias authority before its contribution to lateral execution.
Banked Dynamics	k6_BD	**0.60**	Scales supervisory yaw-rate authority.
Lateral Controller	k1_LC	**0.55**	Heading-to-bank gain; determines the strength of the bank request generated by heading error.
Lateral Controller	k2_LC	**0.80**	Scales the Banked Dynamics correction within the commanded bank angle.
Lateral Controller	k3_LC	**1.62 s^{-1}**	Roll-response frequency parameter; governs how rapidly bank error generates rotational demand.
Lateral Controller	k4_LC	**0.81**	Roll damping parameter; shapes the rate-feedback contribution. The full closed-loop response also depends on the plant and limits.
Lateral Controller	k5_LC	**1.35 s^{-1}**	Yaw-rate feedback gain; determines correction strength for yaw-rate mismatch.
Lateral Controller	k6_LC	**0.90 s^{-2}**	Sideslip feedback gain in the yaw-acceleration demand.
Lateral Controller	k7_LC	**0.225**	Scales the supervisory yaw bias entering the yaw-rate reference.
Vertical Look-Ahead	VLA_settings.K_liftSchedule	**0.75**	Scales the lift-preparation schedule, controlling its contribution to the vertical reference.
Vertical Look-Ahead	VLA_settings.KH_nominal	**0.0032 rad/m**	Converts altitude error into a flight-path correction before subsequent supervision and limits.
Vertical Look-Ahead	VLA_settings.previewFraction	**0.20**	Introduces a partial contribution from the future load requirement, balancing early lift preparation against premature vertical response.

Figure 19 Banked Dynamics, lateral control and vertical look-ahead tuning

Table 02, figure 19 presents the parameters that translate guidance requirements into aircraft motion. Banked Dynamics combines tracking corrections with bank and roll-rate feedback, while the lateral controller determines the strength and damping of the roll–yaw response. Vertical look-ahead introduces lift preparation as the required bank increases. These settings must be considered together because a stronger lateral response can reduce cross-track error while increasing the lift and propulsion needed to retain altitude. Their suitability is therefore assessed through the combined tracking, altitude, airspeed and angular-response histories, rather than through lateral accuracy alone.

SWARAJ / TABLE 03

Disturbance-protection settings and supplementary control authority

Script parameter	Initialised value	Function and interpretation
ADP_Enable	1	Enables Atmospheric Disturbance Protection. Its state must be recorded separately from the modular GRC mode.
LatFraction	2.5	Experimental lateral correction-authority setting. Despite its name, it is a gain and is not restricted to a fraction between zero and one.
VertFraction	0.0	Initialises supplementary vertical correction authority to zero. The elevator OFF/ON comparison supports retaining the disabled branch in the baseline pending further investigation.
EnergyFraction	1.0	Initialises the energy-compensation authority scale; actual intervention depends on activation and scheduling logic.
SafeAngle	15°	Intended supplementary aileron/rudder increment limit. It is separate from the total surface-deflection limits.
FFScale	0	Disables the direct estimated gust-yaw moment contribution. This remains an experimental configuration choice.
BankScale	1	Retains the bank-response contribution to supplementary yaw-moment demand.
GRC_V3_Mode	0	Initialises the modular GRC in its disabled state. Gust-comparison scripts must explicitly override this value when testing GRC operation.
GRC_EnvelopeBankLimit	40°	Configured GRC bank-planning ceiling, subject to further restrictions in the protection logic. It is not a certified aircraft bank limit.
PLP_Mode	0	Initialised mode selector. Its precise function and active connection require confirmation from the receiving model block.

Figure 20 Disturbance-protection settings

Table 03, figure 20 defines the authority and activation settings for supplementary disturbance response. Separate lateral, vertical and energy gains allow each contribution to be evaluated without changing the entire nominal controller. This supports controlled comparisons and allows an ineffective contribution to be withheld while other functions remain available. In the elevator comparison, enabling full supplementary vertical authority increased altitude excursions, supporting retention of zero vertical authority in the baseline configuration. ADP and modular GRC activation are recorded separately: the initialisation script disables the modular GRC, while the relevant gust-test runs explicitly enable it. Where the model uses literal block constants, those connected values must agree with the reported test configuration.

SWARAJ / TABLE 04

Model limits and reference quantities supporting controller tuning

Parameter	Initialised value	Classification and relevance
deMax	±25°	Configured total elevator-deflection limit.
daMax	±20°	Configured total aileron-deflection limit.
drMax	±30°	Configured total rudder-deflection limit.
dx_cp, dy_cp, dz_cp	0.12844, 0, 0 m	Aerodynamic force-application offsets relative to the centre of gravity; influence calculated moments. These are model parameters, not controller gains.
dThrust_dThrottle	1334.47 N per unit throttle	Effectiveness of the implemented linear thrust model, used to convert thrust correction into throttle movement.
dthrust_dthrottle	Same value	Compatibility alias; not an independent parameter.
Vcmd, Vref	Approximately 67.089963 m/s	Nominal speed references obtained from the initial trim velocity.
H0	100 m	Initial altitude reference for the reported configuration.
el_trim	0.0644583653 rad, approximately 3.69319°	Initial trimmed elevator position.
th_trim	0.831982731	Initial trimmed throttle command, approximately 83.20%.
entryDistance	400 m	Initial distance before the first waypoint along the first-segment extension; affects route acquisition.
mass_stall	2650 × 0.45359237 kg	Mass used specifically in the preliminary stall-reference calculation.
S_stall	174 × 0.09290304 m²	Wing area used in that calculation.
H_stall_MSL	100 m	Atmospheric altitude used to estimate density for the stall reference.
CLmax_stall	1.35	Assumed maximum lift coefficient for the preliminary clean-configuration stall estimate.
VS	Calculated from mass, density, area and $C_{L_{max}}$	Derived speed reference for protection logic; not an independently tuned gain or a certified stall-speed value.

The surface limits above are **workspace settings**. Where an aircraft-database block also supplies surface limits, the values connected to the active controller and final command limiter must be identified before attributing saturation behaviour to these settings.

Figure 21: Model limits and reference quantities

Table 04, figure 21 distinguishes controller tuning from the aircraft-model quantities and operating references on which it depends. Control-surface limits bound the available commands, while aerodynamic offsets and propulsion effectiveness influence the conversion of corrective demands into moments and thrust. Trim settings establish the initial equilibrium, and the preliminary stall-speed calculation supplies a reference for protection logic. These quantities require a physical or modelling basis and should not be adjusted solely to improve

tracking results. Their inclusion also helps explain whether a response is limited by the selected control gains, available command range or assumed aircraft characteristics.

SWARAJ / TABLE 05

Initial test configuration

Parameter	Initialised value	Purpose
aircraft_id	1	Selects the Cessna 182 configuration.
gust_case	0	Selects zero wind at initialisation.
gust_scale	1.0	Applies the nominal magnitude when a non-zero gust case is selected.
valley_azimuth	psi0	Aligns the wind generator's reference direction with the initial route heading.
alt_step	0	Removes commanded altitude-step excitation.
v_step	0	Removes commanded speed-step excitation.
disturbance	0	Initialises the separate behavioural disturbance signal; it is not the physical gust enable.

Figure 22 Initial test configuration

Table 05 figure 22 records the starting conditions for reproducible comparisons. The initial configuration selects the C182, zero wind and no commanded altitude or speed steps. Gust cases and controller modes are then changed explicitly for the relevant assessment. Holding the remaining settings constant allows differences in performance to be associated with the intervention being investigated. The separate behavioural disturbance signal is distinguished from the physical wind input so that the source of each disturbance remains clear.

Together, these tables establish a traceable link between controller design and the reported response. The results section can refer to the relevant parameter group when discussing delayed turn preparation, overshoot, altitude loss, speed recovery or command saturation. This provides a basis for explaining why a setting was retained or revised, while recognising that extending SWARAJ to another aircraft or route requires further tuning and validation.

5.4 Development From CDR

Development from the Critical Design Review (CDR) has focused on connecting route-following decisions more closely to the aircraft's physical response. The present framework uses a nonlinear six-degree-of-freedom model to represent coupled translational and rotational motion. This allows banking, sideslip and pitch response to influence the assessment of tracking, altitude and speed. Aircraft properties are supplied through a parameter database, supporting the intended extension of SWARAJ beyond the C182 benchmark.

The control methodology now gives greater emphasis to anticipation and coordination. Near and far look-ahead interpret approaching route geometry, while Banked Dynamics relates the requested manoeuvre to the aircraft's developing response. Vertical look-ahead prepares for changing lift requirements, and longitudinal energy supervision coordinates elevator and throttle demands. These connections establish a continuous feedback process in which route preparation, manoeuvre execution and recovery are assessed together.

Disturbance assessment has also progressed from displacing the route centreline to applying external wind while retaining a fixed path. The GRC supplements the primary controllers through disturbance estimation, predicted tracking response and bounded corrective commands. Explicit tuning records and controlled OFF/ON comparisons strengthen the connection between design choices and observed performance, including identifying

supplementary functions that require further development before inclusion in the baseline configuration.

5.5 Role in autonomy

SWARAJ's autonomous capability arises from the continuous interaction between route interpretation, manoeuvre assessment, control execution and aircraft feedback. The environment map supplies a fixed route, while the control architecture determines how the aircraft should follow it under its changing flight condition. The operator specifies the route and operating objectives; the system generates and revises the heading, attitude and control commands required to pursue them. This establishes autonomy at the level of route following and disturbance recovery within the simulated operating conditions.

Look-ahead guidance provides anticipation by examining the path over near and far time horizons. The relative importance of these horizons changes with approaching curvature, tracking error and the aircraft's response. Consequently, the system can prepare for a bend, prioritise return towards the centreline or manage a turn reversal without requiring a prescribed sequence of control movements. Banked Dynamics connects these geometric requirements to the aircraft's current motion. Its feedback indicates whether the aircraft is turning in the required direction and whether the estimated turning response is sufficient, allowing guidance and manoeuvre corrections to be reconsidered together.

The lateral and longitudinal controllers translate these decisions into coordinated aircraft motion. Aileron and rudder commands establish and regulate the turning response, while vertical look-ahead anticipates the additional lift requirement associated with banking. Elevator and throttle commands then respond to altitude, vertical motion and airspeed. Energy supervision moderates climb demand when speed is deficient and adjusts propulsion to support recovery. These connections are central to the architecture: following a curve requires simultaneous management of direction, vertical support and speed, rather than independent correction of each variable.

Atmospheric Disturbance Protection and the GRC extend this process to disturbed flight. Wind assessment, estimated aerodynamic changes and predicted cross-track response inform the selection and strength of supplementary corrections. Intervention depends on whether the aircraft is departing from the intended condition or already recovering, and can continue after the initiating disturbance subsides. The nominal controllers remain active throughout. In the baseline configuration, the supplementary GRC elevator branch is disabled, so vertical recovery continues through the established vertical look-ahead and longitudinal control functions.

The resulting behaviour is governed by explicit equations, scheduled priorities and bounded commands. Its responsiveness comes from repeatedly evaluating the evolving aircraft state; it does not imply online learning or automatic retuning. Aircraft geometry, inertia and aerodynamic derivatives enter as parameters, supporting extension to other configurations, although each requires appropriate tuning and validation.

The present implementation demonstrates a framework for autonomous execution of a supplied route. Live terrain perception, obstacle avoidance and navigation during GPS loss require additional sensing and localisation capabilities. Simulation currently supplies aircraft and wind states directly; a physical implementation would require measured or estimated

equivalents. The extent of demonstrated autonomy must therefore be established through route completion, tracking accuracy, altitude and speed retention, control saturation and disturbance-recovery results. Controlled comparisons are needed to determine the contribution of individual supervisory functions.

5.6 Simulation and Validation Methodology

The simulation programme examines nominal route following, responses to commanded altitude and speed changes, and recovery under atmospheric disturbances. The aircraft configuration and controller gains are retained across the tests so that differences in performance can be related to the selected operating condition.

5.6.1 Operating conditions

The operating conditions are selected with mountainous routes in mind, particularly those involving repeated turns, confined valleys and changing wind conditions. The corridor bands provide progressively tighter conditions for assessing route-following performance.

Table 8 Route following conditions

Tracking condition	Permitted centreline deviation	Total tracking band	Valley-width evidence
Broad corridor	±250 m	500 m	Approximately 1,200 m around Leh and an average of approximately 1,500 m in the Kullu study area (Kumar and Srivastava 2018; Saczuk 2006).
Restricted corridor	±100 m	200 m	Approximately 500 m at the Kiari bend of the Indus (*Geomorphology* n.d.:60).
Tight corridor	±50 m	100 m	Approximately 180 m near Mahe (Kumar and Srivastava 2018).

Approximately 50 m is adopted as the desired maximum centreline error, with 55 m as the upper acceptable deviation. The broader corridor references and the tighter tracking target are assessed separately.

The 100 m altitude hold is an arbitrary reference selected for the Cessna 182 as a comparatively large test aircraft. It reflects the project's interest in low-level reconnaissance, with reduced radar exposure forming part of the broader mission motivation. Within the simulation, it provides a reference for examining how banking and coupled roll, yaw and rudder responses affect vertical motion.

The commanded speed is approximately 67.09 m/s, or 240 km/h, following the C182 reference condition used in the MathWorks example based on Roskam's aircraft data (MathWorks n.d.-a). Airspeed obtained from the aircraft model is compared with the commanded reference, allowing the controller to respond to any difference. The assessment considers whether speed can be maintained alongside path and altitude tracking.

Based on the geographical data from table 6 we derive an arbitrary set of design constraints for our simulation mentioned below:

Table 9 System Design Constraints

Design requirement	Acceptance criterion
Route-tracking accuracy	Peak absolute CTE ≤50 m preferred; ≤55 m maximum
Altitude tracking	Maximum absolute altitude error ≤5 m
Minimum airspeed	Airspeed ≥58 m/s
Nominal bank angle	Absolute bank angle ≤30° during active-route tracking
Mission completion	Complete the prescribed route

5.6.2 Tuning sensitivity optimisation

The tuning parameters of the established baseline mentioned in section 4.3 in figures 18 and 19 were further subjected to sensitivity analysis for examining the change of aircraft performance with respect to deviation from base line, it assessed how controller parameters influenced nominal route tracking and whether the resulting responses satisfied the project-defined operating conditions. Sensitivity analysis provides a means of relating parameter changes to model behaviour and design requirements (MathWorks- guidance on sensitivity analysis).

A reference simulation established the initial response. The Behaviour Supervisor, Lateral Manoeuvre Coordinator, Master Look-Ahead and Banked Dynamics were then assessed separately. Within each group, parameters were scaled together from −50% to +50% in 10% increments, while the remaining controller settings and nominal mission conditions were held constant. This produced 44 recorded cases, including repeated reference cases, and allowed performance changes to be attributed to the group being varied.

Peak cross-track error was the primary selection measure, supported by RMS cross-track error, altitude deviation, minimum airspeed, maximum active-route bank angle and mission completion. Candidate settings were assessed against the project's tracking and operating requirements, rather than ranked solely by their lowest error.

The resulting histories were compared to identify changes in turn timing, tracking accuracy and aircraft response. Selected settings were checked using the nonlinear simulation, consistent with the need to validate tuned controllers beyond their tuning objectives (MathWorks- guidance on validating tuned control systems). This process established a justified baseline for subsequent command-step and gust-response assessments under separate scenarios.

5.6.3 Initialisation and nominal simulation

The initialisation script loads the aircraft selection, verified trim, initial navigation geometry, controller gains, actuator limits and disturbance settings. The aircraft begins approximately 400 m before the first route point, aligned with the first path segment.

Table 10 Simulation setting

Parameter	Simulation setting
Aircraft model	Cessna 182, `SWARAJ_C182V4`
Initial altitude	100 m
Initial airspeed	Approximately 67.09 m/s
Simulation duration	160 s
Solver	ode45
Maximum integration step	0.01 s
Nominal wind condition	Gust Case 0
Nominal command perturbations	Zero altitude and speed steps

The nominal run script executes initialisation, checks the model configuration, updates the model and performs the simulation. It checks the availability of logged signals and whether the expected duration was completed before passing the new output to post-processing.

This run establishes the reference response without wind or commanded altitude and speed changes. The trajectory, tracking errors, aircraft motion and control commands provide the baseline against which the validation cases are compared.

5.6.4 Altitude and speed command validation

Three tests examine the response to changes in the required flight condition. Each command change begins at **30 s**, and the original reference is restored at **90 s**.

Table 11 altitude and speed step

Test	Altitude command	Speed command	Purpose
Altitude only	100 → 105 → 100 m	Nominal reference retained	Examine altitude tracking and its effect on speed and lateral tracking
Speed only	100 m retained	Approximately 67.09 → 62.09 → 67.09 m/s	Examine deceleration and speed recovery while maintaining altitude
Combined	100 → 105 → 100 m	Approximately 67.09 → 62.09 → 67.09 m/s	Examine simultaneous altitude and speed requirements

The positive speed step setting produces the intended speed reduction through the sign convention implemented in the model. Controller gains remain unchanged throughout the three runs.

Each test plots actual and commanded altitude, actual and supervised airspeed, and active-route cross-track error. These plots are examined for response delay, overshoot, remaining tracking error and repeated oscillations. The combined case is particularly useful for identifying whether satisfying one command produces a larger departure in another controlled quantity.

5.6.5 Gust-response validation

Gusts are applied externally while the intended route remains unchanged. This allows the aircraft's response to atmospheric loading to be examined separately from changes in the commanded route. Mountain terrain can channel and accelerate wind and produce turbulence

and strong vertical currents (Federal Aviation Administration n.d.; United States Air Force 2012).

The prescribed wind cases are:

Table 12 Gust Cases

Case	Wind condition	Events at nominal gust scale
0	Calm baseline	No wind
1	Longitudinal wind	5 m/s headwind during 15–35 s; 4 m/s tailwind during 100–116 s
2	Lateral wind	Opposing crosswinds of 6 m/s during 42–62 s and 4 m/s during 82–99 s
3	Vertical wind	2 m/s downdraft during 63–81 s; 1.5 m/s updraft during 84–99 s
4	Combined wind	Longitudinal, lateral and vertical events applied together

The event windows include the gradual introduction and removal of the wind components.

Cases 1–4 are each simulated with GRC OFF and GRC ON, producing eight runs. Altitude and speed command perturbations are set to zero, and the gust scale remains at 1.0. ADP remains enabled in both conditions, so the comparison examines the additional contribution of GRC actuation.

5.6.6 Interpretation of the simulation plots

The plots are examined together because reduced error in one quantity may require greater control activity or produce a departure elsewhere.

Table 13 Examination of simulation Plots

Plot	Features examined
Aircraft trajectory and CTE	Route acquisition, maximum departure, corridor containment and return towards the centreline
Altitude and altitude error	Command following, disturbance-related displacement, overshoot and recovery
Airspeed	Minimum speed, response to the supervised reference and recovery after a command change or gust
Bank angle	Corrective manoeuvre demand, repeated reversals and excessive or sustained banking
Control commands, where available	Abrupt changes, sustained saturation and continued correction after recovery
Route progress	Continued movement along the route and completion of the defined mission

The command-validation plots distinguish the response during the changed reference from the response after restoration at 90 s. Gust comparison plots show GRC OFF and ON responses on the same axes.

An improvement is assessed from the complete response. For example, reduced CTE is considered alongside altitude departure, minimum speed and bank demand, rather than being treated as sufficient evidence of improved overall performance.

5.6.7 Performance measures and comparison

The scripts calculate peak and root-mean-square errors from the logged samples:

Equation 98

$$e_{\text{peak}} = \max_i |e_i|,$$
$$e_{\text{RMS}} = \sqrt{\frac{1}{N}\sum_{i=1}^{N} e_i^2}.$$

Altitude error is calculated as actual altitude minus the required altitude, after aligning the command with the altitude samples. CTE is evaluated between the first and last samples identifying active route following, where route progress lies between zero and one. Altitude error, minimum speed and maximum absolute bank are evaluated over the full simulation.

The recorded measures are:

- Peak and RMS active-route CTE.
- Peak and RMS altitude error.
- Minimum airspeed, maximum absolute bank angle and final route progress.

For gust cases, the percentage reduction in peak and RMS CTE is calculated relative to GRC OFF:

Equation 99

$$\text{Reduction } (\%) = 100\,\frac{E_{\text{OFF}} - E_{\text{ON}}}{E_{\text{OFF}}}.$$

Positive values indicate reduced error; negative values indicate increased error. The gust plots also include the script's reference lines of ±10 m altitude error, 40.5 m/s airspeed and ±60° bank. These are evaluation markers, distinct from the 40° GRC bank-command setting.

5.6.8 Scope of validation

The present assessment concerns closed-loop simulation under defined command changes and repeatable wind profiles. It examines the intended control responses without claiming that the prescribed gusts reproduce every atmospheric condition encountered in mountainous terrain. Future hardware-in-the-loop testing would connect the physical controller to a real-time simulated aircraft, allowing execution timing and hardware interfaces to be assessed (MathWorks n.d.-b). Subsequent physical testing and measured disturbance data would provide further evidence for checking the model assumptions and extending the framework towards reconnaissance and search-and-rescue applications.

In the following section we will look at the simulation results and validate our findings

6. Results

The simulation results presented in this section evaluate the performance of the SWARAJ autonomous flight-control architecture using the nonlinear six-degree-of-freedom Cessna 182 model. The assessment follows a structured validation sequence, beginning with controller parameter sensitivity and progressing towards nominal flight performance, commanded flight-condition changes and atmospheric disturbances. The objective is to establish how effectively the integrated guidance and control architecture maintains trajectory tracking, altitude, airspeed and aircraft stability under the prescribed operating conditions.

The results are presented in four stages:

- **Controller Parameter Tuning and Sensitivity:** The first stage examines the influence of selected guidance and control parameters on nominal valley following. Grouped parameter variations of ±10% to ±50% are assessed against the established operating requirements. The resulting cross-track errors, altitude deviations, airspeed and bank responses provide the basis for selecting a suitable nominal controller configuration.
- **Nominal Flight Performance:** The selected configuration is subsequently evaluated under undisturbed atmospheric conditions. The analysis considers valley-route tracking, longitudinal and lateral aircraft response, supervisory behaviour and control execution. Relevant aircraft performance characteristics are also compared with available Cessna 182 flight-manual data, where applicable, to assess the physical reasonableness of the simulated response.
- **Altitude and Airspeed Command Validation:** The third stage introduces prescribed altitude and speed-reference changes, individually and simultaneously, while retaining the selected controller gains. These tests examine command tracking, supervisory authority, interactions between longitudinal and lateral control, and the ability to maintain route-following performance during changes in the requested flight condition.
- **Atmospheric Gust Validation:** The final stage assesses aircraft response under four prescribed disturbance cases representing longitudinal wind, lateral wind, vertical wind and combined atmospheric disturbances. Each case is examined with Gust Recovery Controller (GRC) intervention disabled and enabled, allowing its contribution to lateral tracking, altitude retention, airspeed management and recovery behaviour to be evaluated.

Together, these stages provide a progressive assessment of SWARAJ, from the selection of controller parameters to the evaluation of integrated closed-loop performance. The results are interpreted using both numerical performance measures and the corresponding time histories, allowing improvements in one aspect of flight behaviour to be considered alongside potential compromises in others.

6.1 Controller Parameter Tuning and Sensitivity

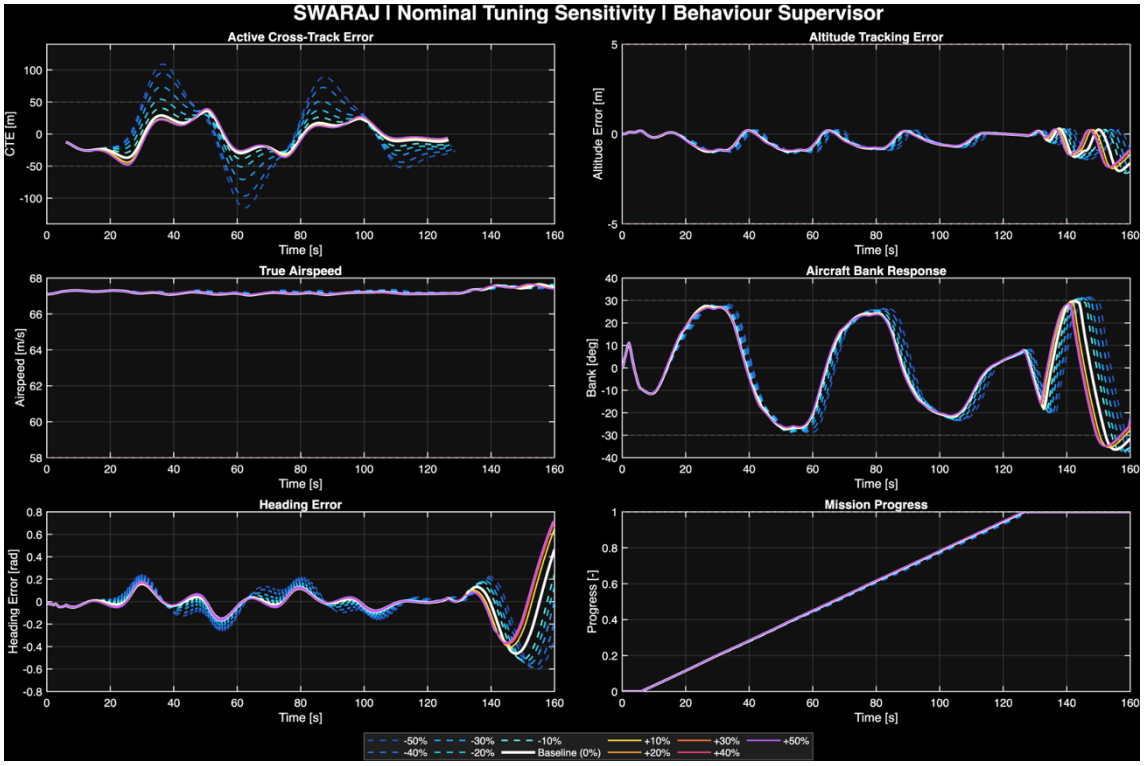


Figure 23 Behaviour Supervisor Tuning

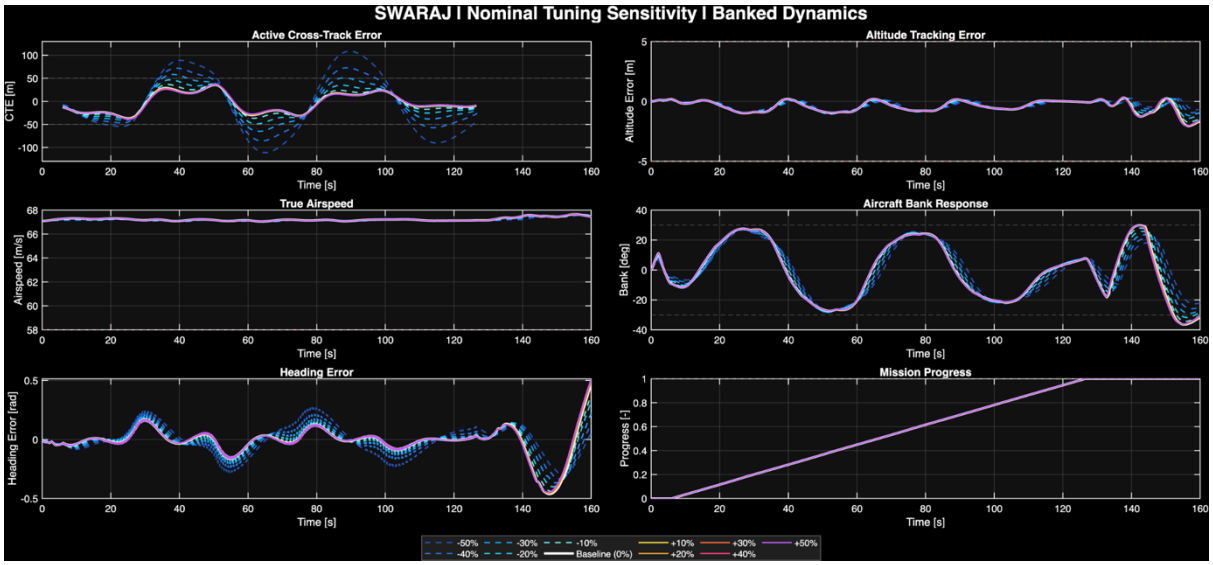


Figure 24 Banked Dynamics tuning

The tuning results show that route-tracking accuracy depended strongly on the controller settings, even when altitude, airspeed and bank remained acceptable. As illustrated in Figures 23 and 24, large reductions in Behaviour Supervisor and Banked Dynamics parameters produced substantial route departures. This supports the importance of turn preparation and coordination established in the methodology: maintaining an acceptable flight condition alone did not ensure accurate navigation.

Table 14 baseline parameter results

Controller group	Parameter variation	Peak CTE (m)	Interpretation
Original reference	0%	37.005	Meets preferred tracking target
Behaviour Supervisor	−50%	114.826	Substantial deterioration
Banked Dynamics	−50%	111.270	Substantial deterioration
Lateral Manoeuvre Coordinator	Complete sweep	36.356–38.201	Comparatively small variation
Master Look-Ahead	−10%	**35.309**	Lowest peak error in the tested sweep

The Lateral Manoeuvre Coordinator responses largely overlapped, indicating comparatively low sensitivity under the tested nominal conditions. Its complete response figure is provided in Appendix A.

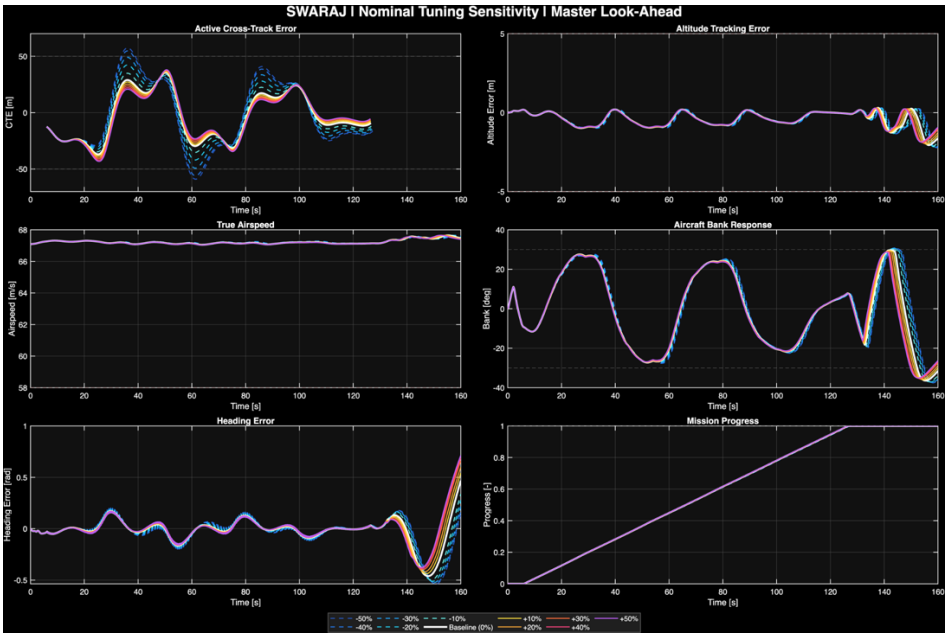


Figure 25 Master Look Ahead Tuning

Master Look-Ahead produced the selected improvement shown in Figure 25. The adjustment reduced the largest centreline departure but increased RMS error. This trade-off was accepted because limiting peak departure was the primary tuning objective.

Table 15 Master Lookahead Tuning

Tracking measure	Original reference	Selected MLA setting
Parameter adjustment	0%	−10%
Peak active-route CTE (m)	37.005	**35.309**
RMS active-route CTE (m)	21.083	22.561
Reduction in peak CTE	—	**4.58%**

The selected configuration also met the remaining project-defined operating conditions.

Table 16 results of tuning sensitivity analysis

Assessment quantity	Selected result	Design condition	Outcome
Peak active-route CTE	35.309 m	≤50 m preferred; ≤55 m maximum	Meets target
Peak full-run altitude error	2.139 m	≤5 m	Meets requirement

Assessment quantity	Selected result	Design condition	Outcome
Minimum full-run airspeed	67.057 m/s	≥58 m/s	Meets requirement
Maximum absolute active-route bank	27.405°	≤30°	Meets requirement
Mission progress	1.000	Route completion	Completed

These findings justify retaining the original settings for the other controller groups and adopting the reduced Master Look Ahead setting as the tuning baseline. More broadly, they demonstrate why tuning must be assessed against tracking accuracy and operating conditions together.

6.2 Nominal Simulation

The final nominal simulation evaluated SWARAJ's ability to follow the prescribed valley route while maintaining altitude, airspeed and aircraft stability. The selected Master Look-Ahead tuning and revised Energy/Throttle Controller were retained, with atmospheric disturbances and external command perturbations disabled. The simulation was terminated at 125 seconds, near the end of the prescribed route, to exclude post-route behaviour for which automatic terminal guidance has not yet been implemented.

Table 17 Nominal simulation conditions

Simulation condition	Setting
Aircraft model	Nonlinear six-degree-of-freedom C182, `SWARAJ_C182V4`
Initial altitude	100 m
Initial airspeed	67.089963 m/s
Initial position	Approximately 400 m before the first route point
Simulation duration	125 s
Solver / maximum integration step	ode45 / 0.01 s
Atmospheric disturbance	Gust Case 0
Altitude and speed command perturbations	Zero
Modular Gust Recovery Controller	Disabled

6.2.1 Route Tracking and Lateral Control

Figure 26 illustrates the aircraft's trajectory relative to the prescribed valley centreline, together with the cross-track error, mission progress and heading guidance. The aircraft follows successive changes in route curvature while maintaining close alignment with the reference path. The largest lateral deviations occur around changes in turn direction, where the controller must anticipate the approaching geometry and establish the required bank before returning towards the centreline.

The maximum absolute active-route cross-track error (CTE) was **34.802 m**, with an RMS error of **22.537 m**, satisfying both the preferred 50 m tracking target and the maximum allowable deviation of 55 m. The actual heading follows the Look-Ahead command with a finite response delay, reflecting the time required for the aircraft to generate and reverse its lateral manoeuvres. Tracking was assessed using the active-route Look-Ahead CTE, rather than the separate map-based error recorded during initial route acquisition.

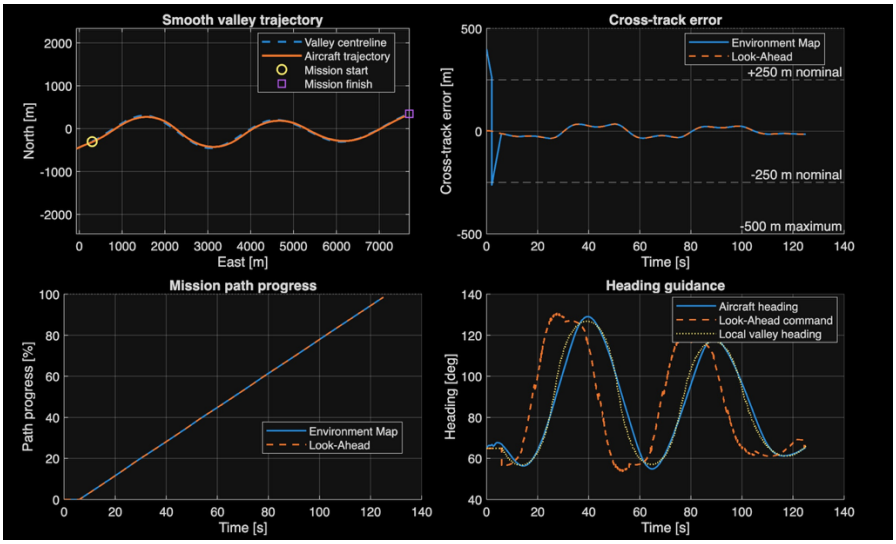


Figure 26 . Nominal valley trajectory, active cross-track error, route progress and heading guidance.

Figure 27 presents the corresponding aircraft bank response, lateral control-surface deflections and yaw-rate behaviour. The maximum absolute bank was **27.347°**, remaining within the nominal 30° design requirement. Aileron and rudder commands vary in response to the alternating turn directions, while the actual bank follows the commanded profile without sustained departures. The control-surface histories remain within the initialized actuator limits, indicating that the selected lateral controller provides sufficient manoeuvring authority for the nominal route.

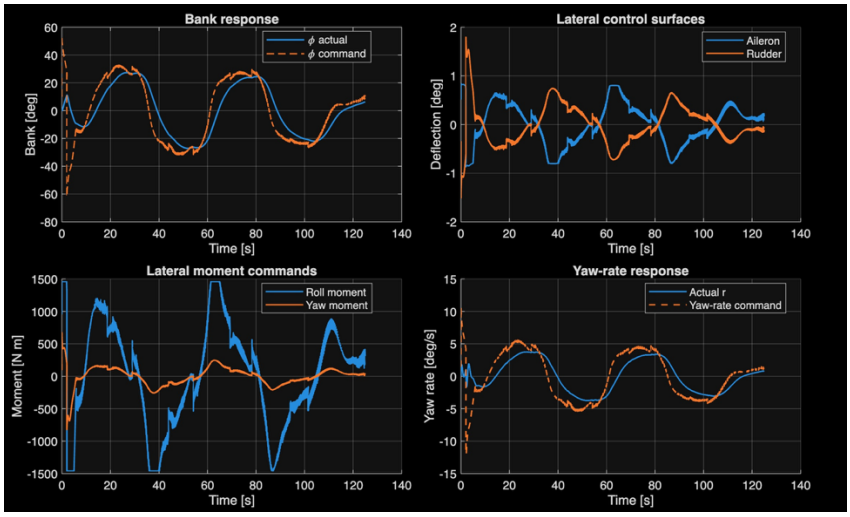


Figure 27 Aircraft bank response, lateral control-surface commands, lateral moments and yaw-rate tracking.

6.2.2 Altitude, Airspeed and Longitudinal Control

Figure 28 shows the aircraft's altitude, flight-path-angle and pitch responses during nominal valley following. Small altitude variations accompany successive turns as the longitudinal controller compensates for changing aircraft attitude and aerodynamic loading. The maximum absolute altitude error was **1.000 m**, with an RMS error of **0.531 m**. These results remain comfortably within the prescribed ±5 m altitude requirement, with no sustained deviation from the 100 m reference.

The pitch and flight-path-angle responses follow their respective commands with small differences during manoeuvre transitions. This demonstrates satisfactory coordination between Vertical Look-Ahead guidance, pitch control and longitudinal supervision under undisturbed conditions.

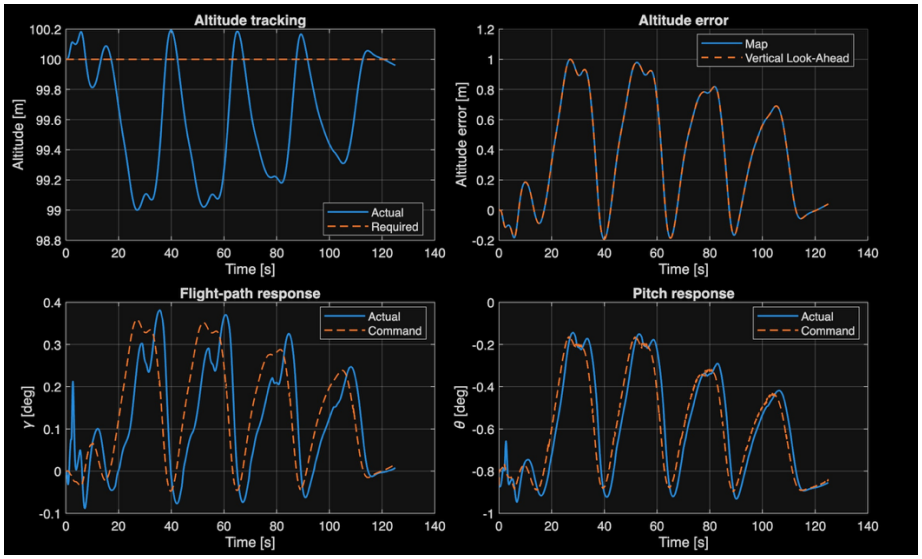


Figure 28 Nominal altitude tracking, altitude error, flight-path-angle response and pitch-command tracking

The aircraft maintained a true airspeed between **66.888 and 67.313 m/s**, remaining above the minimum design speed of 58 m/s. Figure 29 shows temporary reductions in the effective speed command during approaching turns, accompanied by corresponding throttle adjustments. However, the actual speed varies much less than the supervised target, indicating that the response is primarily characterized by nominal airspeed retention rather than precise tracking of anticipatory deceleration commands. The normalized throttle command varied approximately between 0.781 and 0.886 and closely followed its target. No throttle saturation was reported. These results demonstrate satisfactory nominal energy regulation while identifying transient speed-command tracking as an area for further refinement.

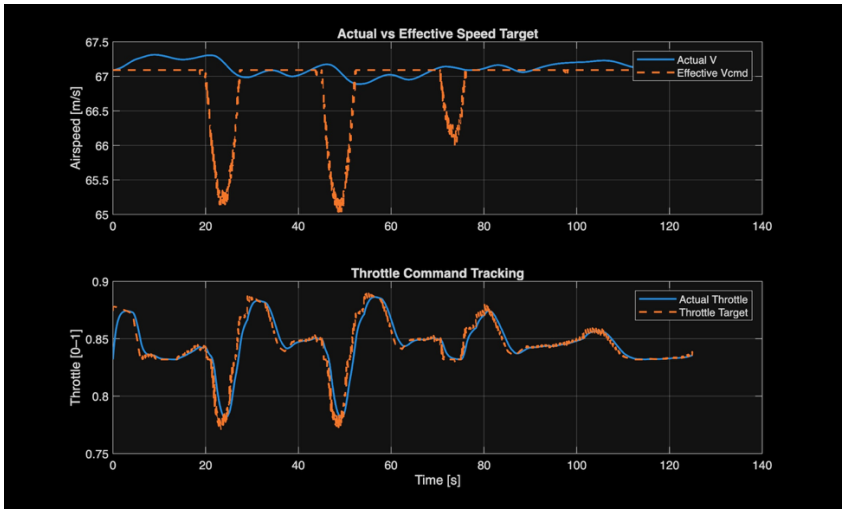


Figure 29 Actual airspeed and effective speed command, with normalized throttle-target tracking.

6.2.3 Design Assessment and Cessna 182 Handbook Comparison

The numerical performance of the selected configuration was assessed against the project-defined requirements, as summarized in Table 16

Table 18 Nominal performance requirement assessment

Assessment quantity	Recorded result	Project requirement	Assessment
Peak absolute active CTE	34.802 m	≤50 m preferred; ≤55 m maximum	Met
RMS active CTE	22.537 m	Supporting measure	Reported
Peak absolute altitude error	1.000 m	≤5 m	Met
RMS altitude error	0.531 m	Supporting measure	Reported
Minimum airspeed	66.888 m/s	≥58 m/s	Met
Maximum absolute bank	27.347°	≤30°	Met
Normalized throttle	0.781–0.886	Within 0–1	Met
Mission completion	Flag = 1; progress = 98.51%	Complete prescribed route	Qualified

For an additional aircraft-level assessment, the simulated performance was compared with the Cessna 182N Owner's Manual (Cessna Aircraft Company 1970, pp. 4-1–4-2). The handbook specifies operating airspeeds in calibrated airspeed (CAS), whereas the simulation records true airspeed (TAS). The published speed limits have therefore been converted into metres per second for consistency with the simulation.

Table 19 Cessna 182N operating references converted to SI units

Operating reference	Handbook value	Converted value
Normal-category turn bank reference	≤60°	≤60°
Manoeuvring speed, V_A	131 mph CAS	**58.56 m/s CAS**
Maximum structural cruising speed, V_NO	160 mph CAS	**71.53 m/s CAS**
Never-exceed speed, V_NE	198 mph CAS	**88.51 m/s CAS**
Positive manoeuvring load factor, flaps up	+3.8 g	+3.8 g
Simulated operating airspeed	66.888–67.313 m/s TAS	66.888–67.313 m/s TAS

The Cessna 182N handbook permits normal-category turns up to 60° of bank, subject to the applicable operating and structural limitations. The simulated maximum of 27.347° is therefore below both the handbook's normal-category bank reference and the more restrictive 30° limit established for SWARAJ. For an approximately coordinated, level turn, the bank angle can be related to normal load factor through:

Equation 100

$$n = 1/cos(\varphi)$$

At the maximum simulated bank of 27.347°, this gives an estimated load factor of **1.126 g**, substantially below the handbook's positive manoeuvring limit of +3.8 g. This is a theoretical steady-turn estimate rather than a measurement of instantaneous structural loading. Aircraft-specific aerodynamic loads and transient control inputs would require separate evaluation.

The more significant operating consideration is the relationship between manoeuvring speed and the selected nominal flight speed. Using the initialized air density of 1.2133 kg/m³ and standard sea-level density of 1.225 kg/m³, the equivalent airspeed can be estimated from:

Equation 101

$$V_E AS = V_T AS \times \sqrt{(\rho/\rho_0)}$$

The simulated TAS range of 66.888–67.313 m/s corresponds to an estimated 66.57–66.99 m/s EAS**.** At the low altitude and speed considered, EAS is expected to be close to CAS, although a precise CAS value requires the appropriate compressibility and air-data calibration corrections. This places the nominal operating speed approximately 8.0–8.4 m/s above the handbook's published manoeuvring-speed reference of 58.56 m/s CAS, while remaining below the 71.53 m/s maximum structural cruising-speed reference.

Exceeding manoeuvring speed does not itself prohibit normal flight or controlled turns. Rather, it reduces the assurance associated with abrupt control inputs and requires the resulting aerodynamic and structural loads to be considered. Although the simulated bank angle is moderate and the lateral-control histories are bounded, the present results do not independently establish structural compliance during all manoeuvre transitions.

Consequently, speed-dependent manoeuvring authority should be introduced in subsequent design development**,** allowing the guidance and bank-control functions to coordinate turn severity, airspeed and available load-factor margin. The present 30° nominal bank limit provides a conservative operational target but is not, by itself, sufficient to demonstrate compliance with the aircraft's complete manoeuvring envelope.

The nominal simulation demonstrates that SWARAJ maintains satisfactory valley-centreline tracking, altitude regulation and minimum airspeed while completing the prescribed sequence of turns. The numerical tracking and flight-condition requirements are satisfied throughout the evaluated interval. Comparison with the Cessna 182N handbook supports the physical reasonableness of the observed bank and airspeed response, while identifying manoeuvring-speed coordination as an important outstanding design consideration. The simulation therefore establishes a satisfactory preliminary nominal-performance baseline, rather than demonstrating complete aircraft flight-envelope or structural compliance. These results form the reference for the subsequent altitude and speed command-validation tests and atmospheric gust assessments.

6.3 Altitude and Speed Command-Deflection Validation

The command-deflection tests evaluated SWARAJ's response to changes in the required flight condition while maintaining the prescribed valley trajectory. Three simulations examined altitude and airspeed changes independently and simultaneously, retaining the selected guidance gains and revised Energy/Throttle Controller. The commanded changes were introduced at 30 seconds and restored at 90 seconds, with atmospheric disturbances disabled throughout.

The response was assessed through command tracking, transient deviation, recovery behaviour and the effect on lateral-route tracking. These measures follow established time-domain control-system assessment principles (MathWorks, *Step-response characteristics*). The combined test additionally examines the interaction between speed and altitude regulation, consistent with the coordinated longitudinal-control principles discussed by Lambregts (1984).

Table 20 Altitude and airspeed command-validation conditions

Test	Altitude command	Airspeed command
Altitude only	100 → 105 → 100 m	67.09 m/s retained
Speed only	100 m retained	67.09 → 62.09 → 67.09 m/s
Combined	100 → 105 → 100 m	67.09 → 62.09 → 67.09 m/s

6.3.1 Altitude Command Response

Figure 30 illustrates the aircraft's response to a 5 m increase in commanded altitude. Following the step at 30 seconds, the aircraft begins climbing towards the revised reference and approaches 105 m at approximately 40 seconds. The response exhibits small oscillations around the commanded altitude during successive route manoeuvres, before returning towards 100 m following reference restoration at 90 seconds.

The initial command change produces a temporary altitude-tracking error because the reference changes instantaneously while the aircraft responds through its finite longitudinal dynamics. This transient should be distinguished from the nominal altitude-hold requirement, which was evaluated separately in Section 5.2. The subsequent response demonstrates successful command recognition and altitude recovery, although repeated variations indicate that transient damping remains an area for refinement.

Airspeed remains close to its nominal operating condition, with temporary changes associated with the existing turn-speed supervision. The cross-track error follows a pattern similar to the nominal response, indicating that the commanded climb does not introduce a substantial deterioration in lateral tracking. This supports the ability of the vertical guidance and longitudinal controller to accommodate an altitude change without disrupting the primary route-following function.

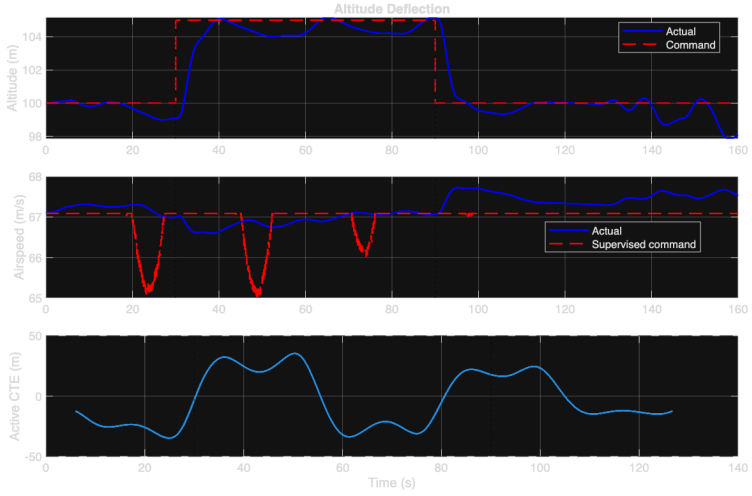


Figure 30 Altitude-command response showing actual and required altitude, supervised airspeed and active cross-track error.

6.3.2 Airspeed Command Response

Figure 31 presents the aircraft's response to a commanded airspeed reduction from approximately 67.09 to 62.09 m/s, while retaining the 100 m altitude reference. Following the

command at 30 seconds, the aircraft progressively decelerates, reaching approximately 63.7–63.8 m/s before the original speed reference is restored.

The response demonstrates that the revised Energy/Throttle Controller provides meaningful deceleration authority. However, the aircraft does not fully reach the reduced reference during the 60-second interval, retaining a speed-tracking difference of approximately 1.6–1.7 m/s near the end of the commanded reduction. Following reference restoration at 90 seconds, the aircraft gradually accelerates towards the nominal operating speed. This attributed to the control logic implied that prioritises throttle and speed management over response to speed deviations that is it manages to efficiently maintain command speed with its best ability during turns as well as other commands.

Temporary reductions in the supervised speed command are also observed during approaching turns, although the actual aircraft response is slower and smaller in magnitude. This indicates that reference generation and commanded-speed execution remain distinguishable functions.

Altitude remains close to the prescribed 100 m reference despite the change in airspeed, while active-route CTE remains comparable to the nominal response. This is consistent with established aircraft-control principles in which speed and flight-path regulation must be coordinated because of their shared influence on aircraft energy (Stevens, Lewis and Johnson 2015; Lambregts 1984).

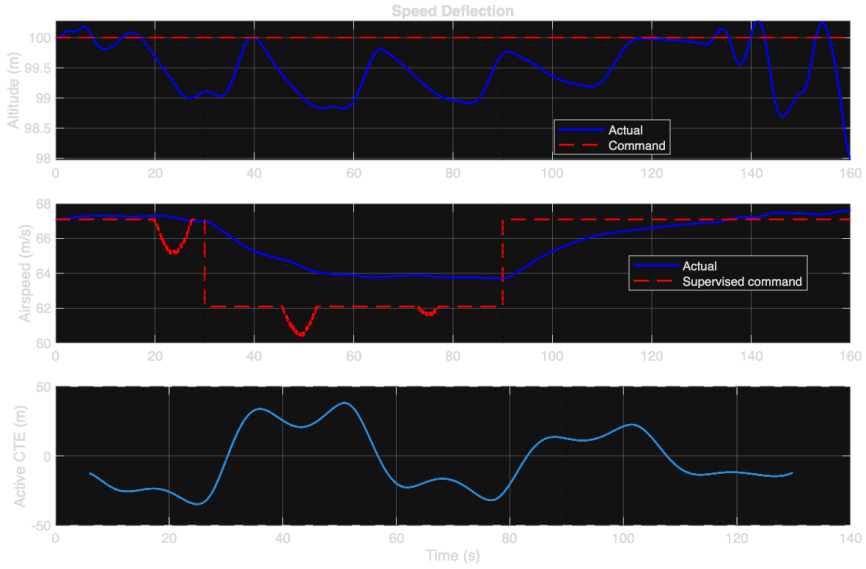


Figure 31 Airspeed-command response showing altitude retention, actual and supervised speed, and active cross-track error.

6.3.3 Combined Command Response

Figure 32 examines simultaneous changes in altitude and airspeed. At 30 seconds, the aircraft is commanded to climb from 100 to 105 m while reducing airspeed from approximately 67.09 to 62.09 m/s. Both references return to their nominal values at 90 seconds.

The aircraft initiates the climb and deceleration together, with altitude approaching 105 m and airspeed decreasing to approximately 63.7–63.8 m/s. The altitude response remains similar to the altitude-only experiment, while the speed response closely resembles the independent deceleration test. Following restoration, the aircraft descends towards 100 m and progressively accelerates towards its original speed.

These responses are significant because climbing increases aircraft potential energy while deceleration reduces kinetic energy. The combined test therefore examines whether the Energy/Throttle Supervisor can coordinate competing longitudinal requirements rather than responding to either quantity independently. This interpretation is consistent with the energy-based longitudinal-control principles established by Lambregts (1984), although SWARAJ retains its own supervisory logic and controller implementation.

The resulting active cross-track response remains broadly consistent with the nominal trajectory. This indicates that simultaneous longitudinal commands can be accommodated without a major observable deterioration in valley following.

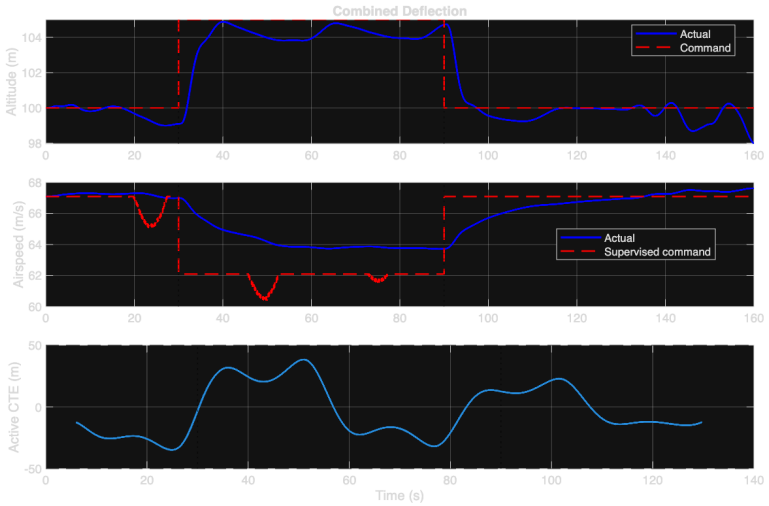


Figure 32 . Combined altitude and speed command-deflection response showing longitudinal coordination and active cross-track error.

6.3.4 Performance Assessment

The three simulations demonstrate that SWARAJ responds to changes in the requested flight condition while retaining its primary route-following behaviour. The observed responses are summarized in Table 19

Table 21 Command-deflection performance assessment

Assessment	Altitude only	Speed only	Combined
Altitude response	Approaches 105 m	Remains near 100 m	Approaches 105 m
Speed response	Nominal speed maintained	Decelerates to ~63.7–63.8 m/s	Decelerates to ~63.7–63.8 m/s
Command tracking	Achieved with oscillation	substantial speed tracking	Altitude achieved and substantial speed tracking
Lateral tracking	Nominal-like CTE response	Nominal-like CTE response	Nominal-like CTE response
Reference restoration	Altitude recovery	Speed recovery	Both quantities recover
Identified limitation	Minor altitude oscillation	Speed-response lag	coordination requires refinement

The altitude-only test demonstrates effective vertical command execution, the speed-only test confirms that a relaxed throttle authority produces substantial deceleration. The combined test provides further evidence that the longitudinal control functions can operate simultaneously without a major observable loss of lateral-route tracking. These results support the preliminary integration of SWARAJ's guidance, vertical control and energy-supervision functions.

However, the experiments do not establish complete independence between the control channels, nor do they demonstrate optimum transient performance. Further development would therefore focus on refined speed recovery and altitude damping while retaining the nominal valley-tracking performance established in Section 5.2.

Overall, the command-deflection validation demonstrates satisfactory preliminary control coordination under changing flight requirements, providing a basis for evaluating the integrated system under the atmospheric disturbances considered in Section 5.4.

6.4 Atmospheric Gust Validation

The atmospheric gust validation assessed SWARAJ's ability to maintain valley-route tracking, altitude and airspeed under prescribed wind disturbances. Four gust cases were examined, representing longitudinal, lateral, vertical and combined wind conditions. Each case was simulated with the Gust Recovery Controller (GRC) disabled and enabled, while retaining the selected guidance gains and revised Energy/Throttle Controller. The Autonomous Disturbance Predictor (ADP) remained enabled in both configurations, allowing the additional contribution of GRC intervention to be evaluated.

The response was assessed through active-route cross-track error (CTE), altitude deviation, minimum airspeed, aircraft bank response and recovery behaviour. These comparisons follow the time-domain disturbance-response approach discussed by Wright and Cooper (2015). The selected wind conditions reflect the atmospheric disturbances associated with mountainous terrain, including abrupt changes in wind direction and vertical air movement (Federal Aviation Administration n.d.).

6.4.1 Longitudinal Wind Disturbance

Figure 33 presents the aircraft's response to longitudinal wind disturbances, comprising a 5 m/s headwind followed by a 4 m/s tailwind. Both configurations exhibit changes in airspeed as the wind conditions vary, while maintaining the prescribed valley trajectory.

With GRC disabled, the maximum active-route CTE was 42.307 m. Enabling GRC reduced this value to **37.820 m**, representing a 10.61% improvement. The peak altitude error also decreased from 2.140 m to 1.744 m, while minimum airspeed remained approximately 64 m/s in both configurations.

The lateral tracking responses remain broadly similar, indicating that the existing guidance and energy-control functions already accommodate much of the longitudinal disturbance. Although GRC improves peak CTE, the RMS CTE increases slightly from 21.625 m to 21.960 m. hence suggesting a trade off between large error and tracking accuracy.

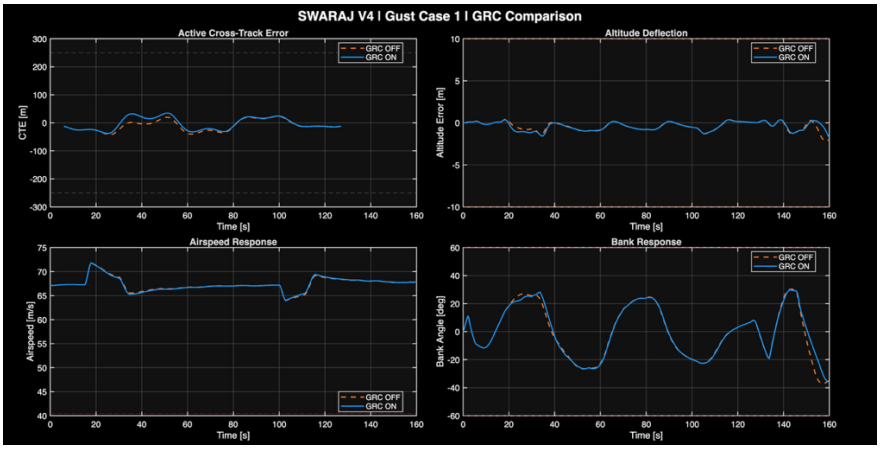


Figure 33 . Longitudinal gust response comparing GRC OFF and ON through active CTE, altitude deviation, airspeed and aircraft bank.

6.4.2 Lateral Wind Disturbance

Figure 34 examines the response to opposing crosswind disturbances of 6 m/s and 4 m/s. Compared with the longitudinal case, the lateral wind produces a substantially larger change in route-tracking performance, particularly when GRC intervention is disabled.

Without GRC, the aircraft reaches a maximum active CTE of 83.104 m, exceeding the project's 55 m maximum tracking criterion. With GRC enabled, the peak error decreases to **44.979 m**, representing a **45.88% reduction**. RMS CTE also decreases from 34.498 m to 27.480 m, an improvement of 20.34%.

The bank response demonstrates that GRC modifies the timing and magnitude of corrective lateral manoeuvres, allowing the aircraft to contain the larger cross-track departures and recover towards the centreline. This supports the intended role of GRC as a supplementary disturbance-response layer operating alongside the primary lateral controller.

The improved lateral response is accompanied by an increase in peak altitude error from 1.866 m to 3.354 m. Although this remains within the prescribed ±10 m gust-altitude

criterion, it indicates that increased lateral corrective activity can be associated with greater longitudinal deviation, which still showcases aircrafts tendency to recover as seen in figure 34.

Minimum airspeed remains approximately 62.6 m/s with GRC enabled, preserving the required speed margin. The results therefore demonstrate a substantial improvement in crosswind tracking while identifying a moderate altitude-control trade-off.

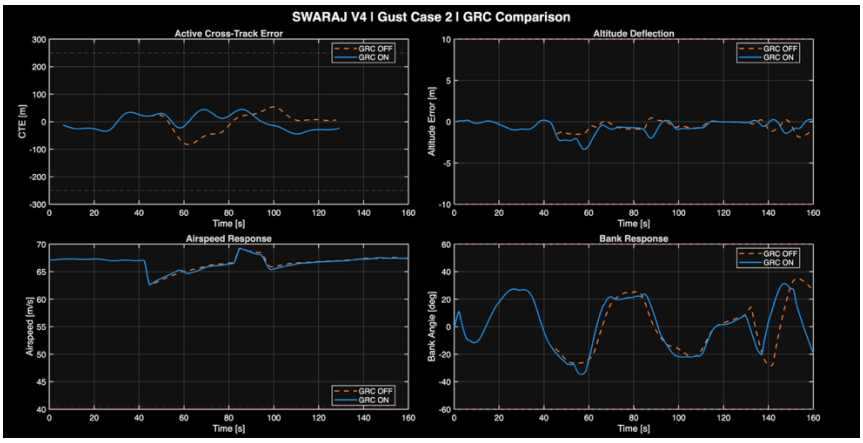


Figure 34 Lateral gust response demonstrating improved cross-track containment with GRC enabled, alongside altitude, airspeed and bank responses.

6.4.3 Vertical Wind Disturbance

Figure 35 presents the aircraft response to a 2 m/s downdraft followed by a 1.5 m/s updraft. Unlike the lateral wind case, the principal effect is observed in altitude tracking rather than route-following accuracy.

The active CTE histories remain almost identical between GRC OFF and ON, with a maximum recorded error of 34.802 m in both configurations. This indicates that the additional GRC intervention provides no meaningful improvement in peak lateral tracking under the prescribed vertical disturbance.

The altitude response shows a more pronounced difference. With GRC disabled, the maximum absolute altitude error was 7.897 m, increasing to **10.502 m** when GRC was enabled. The RMS altitude error similarly increased from 2.442 m to 3.094 m.

During the downdraft, the aircraft descends below its commanded altitude before recovering through the subsequent updraft and controller response. The GRC ON configuration exhibits a larger initial downward excursion, followed by a positive altitude overshoot of approximately 5–6 m. The aircraft subsequently returns towards the nominal 100 m reference rather than developing a sustained divergence.

The recorded peak exceeds the ±10 m gust-altitude criterion by 0.502 m. Although the deviation is transient and recovery is demonstrated, this remains a minor trade off of the specified assessment boundary.

The selected GRC configuration retains zero direct supplementary elevator authority. Consequently, the increased altitude response may be associated with coupling between lateral intervention, aircraft dynamics and the existing longitudinal controller, although the present plots do not independently establish the exact mechanism.

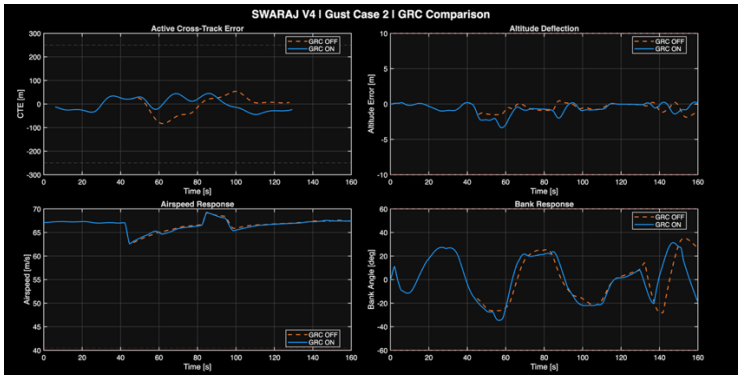


Figure 35 Vertical gust response showing altitude displacement and recovery, with largely unchanged lateral tracking between GRC configurations.

6.4.4 Combined Atmospheric Disturbance

Figure 36 evaluates the simultaneous application of longitudinal, lateral and vertical wind disturbances. This represents the most demanding integrated case in the validation programme, requiring SWARAJ to maintain lateral guidance while responding to changes in airspeed and altitude.

With GRC disabled, peak active CTE reaches 91.526 m, exceeding the prescribed maximum tracking deviation. Enabling GRC reduces this to **37.820 m**, corresponding to a **58.68% improvement**. RMS CTE also decreases substantially, from 35.046 m to 20.862 m, representing a 40.47% reduction.

The cross-track histories show that GRC contains the largest disturbance-induced departures while preserving progression along the prescribed route. The bank response reflects additional corrective manoeuvres, particularly during the crosswind events, although the recorded aircraft response remains bounded.

The longitudinal response reveals a trade-off similar to the vertical-only case. Peak altitude error increases from 8.672 m with GRC OFF to **11.333 m with GRC ON**, exceeding the ±10 m criterion by 1.333 m. Nevertheless, the aircraft demonstrates recovery towards its commanded altitude following the principal disturbance events, rather than an accumulating altitude departure.

Minimum airspeed with GRC enabled was 61.507 m/s, remaining above the 58 m/s design reference. The simulation also recorded full route progress, demonstrating continued mission execution despite the combined disturbance inputs.

These results indicate that GRC provides substantial lateral tracking benefits under compound atmospheric loading, although further coordination with longitudinal control is required to reduce transient altitude excursions.

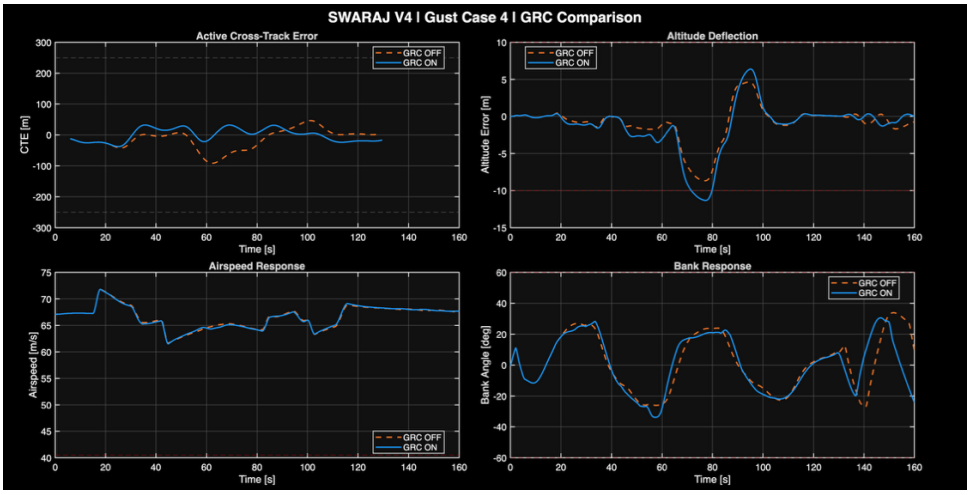


Figure 36 Combined gust response comparing GRC OFF and ON, illustrating improved lateral containment and the associated altitude-response trade-off.

6.4.5 Gust-Response Performance Assessment

The numerical results from all four gust cases are summarized in Table 5.x. The improvement percentages are calculated relative to GRC OFF, with positive values representing reduced tracking error.

Table 22 Gust Recovery Controller performance comparison

Gust case	Peak CTE OFF → ON (m)	Peak CTE reduction	RMS CTE OFF → ON (m)	Peak altitude error OFF → ON (m)
1 — Longitudinal	42.307 → 37.820	10.61%	21.625 → 21.960	2.140 → 1.744
2 — Lateral	83.104 → 44.979	**45.88%**	34.498 → 27.480	1.866 → 3.354
3 — Vertical	34.802 → 34.802	0.00%	22.219 → 22.198	7.897 → 10.502
4 — Combined	91.526 → 37.820	**58.68%**	35.046 → 20.862	8.672 → 11.333

The most significant improvements occur in Cases 2 and 4, where GRC substantially reduces peak and RMS CTE under lateral and combined disturbances. All four GRC ON configurations maintain peak active CTE below the preferred 50 m requirement, while the minimum recorded airspeed across the cases remains above 58 m/s.

The maximum recorded bank angles with GRC enabled range from approximately 33.86° to 36.16° over the complete simulations. These values exceed the 30° nominal tracking target, but gust-recovery operation is assessed separately from nominal manoeuvring.

Altitude performance remains a principle trade off. Cases 3 and 4 marginally exceed the specified ±10 m maximum altitude-error criterion with GRC enabled by maximum of 1.3m. this can be neglected given size and mass of the aircraft accounting to minor modal differences but the overall response is quite satisfactory given the aircraft shows tendency to maintain the altitude rather than diverging to error of 10+ m in one direction that is, when it faces a dip in altitude in case 3 and 4 it tends to restore its altitude thereby shooting above 100m and then restoring its motion.

The distinction between reducing tracking error and alleviating aerodynamic or structural gust loads is also important. Aircraft gust-alleviation research commonly evaluates forces, moments and structural response alongside controller activity (Wright and Cooper 2015; Capello, Guglieri and Quagliotti 2014). The present SWARAJ assessment demonstrates improvements in trajectory containment but does not independently establish reductions in structural gust loading.

6.4.6 Gust Validation Outcome

The four gust-validation cases demonstrate that SWARAJ maintains prescribed route progression and sufficient airspeed under the tested disturbances. Enabling GRC produces the strongest improvements in lateral and combined wind conditions, reducing peak active CTE by 45.88% and 58.68%, respectively. These results support the effectiveness of the supplementary recovery controller in improving lateral containment under disturbed flight conditions.

The vertical-response results also identify an important limitation. While altitude recovery is demonstrated, the selected GRC configuration marginally exceeds the prescribed gust-altitude criterion in Cases 3 and 4. Accordingly, the present results establish satisfactory preliminary lateral gust-compensation performance, with vertical-response refinement remaining necessary.

Overall, the gust-validation programme provides evidence that the integrated guidance and recovery architecture can respond to multiple atmospheric disturbance directions while preserving route-following capability. Future development should concentrate on improving the coordination between lateral gust intervention, aircraft energy management and vertical recovery, while maintaining the demonstrated improvements in cross-track containment.

6.5 Development Progress from Conceptual Design

The Conceptual Design Review (CDR) established the feasibility of SWARAJ's behavioural supervisory architecture using simplified three-degree-of-freedom aircraft dynamics. The Preliminary Design Review (PDR) advances this concept through integration with a nonlinear six-degree-of-freedom Cessna 182 model, incorporating coupled aircraft motion, aerodynamic forces and moments, propulsion response and coordinated flight-control functions.

The progression in model fidelity, guidance performance, operating conditions and validation scope is summarized in Table 23.

Table 23 Comparison of conceptual and preliminary design performance

Assessment parameter	Conceptual Design Review	Preliminary Design Review
Aircraft representation	Simplified 3-DoF UAV model	Nonlinear 6-DoF Cessna 182 model
Guidance and supervision	Adaptive look-ahead, behavioural supervision, lateral and longitudinal control	Predictive look-ahead, behavioural supervision, coordinated lateral and longitudinal control, energy management and GRC
Valley corridor conditions	Preferred ±250 m; maximum permitted ±500 m	Broad ±250 m; restricted ±100 m; tight ±50 m
Nominal peak CTE	175.5 m (sampled)	34.802 m (active route)
Tracking target	Maintain preferred corridor containment	Desired peak CTE ≤50 m; upper acceptable 55 m
Nominal altitude	Approximately 100 m	100 m reference; peak error 1.000 m
Speed evaluation	Approximately 45.03–65.00 m/s; limited response to altered speed priorities	66.888–67.313 m/s nominal; separate speed-command tracking tests
Disturbance assessment	One predefined environmental disturbance	Four atmospheric gust cases, each evaluated with GRC disabled and enabled
Disturbance compensation	Supervisory priority modification and controlled climb	Peak CTE reductions of 45.88% under lateral gusts and 58.68% under combined gusts
Validation outcome	Behavioural-control concept demonstrated	Integrated nonlinear control operation and quantitative performance demonstrated

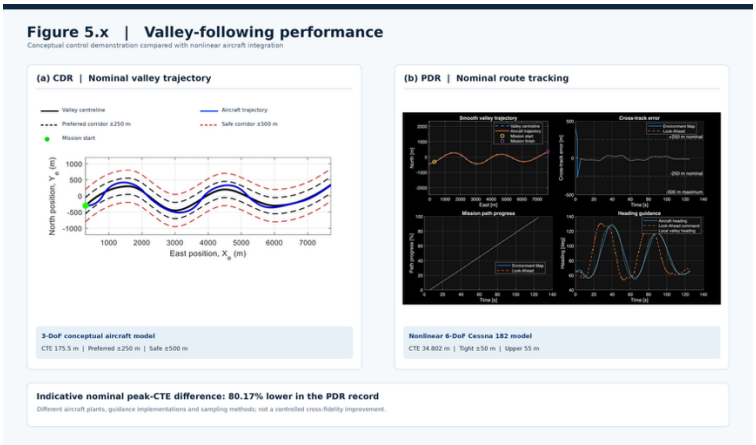


Figure 37 Comparison of nominal valley-following trajectories during CDR and PDR.

The recorded nominal peak CTE decreased from 175.5 m to 34.802 m, an indicative reduction of **80.17%**. The PDR also demonstrated tracking within the tight ±50 m corridor, compared with the broader ±250 m preferred condition used during conceptual validation. These illustrate development progress rather than a controlled comparison between identical aircraft models, routes and sampling methods.

Beyond improved tracking, the principal advancement is the evaluation of behavioural-control decisions through nonlinear coupled aircraft dynamics. Bank, heading, altitude, airspeed and actuator responses are now assessed within a common aircraft model, addressing the principal aerodynamic and modelling limitations identified during the CDR. The expanded command-deflection and gust-validation studies further establish the effectiveness and remaining limitations of the integrated control architecture.



Figure 38 Nonlinear aircraft response during nominal valley following.

Overall, the PDR demonstrates the progression from conceptual behavioural-control feasibility to quantitative validation using a more representative aircraft model, while establishing tighter tracking requirements and broader disturbance-testing conditions.

These results help us validate our model for the scope of this report and thus further provide us with a summary of objectives that were defined in section 2.2 table 2, thus establishing a comprehensive summary of what objectives have been achieved and what remains out of the scope of the current report.

7. Objectives Achieved

7.1 Objectives Achieved

The research objectives established in Section 2.2 were formulated to address the identified gaps in behavioural flight-control development, quantitative control evaluation and nonlinear aircraft validation. The Conceptual Design Review (CDR) established the feasibility of the proposed architecture using simplified three-degree-of-freedom aircraft dynamics, demonstrating that environmental conditions could influence flight-control priorities and produce corresponding aircraft responses. However, the conceptual simulations also highlighted limitations in aerodynamic coupling, propulsion and speed response, and the representation of aircraft motion.

The Preliminary Design Review (PDR) addressed these limitations through the integration of SWARAJ with a validated nonlinear six-degree-of-freedom Cessna 182 aircraft model. The expanded architecture was subsequently evaluated through controller sensitivity analysis, nominal valley following, commanded altitude and speed changes, and atmospheric disturbance simulations. The resulting achievements against the original research objectives are summarized in Table 24.

Table 24 Research objectives and achieved deliverables

Research objective	Deliverables achieved	Supporting evidence	Assessment
Objective 1 — Behavioural Flight-Control Development	Integrated behavioural supervision, predictive guidance, coordinated lateral and longitudinal control, propulsion management and supplementary disturbance recovery, with autonomous control-surface and throttle commands.	Nominal valley following achieved a peak active CTE of 34.802 m, maximum altitude error of 1.000 m and maximum bank angle of 27.347°.	Achieved
Objective 2 — Quantitative Behavioural-Control Evaluation	Defined performance measures, controller sensitivity analysis, actuator-response assessment, command-deflection evaluation and comparative disturbance-rejection testing.	Four atmospheric gust cases assessed with GRC disabled and enabled. Peak CTE reductions of 45.88% and 58.68% were obtained under lateral and combined disturbances respectively.	Achieved
Objective 3 — Progressive Nonlinear Aircraft Validation	Successful integration of the behavioural-control architecture with a validated nonlinear 6-DoF Cessna 182 aircraft model, incorporating coupled aerodynamic, propulsion and rigid-body responses.	Aircraft trim and dynamic validation, nominal manoeuvring, commanded flight-condition changes and atmospheric disturbance simulations.	Achieved at nonlinear simulation level

Achievement of Objective 1 — Behavioural Flight-Control Development

The first objective was initially demonstrated during the CDR through the successful interaction between environmental interpretation, behavioural supervision and aircraft-control commands. The PDR extended this concept by integrating predictive Look-Ahead guidance, behavioural supervision, lateral manoeuvre coordination, Banked Dynamics, longitudinal control, energy management and supplementary gust recovery within a common feedback architecture.

The nominal simulation demonstrated autonomous progression along the prescribed valley trajectory while maintaining the established altitude, airspeed and bank requirements. The

subsequent command-deflection experiments confirmed that the architecture could accommodate changes in altitude and speed requirements without substantial deterioration in lateral route tracking. Atmospheric disturbance testing further demonstrated the ability of supplementary supervisory intervention to modify corrective aircraft behaviour while preserving route progression.

These results establish that the behavioural-control concept developed during the CDR remains functional within the nonlinear aircraft simulation, fulfilling the first research objective.

Achievement of Objective 2 — Quantitative Behavioural-Control Evaluation

The second objective was addressed through a structured assessment of controller sensitivity, trajectory accuracy, aircraft-state response, actuator activity and atmospheric disturbance recovery.

The controller sensitivity analysis identified the influence of guidance and supervisory parameters on route-tracking performance, while the selected nominal configuration satisfied the prescribed cross-track error, altitude, airspeed and bank requirements. Independent and combined command-deflection tests demonstrated effective altitude-command execution and substantial airspeed response under changing flight requirements.

The atmospheric gust simulations provided further quantitative evidence of controller effectiveness. Enabling the Gust Recovery Controller reduced peak active CTE by 45.88% under lateral disturbances and 58.68% under combined disturbances. All four GRC-enabled cases maintained peak active CTE below the preferred 50 m tracking requirement, while minimum airspeed remained above the prescribed 58 m/s threshold.

The tests also established the influence of corrective lateral intervention on longitudinal aircraft behaviour. Vertical and combined gust cases produced temporary altitude deviations beyond the specified 10 m criterion, followed by recovery towards the commanded altitude. These results provide a measurable assessment of the control architecture's response and identify opportunities for further refinement of longitudinal recovery coordination.

The resulting validation programme demonstrates the effectiveness and operating characteristics of the integrated controller across the prescribed flight conditions, fulfilling the second research objective.

Achievement of Objective 3 — Progressive Nonlinear Aircraft Validation

The third objective was addressed through the progression from the simplified conceptual aircraft representation to a validated nonlinear six-degree-of-freedom Cessna 182 model.

Prior to integrating SWARAJ, the aircraft plant was evaluated through nonlinear trim verification and longitudinal and lateral-directional dynamic assessments against available reference characteristics. The model incorporates aerodynamic forces and moments, propulsion response, rigid-body inertia and coupled translational and rotational motion.

The integrated simulations subsequently demonstrated coordinated bank, heading, pitch, altitude and airspeed responses during nominal route following, command-deflection manoeuvres and prescribed atmospheric disturbances. This provided a more representative basis for examining the interaction between behavioural decisions, aircraft-control commands and the resulting flight response.

The successful implementation establishes the principal nonlinear aircraft-validation milestone defined in the original research objectives, demonstrating the developed architecture's operation within the validated C182 simulation environment.

Resolution of Limitations Identified During Conceptual Design

The principal limitations identified during conceptual design were addressed through the integration of validated nonlinear six-degree-of-freedom aircraft dynamics, replacing simplified aerodynamic and rigid-body relationships with coupled aircraft motion. Revised energy and propulsion supervision improved the physical response to commanded airspeed changes, while predictive guidance and coordinated control strengthened route-following performance. Controller sensitivity studies and expanded atmospheric disturbance simulations provided quantitative evidence of control effectiveness and recovery behaviour. Together, these developments establish that the behavioural-control concept demonstrated during the CDR can be implemented and evaluated under more representative aircraft dynamics.

7.2 Overall Research Achievement

The combined findings demonstrate that the three research objectives established in the original literature review have been achieved within the defined scope of simulation-based development and validation. The CDR established the feasibility of behavioural modification of aircraft-control priorities, while the PDR demonstrated the integration and quantitative evaluation of this concept using nonlinear six-degree-of-freedom aircraft dynamics.

The principal contribution is the successful coordination of environmental guidance, behavioural supervision, aerodynamic and propulsion-control functions, and supplementary disturbance recovery within an integrated autonomous flight-control architecture. The nominal, command-deflection and atmospheric validation results provide measurable evidence of its functionality and performance under the prescribed operating conditions.

The research therefore establishes a validated preliminary foundation for SWARAJ's autonomous flight-control architecture, with the principal modelling and integration challenges identified during conceptual development satisfactorily addressed. The resulting framework provides a basis for subsequent improvements in transient-response characterization, aircraft-specific manoeuvring-envelope management, autonomous environmental map generation and complete initial and terminal mission-command management.

8. Gaps Highlighted and Future Work

The Preliminary Design Review has demonstrated the successful integration of SWARAJ's behavioural supervisory architecture with nonlinear six-degree-of-freedom Cessna 182 aircraft dynamics. The simulation results establish satisfactory nominal trajectory tracking, coordinated aircraft-control responses and improved lateral disturbance rejection. Nevertheless, the evaluation has identified specific areas where further refinement could improve control precision, recovery performance and operational adaptability. These findings provide the basis for the next stage of development.

8.1 Gaps Highlighted

- **Route Definition:** Further refinement required to accommodate different path geometries and configurations.
- **Lateral Throttle Requirements:** Improve throttle coordination to address deficiencies in anticipated turn deceleration.
- **Gust-Induced Altitude Damping:** Optimise GRC elevator functions; investigate possible influence of 10% Master Look-Ahead reduction.
- **Structural and Load-Factor Supervision:** Incorporate aircraft-specific structural limits and load-factor requirements during banking and gust manoeuvres.
- **Predictive GRC Integration:** Incorporate established disturbance prediction with statistically calibrated uncertainty estimates.

8.2 Future Work

The gaps identified during the PDR provide the basis for SWARAJ's future development. The proposed improvements focus on refining route perceivance lateral throttle coordination, gust recovery and aircraft-specific control supervision. Beyond these refinements, future work will extend the architecture towards complete autonomous management, adaptable geographic mapping, aircraft-dependent parameter tuning and hardware-based validation.

• **Pre-Valley and Post-Valley Mission Management:** The present research concentrates on autonomous flight within the prescribed valley route. Pre-valley route acquisition, initial mission transitions and post-valley guidance were not included in the current validation scope. Future development will introduce dedicated functions to manage these flight phases and provide a more complete autonomous mission sequence.

• **Multi-Radius Cornering and Adaptive Banked Dynamics:** Future refinement of Look-Ahead guidance and Banked Dynamics will investigate multi-radius cornering strategies, improved curvature analysis and adaptive manoeuvre preparation. The objective is to improve the timing and coordination of successive turns while further reducing cross-track errors.

• **Dedicated Lateral Throttle Supervision and Coordinated Energy Management:** A proposed development is to introduce a dedicated throttle-control function within the lateral guidance architecture, operating under intelligent supervisory control. This function would determine the throttle and speed requirements associated with turn preparation, banking and

lateral recovery. Its demands would subsequently be coordinated through the central Energy Management System, which would reconcile lateral requirements with longitudinal throttle demands for altitude retention and recovery. The intended result is more effective coordination between manoeuvre preparation, propulsion response and aircraft energy distribution.

- **Generalised Geographic Mapping and Coordinate Integration:** Future development will establish a more adaptable mapping system capable of accepting different mission geometries while accounting for the physical and operational requirements of the selected aircraft. This will include converting geographical start and end coordinates into a suitable Earth-referenced coordinate system, followed by the transformations required for aircraft-relative guidance and control. Such an approach would support the integration of SWARAJ with a wider range of mission maps without fundamentally altering its core guidance architecture.

- **Aircraft-Specific Controller Parameterisation:** A systematic tuning methodology will be developed to determine suitable guidance, supervisory and control parameters according to the characteristics of the selected aircraft. This would allow SWARAJ to accommodate differences in aerodynamic response, inertia, propulsion and manoeuvring capability while retaining its fundamental control architecture and behavioural principles.

- **Sensor Integration, Hardware-in-the-Loop Validation and Wider Applications:** Subsequent development will progressively incorporate onboard sensing, environmental perception and obstacle detection and avoidance. Hardware-in-the-loop simulations will provide an intermediate stage between the present software-based validation and future physical implementation. Additional simulation studies using different aircraft configurations, mission conditions and operating environments will further evaluate the framework's suitability for broader autonomous aviation applications.

The identified gaps primarily concern the refinement and extension of capabilities already demonstrated during the PDR. The next development stage will therefore focus on improving predictive manoeuvre coordination, longitudinal and lateral energy management, and disturbance recovery, while extending SWARAJ towards more flexible mapping and aircraft-specific operation.

Together, these developments provide a structured pathway from the present validated nonlinear simulation towards a more adaptable autonomous flight-control framework, without changing the fundamental behavioural supervisory principles established through the current research.

9. Conclusion

The name SWARAJ draws inspiration from the historical legacy of Hindavi Swarajya associated with Chhatrapati Shivaji Maharaj, reflecting the principles of self-governance and independent authority (Gordon, 1993). Within this research, that philosophy takes an engineering form: developing an aircraft-control system capable of interpreting its operating conditions, making independent control decisions and responding appropriately without continuous external intervention.

The development of such a system involves more than enabling an aircraft to follow a predefined route. It requires the coordination of environmental information, predictive guidance, aircraft dynamics and flight-control priorities so that an aircraft can respond to changing conditions while maintaining its intended flight behaviour. This objective forms the foundation of SWARAJ (System for Warfare Autonomous Reconnaissance and Adaptive Judgment). Within its broader development vision, the present research principally establishes the Autonomous Adaptive Judgment component, while autonomous reconnaissance and information-oriented warfare applications remain objectives for subsequent development.

The research began by investigating whether environmental conditions could influence not only the direction of aircraft motion but also the priorities governing its aerodynamic and propulsion-control responses. The Conceptual Design Review demonstrated the feasibility of this approach using simplified three-degree-of-freedom dynamics. However, limitations in aerodynamic representation, propulsion response and coupled aircraft motion highlighted the need for a more representative simulation environment. The Preliminary Design Review addresses these limitations through the development and integration of SWARAJ with a nonlinear six-degree-of-freedom Cessna 182 aircraft model, supported by trim verification and comparisons with published aircraft dynamic characteristics. This progression follows established principles of nonlinear aircraft modelling and flight-control evaluation (Stevens, Lewis and Johnson, 2015).

The resulting architecture integrates a mathematical Environment Map, predictive Look-Ahead guidance, behavioural supervision, Banked Dynamics, lateral and longitudinal control, energy management and supplementary gust recovery. The guidance methodology draws upon established principles of anticipatory trajectory tracking (Park, Deyst and How, 2004), while the coordination of altitude, speed and propulsion requirements is consistent with energy-based flight-control principles (Lambregts, 1984). SWARAJ develops these foundations through its own supervisory logic, allowing information about approaching route geometry and aircraft response to influence the control priorities ultimately executed through the aerodynamic surfaces and throttle.

The simulation results provide quantitative evidence of the effectiveness of this integration. Controller sensitivity studies identified the influence of guidance and supervisory parameters, leading to the selection of the final Master Look-Ahead configuration. Under nominal valley-following conditions, SWARAJ achieved a maximum active cross-track error of 34.802 m and an RMS error of 22.537 m, satisfying the preferred 50 m tracking requirement. The maximum altitude deviation was 1.000 m, while the aircraft maintained airspeed between 66.888 and 67.313 m/s and remained below the prescribed 30° nominal bank limit. The evaluated 125-second simulation recorded 98.51% route progress with the completion flag

activated. Compared with the CDR's maximum sampled cross-track error of 175.5 m, these findings illustrate the progression towards tighter trajectory containment under more representative aircraft dynamics, although the different simulation configurations prevent a direct performance-equivalent comparison.

The command-deflection experiments extended this assessment beyond nominal flight by examining independent and simultaneous changes in altitude and airspeed. The aircraft successfully approached the commanded altitude of 105 m and demonstrated substantial deceleration towards the reduced speed reference. Although the resulting airspeed of approximately 63.7–63.8 m/s remained above the commanded 62.09 m/s, the aircraft retained lateral tracking comparable to the nominal response. These results demonstrate meaningful coordination between flight-path control and aircraft energy management while identifying opportunities to improve transient speed-command tracking.

The atmospheric validation programme further demonstrated the importance of adaptive control intervention. Four prescribed disturbance conditions were evaluated with the Gust Recovery Controller enabled and disabled. Under lateral and combined gust conditions, enabling GRC reduced peak cross-track error by 45.88% and 58.68%, respectively. All four GRC-enabled cases maintained peak active cross-track error below 50 m and minimum airspeed above the prescribed 58 m/s threshold. These results establish the effectiveness of supplementary recovery intervention in improving lateral containment under the tested atmospheric conditions. The vertical and combined gust cases also revealed transient altitude deviations of 10.502 m and 11.333 m, marginally exceeding the specified 10 m criterion before recovery towards the reference altitude. This identifies an important area for further coordination between gust recovery, longitudinal control and energy management.

Together, the nominal, command-deflection and atmospheric validation results demonstrate the achievement of the three principal research objectives within the defined nonlinear simulation scope: behavioural flight-control development, quantitative evaluation of supervisory responses and progressive nonlinear aircraft validation. More importantly, they establish the central contribution of this research: the successful coordination of environmental guidance, adaptive supervisory decisions and aircraft-control execution within a common autonomous architecture. The aircraft does not simply respond to deviations after they occur; its supervisory functions also consider approaching manoeuvres, changing aircraft states and recovery requirements when determining the appropriate control response.

The next stage of SWARAJ will build upon these achievements through improved route definition, multi-radius manoeuvre prediction, refined lateral throttle coordination, predictive gust recovery and aircraft-specific structural and load-factor supervision. Further developments in geographic mapping, aircraft-dependent controller tuning, mission management, onboard sensing and hardware-in-the-loop validation will support the gradual extension of the architecture towards more complete autonomous operation. These developments will also provide the foundation for incorporating reconnaissance and information-processing capabilities into the broader SWARAJ framework.

Ultimately, this Preliminary Design Review establishes more than the ability of a simulated aircraft to negotiate a valley trajectory. It demonstrates how environmental information, aircraft physics and behavioural supervision can be brought together to produce coordinated autonomous decisions under changing flight requirements. The present achievements remain specific to guidance and flight control, but they represent an important foundation for

SWARAJ's wider objectives. In returning to the principle from which its name originates, the purpose of this research becomes clear: to advance autonomy through the ability to interpret, decide and act independently, while remaining accountable to the physical principles and engineering requirements that govern safe and dependable flight.

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